Nucleus and Chromosomes

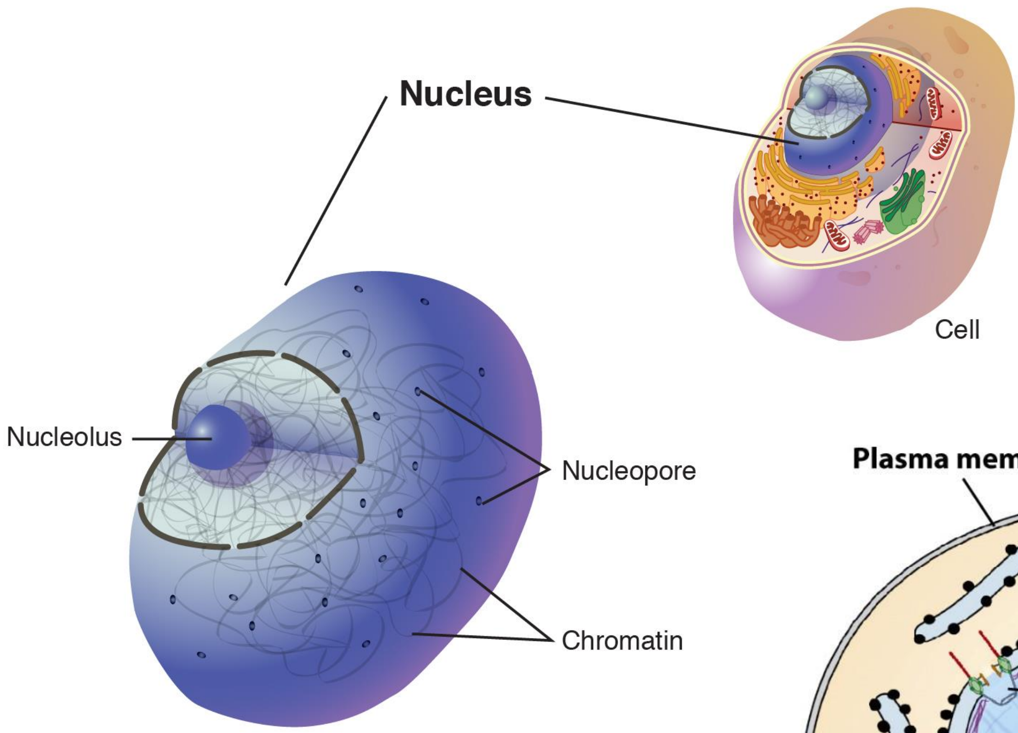
(Albert's textbook Chapter 4)

2026/04/16

Topics

- The Nucleus
- The structure and function of DNA
- DNA packaging in chromosomes
- The high-order structure of chromosomes
- Chromosome modifications
- How genomes evolve

NUCLEUS



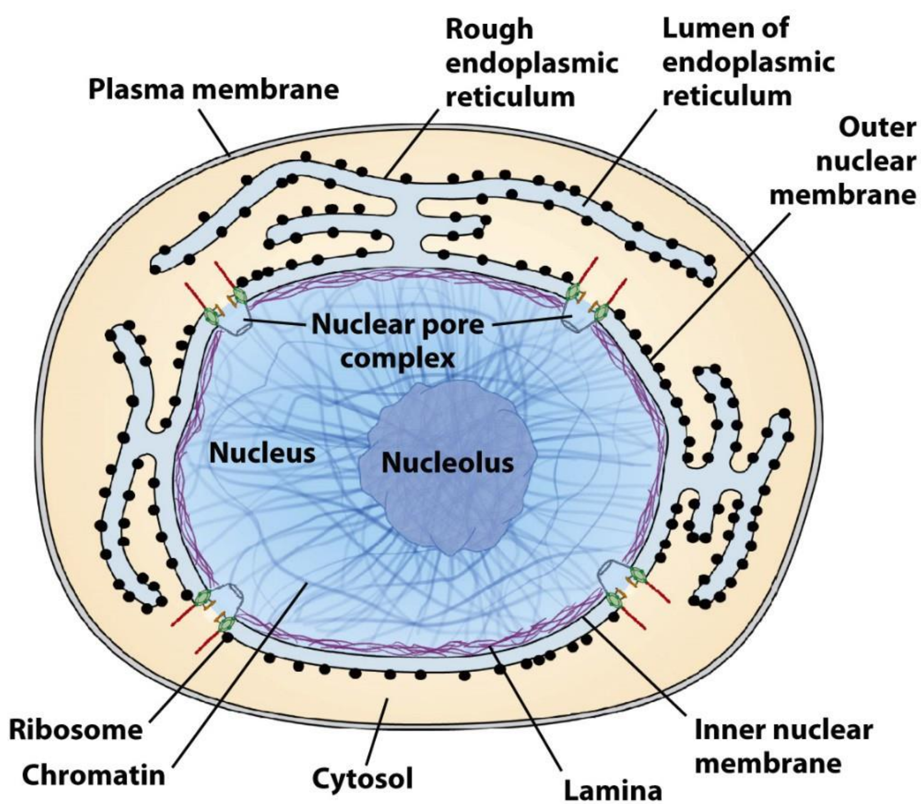
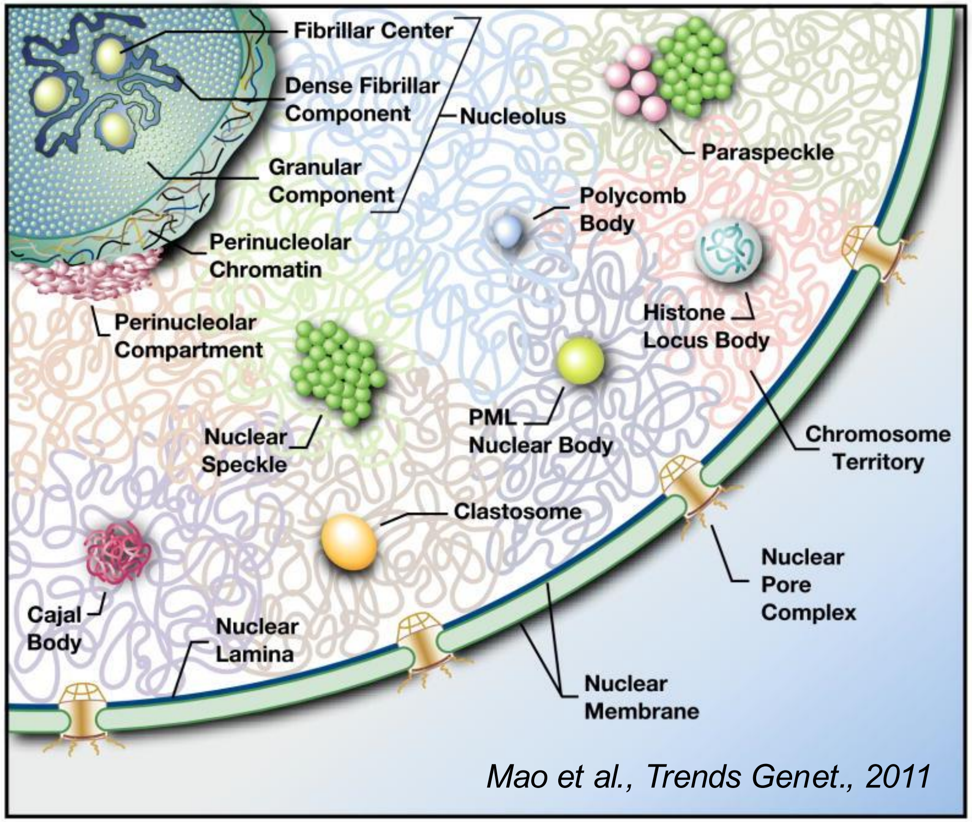


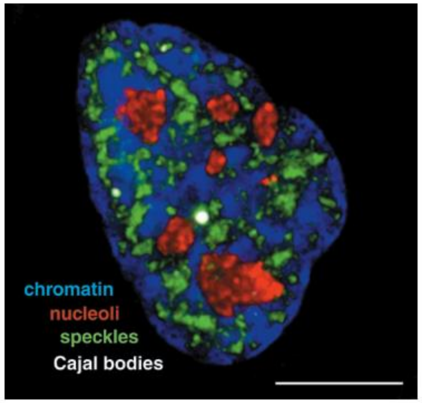
Figure 1-15a
Molecular Cell Biology, Eighth Edition

https://www.genome.gov/genetics-glossary/Nucleus

Nuclear Bodies——Functional Compartmentalization of The Nucleus



Mao et al., Trends Genet., 2011



Membrane less structures and highly dynamic.

Create distinct biochemical environments by immobilizing select groups of macromolecules to achieve great efficiency through complex reaction pathways.

.

The largest is nucleolus, which is to synthesize rRNA and assemble ribosomes.

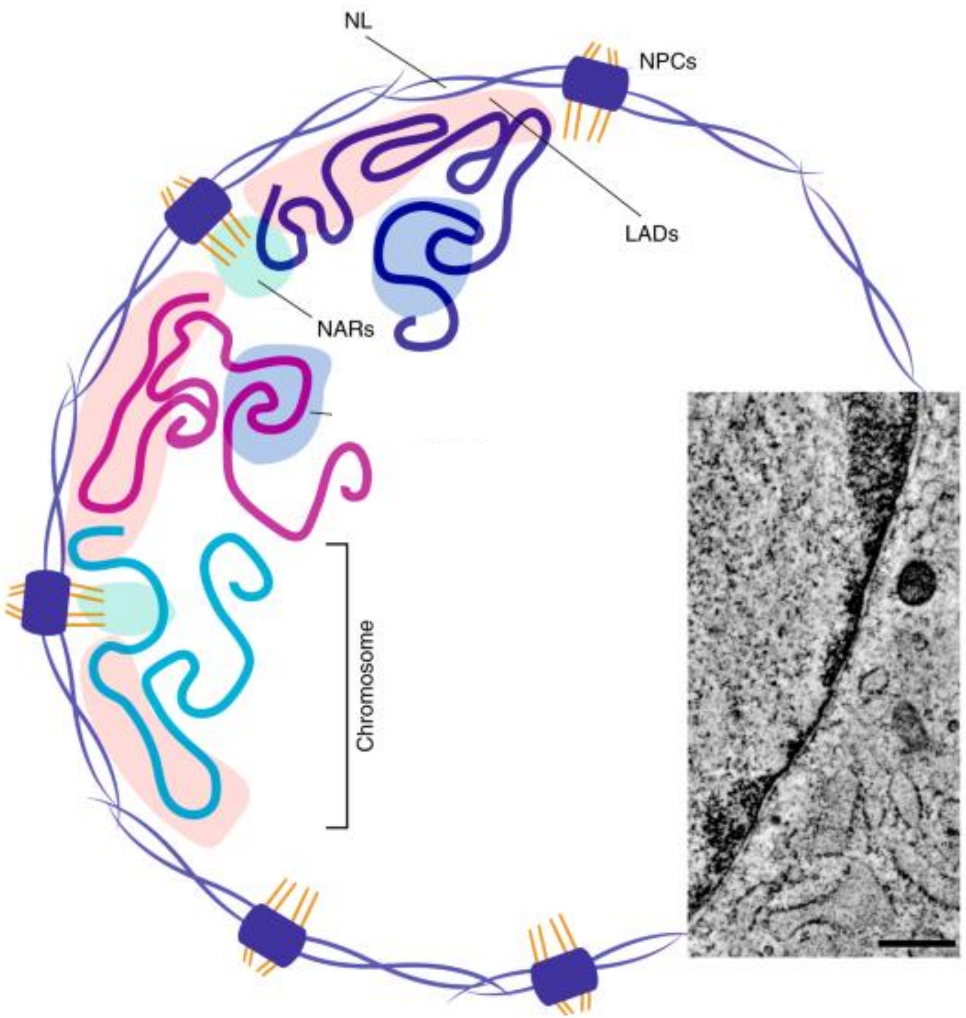
CBs are relating to RNA processing, specifically small nucleolar RNA (snoRNA) and small nuclear RNA (snRNA) maturation.

Speckles are subnuclear structures that are enriched in pre-messenger RNA splicing factors.

Refer to Chapter 6 for more.

Nuclear Lamina

The NL is a dense (~30 to 100 nm thick) fibrous meshwork.

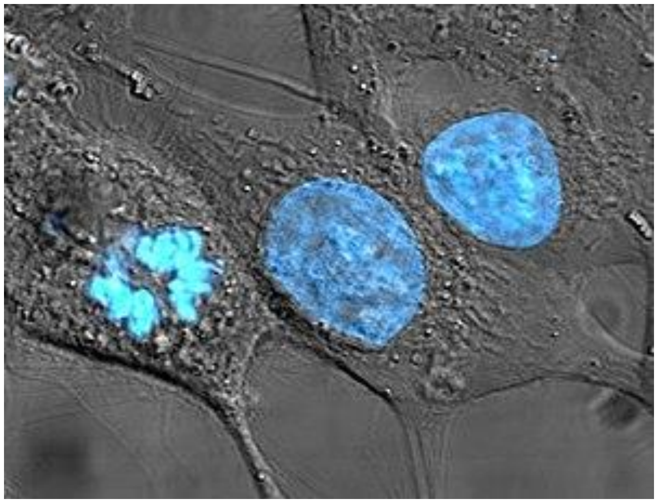


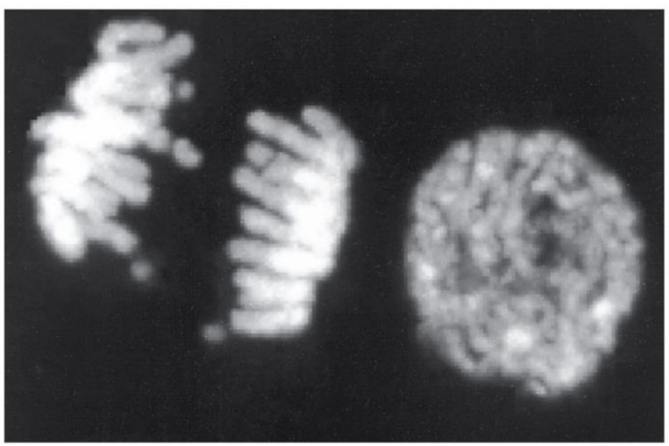
Karoutas et al., Nat Cell Biol (2021)

- The nuclear lamina consists of two components, lamins and nuclear lamin-associated membrane proteins.
- The lamins are type V intermediate filaments which can be categorized as either A-type (lamin A, C) or B-type (lamin B1, B2).
- Besides providing mechanical support, the nuclear lamina regulates important cellular events such as DNA replication and cell division.
- NL participates in chromatin organization and it anchors the nuclear pore complexes embedded in the nuclear envelope.

In Eukaryotes, DNA Is Enclosed in a Cell Nucleus

HeLa cells were stained with the blue fluorescent Hoechst dye.





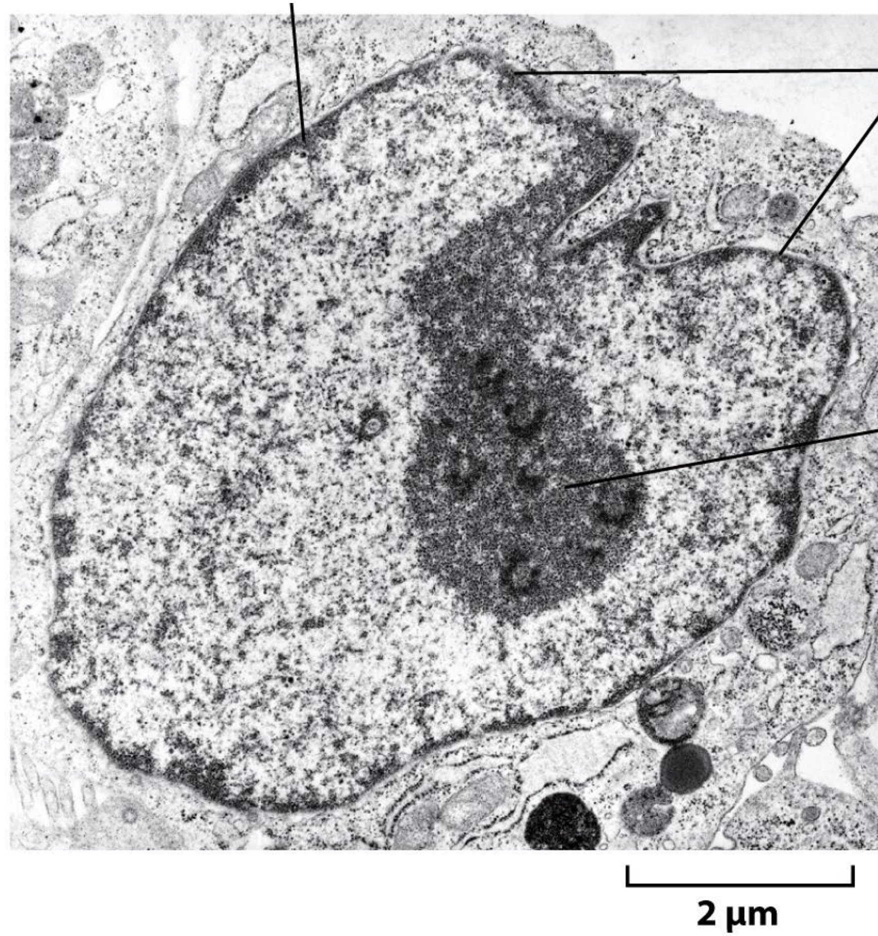


Figure 4-1, 9 *Molecular Biology of the Cell* (© Garland Science 2015)

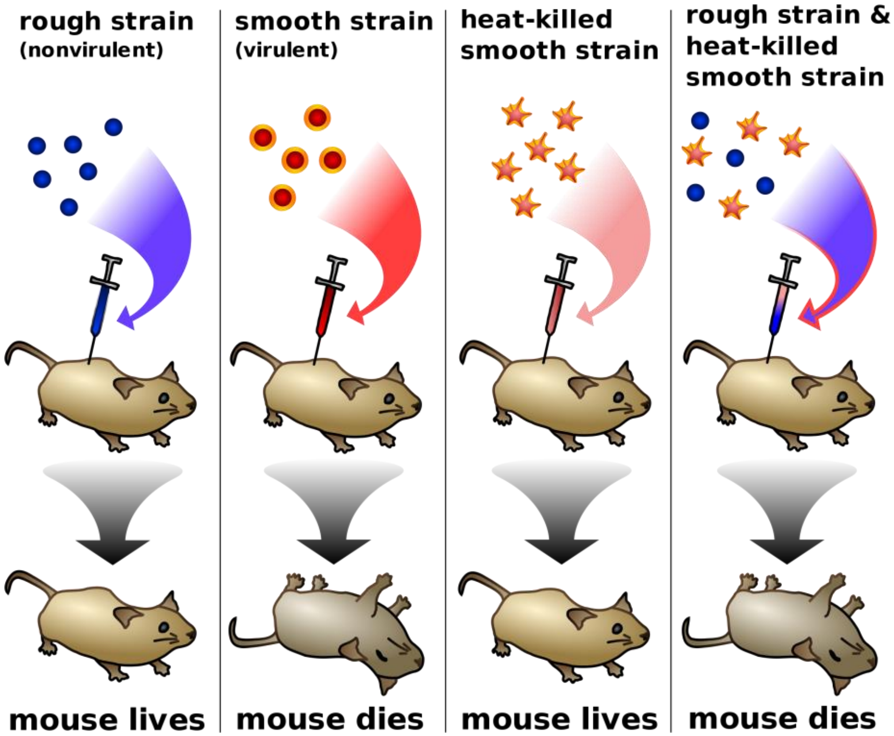
THE STRUCTURE AND FUNCTION OF DNA

- DNA Is The Hereditary Material.
- A DNA Molecule Contains Two Complementary Chains of Nucleotides.
- Double Helix Structure Of DNA

DNA is the Particular Transforming Principle

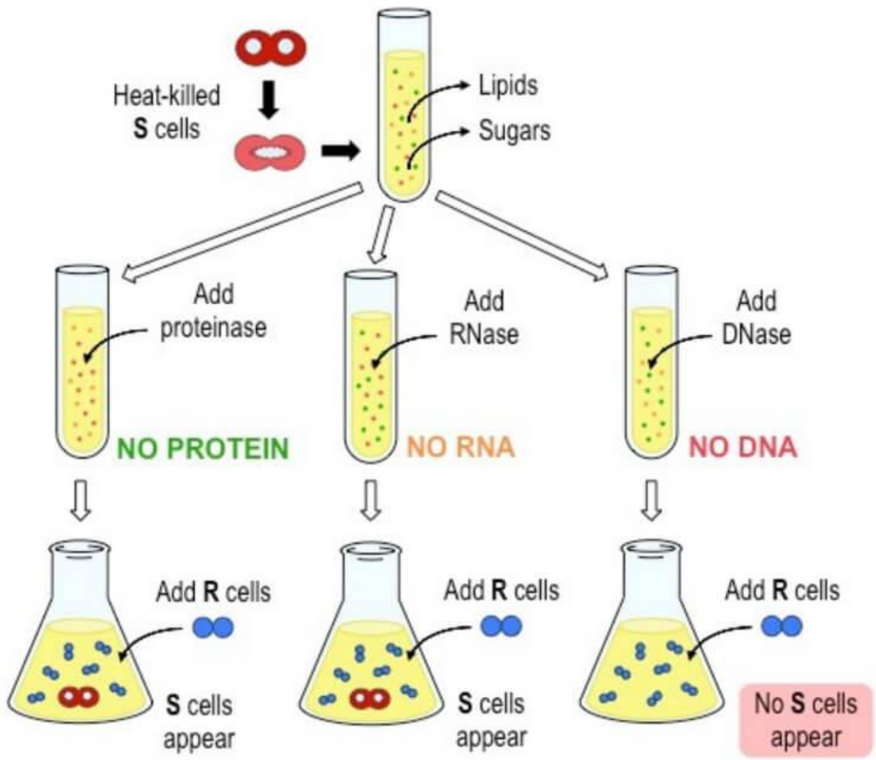
Griffith's Transformation Experiment (1928)

Colonies of Rough (top) and Smooth (bottom) strains of S. pneumoniae.



The transforming principle

Avery, McCarty, and MacLeod Experiment

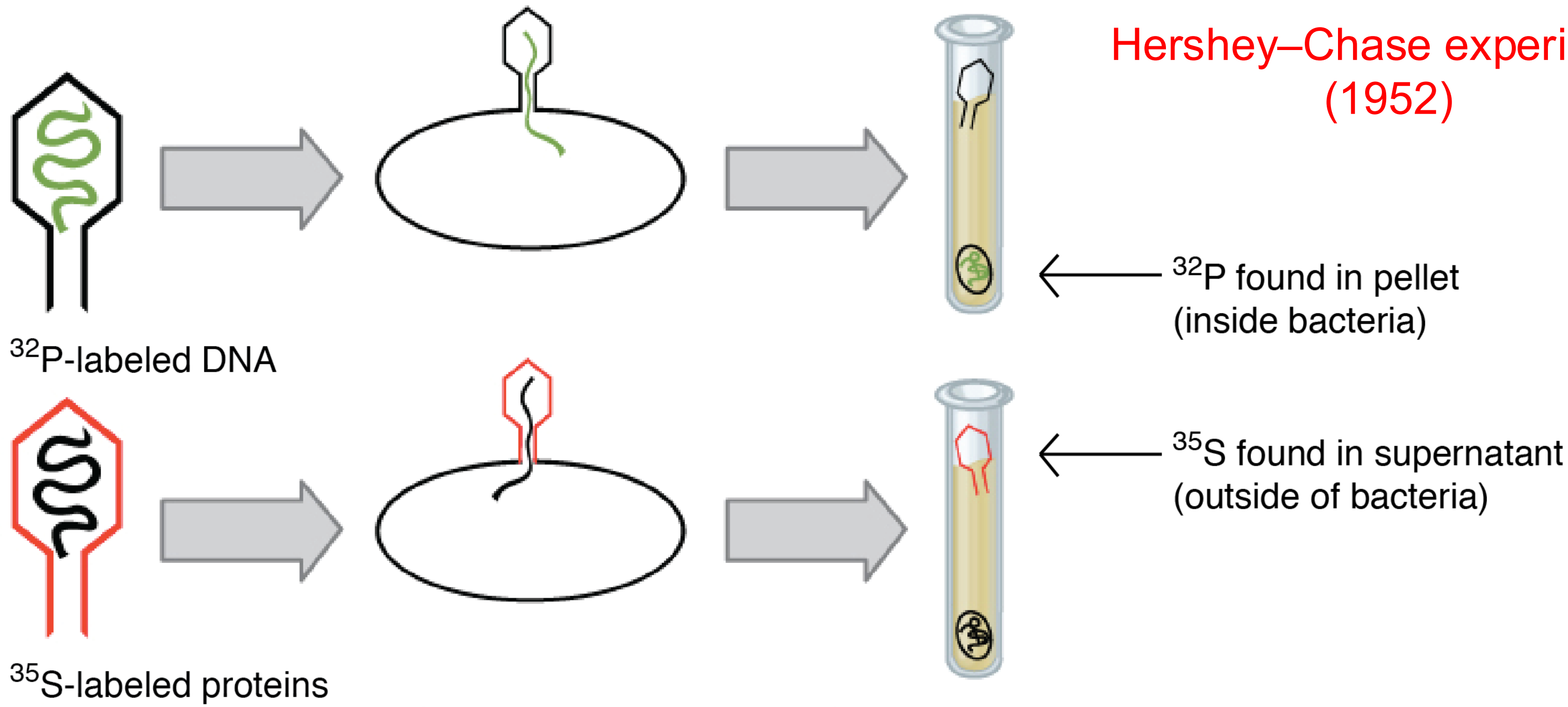


The biochemical nature of Griffith's transforming principle

https://microbenotes.com/dna-experiments-griffith-avery-mccarty-macleod-hershey-chase/

DNA Is The Hereditary Material

Hershey–Chase experiment (1952)



1 One batch of phage was labeled with ^{35}S, which is incorporated into the protein coat. Another batch was labeled with ^{32}P, which is incorporated into the DNA.

2 Bacteria were infected with the phage.

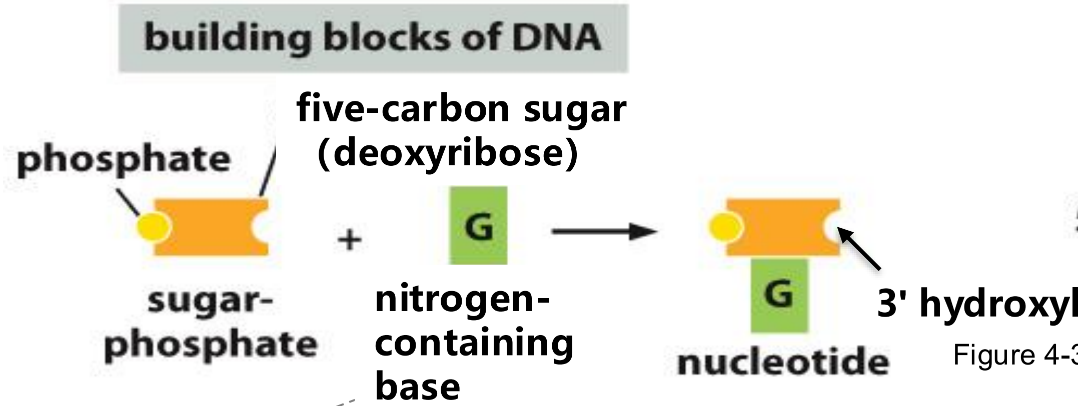
3 The cultures were blended and centrifuged to separate the phage from the bacteria.

4 Radioactivity was measured in the pellet and liquid (supernatant) for each experiment.

https://openstax.org/books/biology/pages/14-1-historical-basis-of-modern-understanding

A DNA Molecule is Built with Nucleotides

Deoxyribonucleic acid (DNA)



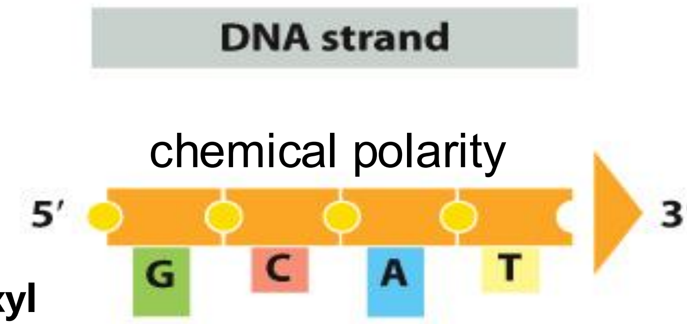
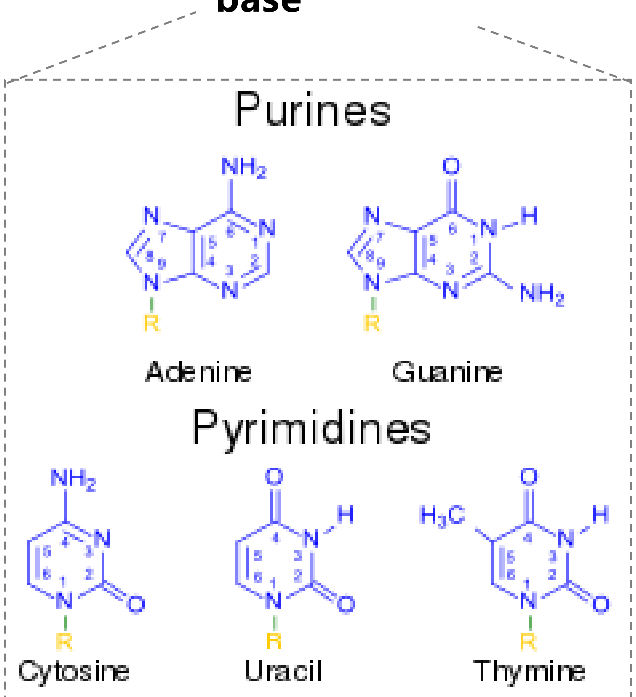
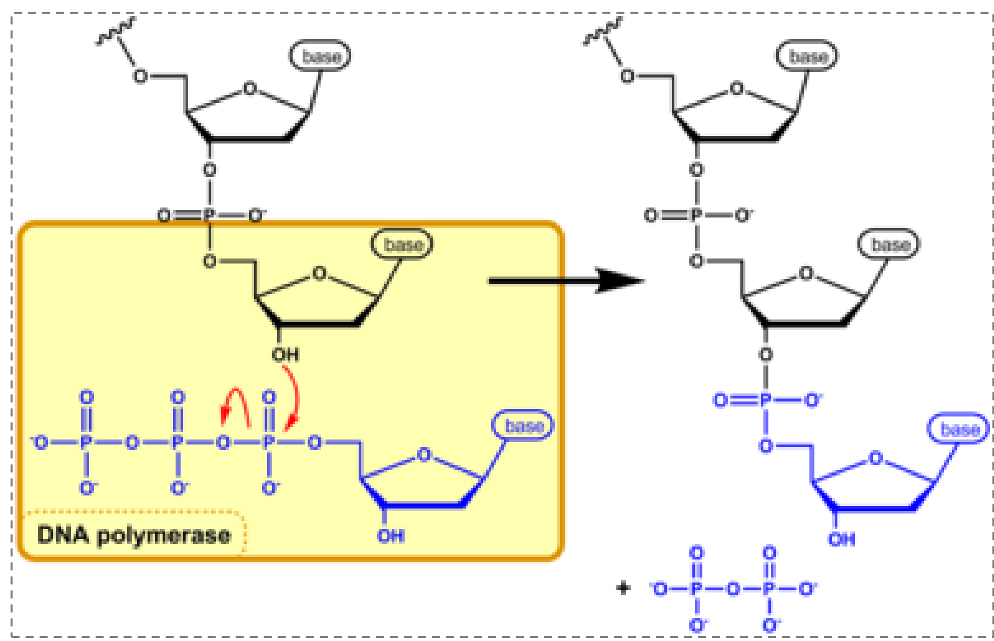


Figure 4-3 *Molecular Biology of the Cell* (© Garland Science 2015)

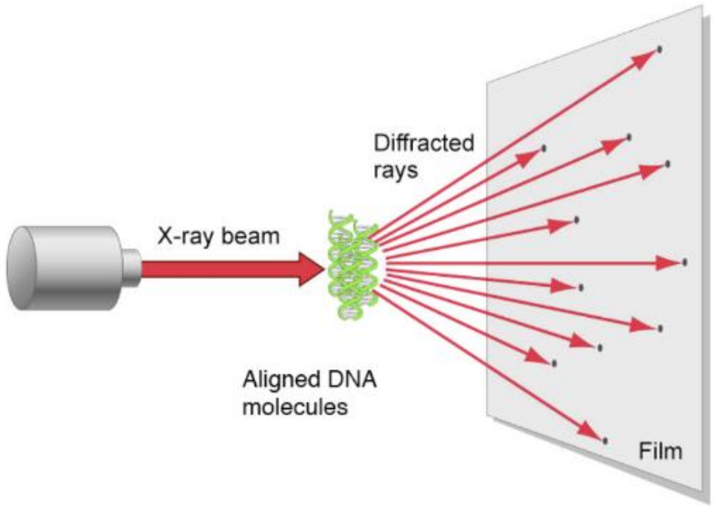


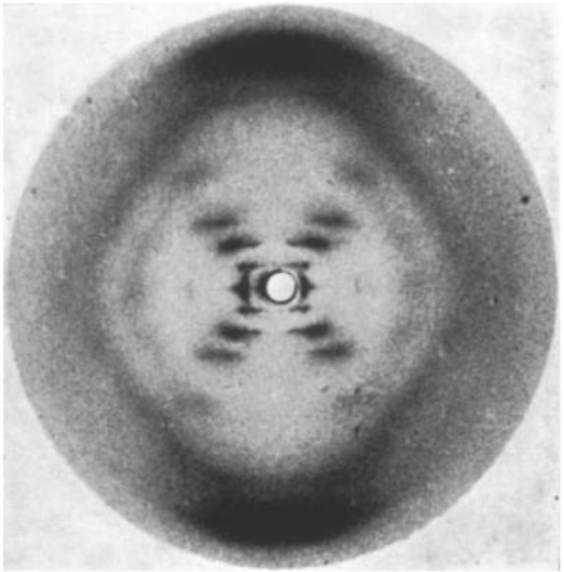


https://en.wikipedia.org/wiki/Nucleoside_triphosphate

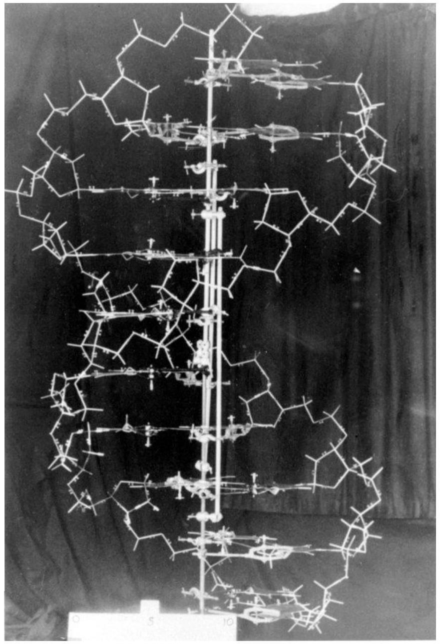
Figure 4-3 *Molecular Biology of the Cell* (© Garland Science 2015)

Double Helix Structure Of DNA ——A Turning Point In Science



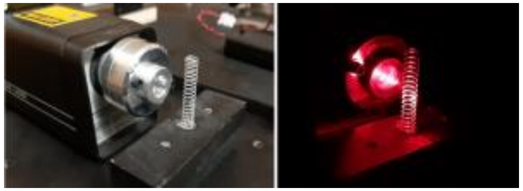


X-ray diffraction photo of DNA
1951



1953

1962 Nobel Prize in Physiology or Medicine



https://physicsopenlab.org/2019/10/01/double-helix-optical-diffraction-pattern/

https://www.youtube.com/watch?v=2tMuMRY1oDo

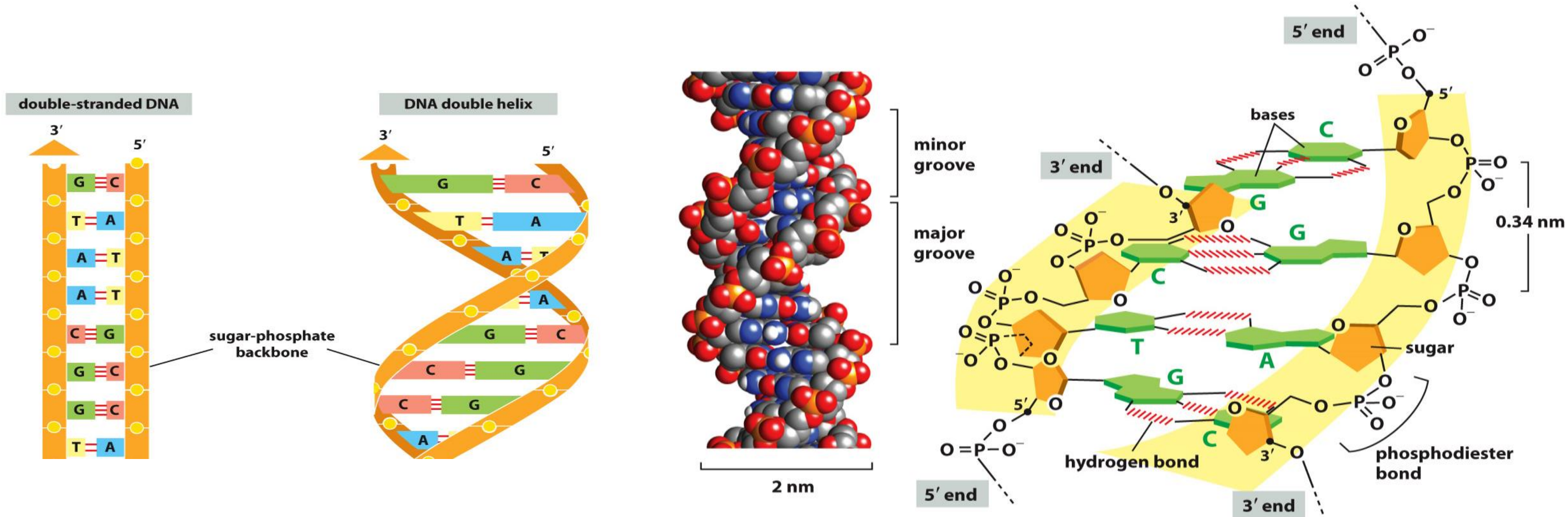
Rosalind Franklin
(1920–1958)

Maurice Wilkins
(1916–2004)

James Watson
（1928-）

Francis Crick
(1916–2004)

A DNA Molecule Contains Two Complementary Chains of Nucleotides



- The two complementary strands wound around each other to form a double helix, which is stabilized by weak hydrogen bonds.
- Bases are on the inside and the sugar-phosphate backbones are on the outside.
- Purines pair with pyrimidines: A always pairs with T, and G with C. Each base pair is of similar width and is the energetically most favorable arrangement.
- The two strands of the helix are antiparallel——the polarity of one strand is opposite to the other strand.

Figure 4-3, 4-5 *Molecular Biology of the Cell* (© Garland Science 2015)

The Structure of DNA Provides a Mechanism for Heredity

Two most fundamental questions about heredity:

1. How could the functional information be carried in a chemical form?
2. How could this information be passed from generation to generation?

The Structure of DNA Provides a Mechanism for Heredity

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1. How could the functional information be carried in a chemical form?
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Sequence and Structure

DNA is a linear polymer of four different kinds of monomers in a defined sequence.

The double-stranded structure

DNA as a template for its own duplication.

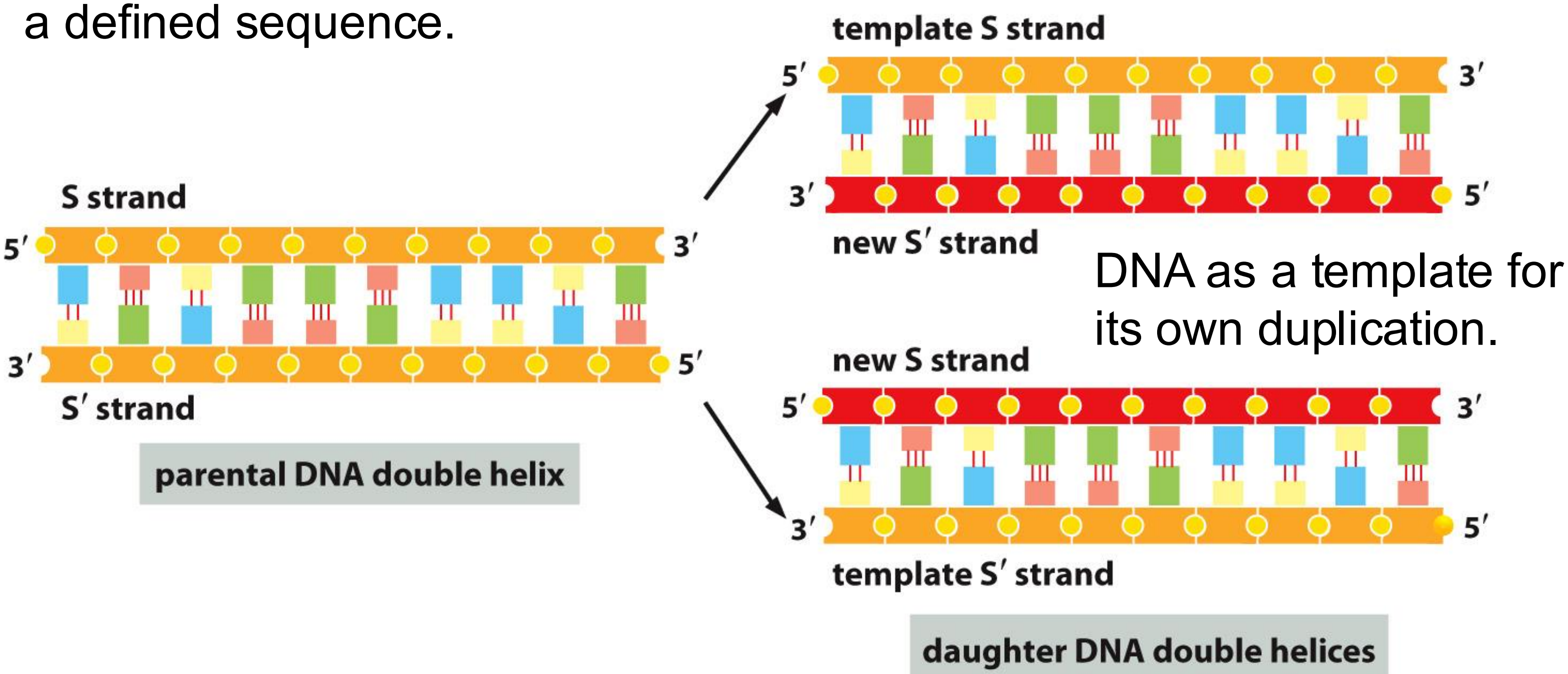


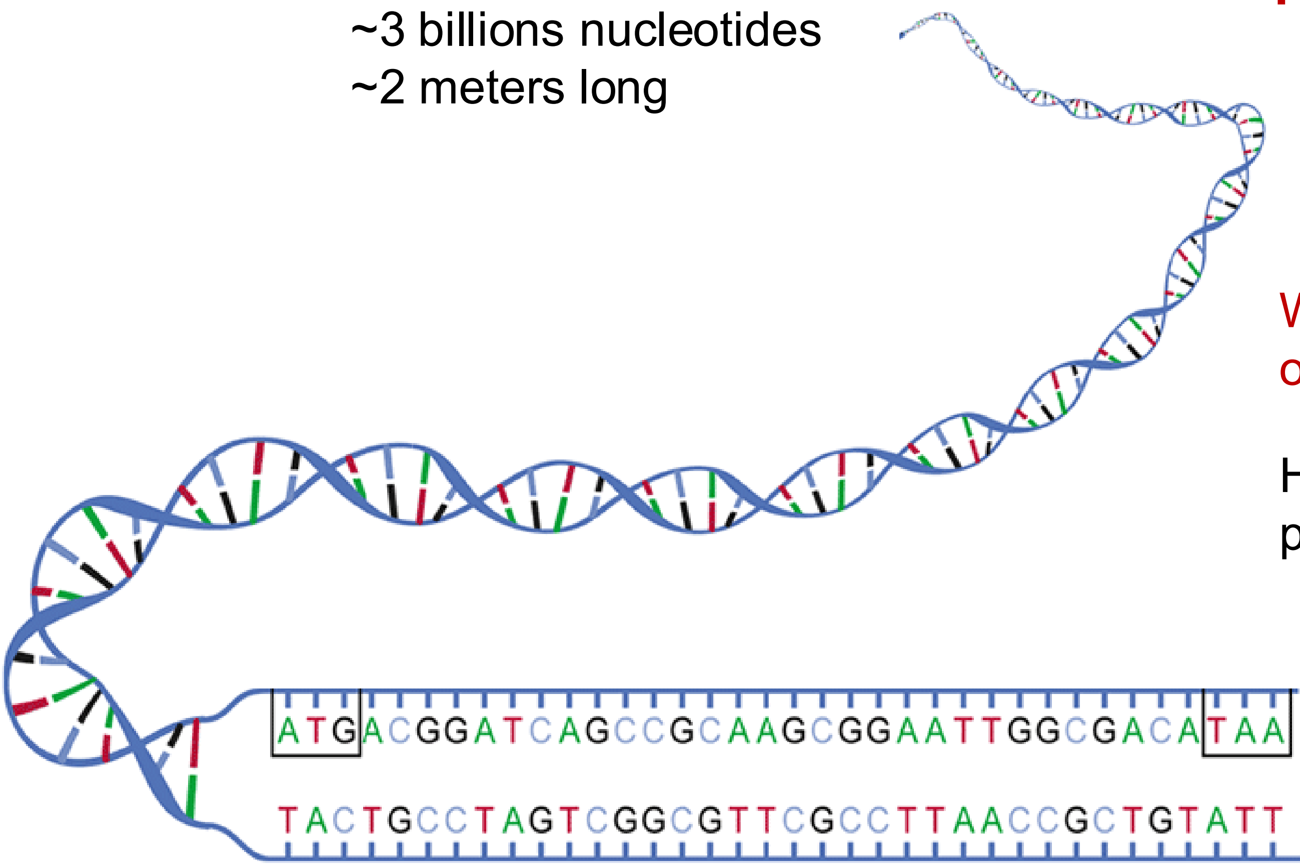
Figure 4-6 Molecular Biology of the Cell 6e (© Garland Science 2015)

From DNA To Genome

The complete store of information in an organism's DNA is called its genome

Human Genome:
~3 billions nucleotides
~2 meters long

Sequence and Structure



genome.gov

What's the DNA sequence of a human genome?

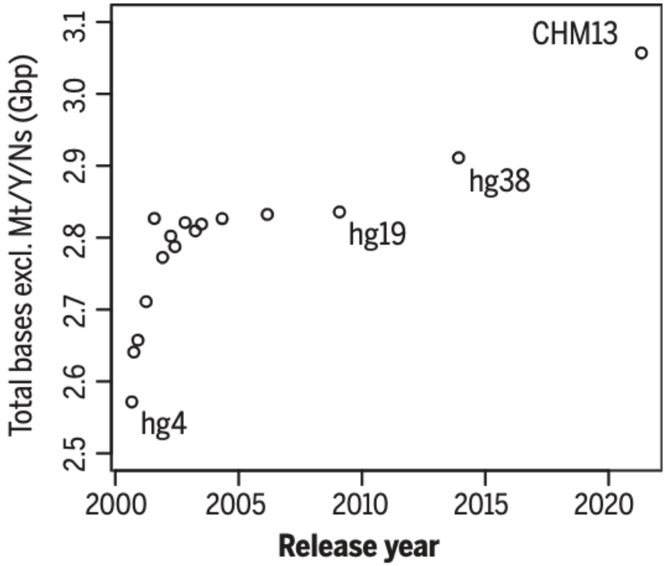
How the DNA is packed in the nucleus?

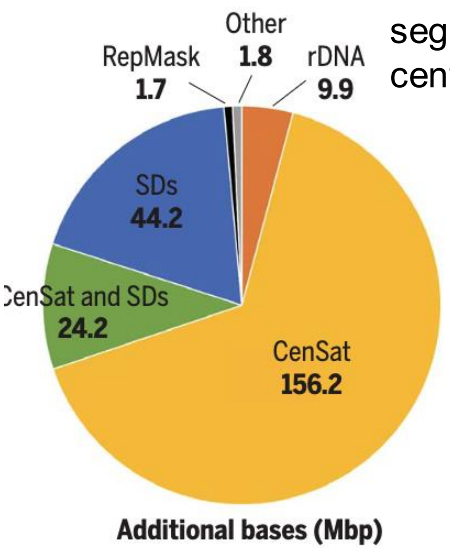
The Complete Sequence Of A Human Genome

Telomere-to-Telomere

newly completed regions

~8% of the human genome





segmental duplications (SDs)
centromeric satellites (CenSat)

Nurk et al., Science, 2022

The Nucleotide Sequence of the Human Genome

```
CCCTGTGGAGCCACACCCTAGGGTTGGCCA
ATCTACTCCCAGGAGCAGGGAGGGCAGGAG
CCAGGGCTGGGCATAAAAGTCAGGGCAGAG
CCATCTATTGCTTACATTTGCTTCTGACAC
AACTGTGTTCACTAGCAACTCAAACAGACA
CCATGGTGCACCTGACTCCTGAGGAGAAGT
CTGCCGTTACTGCCCTGTGGGGCAAGGTGA
ACGTGGATGAAGTTGGTGGTGAGGCCCTGG
GCAGGTTGGTATCAAGGTTACAAGACAGGT
TTAAGGAGACCAATAGAAACTGGGCATGTG
GAGACAGAGAAGACTCTTGGGTTTCTGATA
GGCACTGACTCTCTCTGCCTATTGGTCTAT
TTTCCCACCCTTAGGCTGCTGGTGGTCTAC
CCTTGGACCCAGAGGTTCTTTGAGTCCTTT
GGGGATCTGTCCACTCCTGATGCTGTTATG
GGCAACCCTAAGGTGAAGGCTCATGGCAAG
AAAGTGCTCGGTGCCTTTAGTGATGGCCTG
GCTCACCTGGACAACCTCAAGGGCACCTTT
GCCACACTGAGTGAGCTGCACTGTGACAAG
CTGCACGTGGATCCTGAGAACTTCAGGGTG
AGTCTATGGGACCCTTGATGTTTTCTTTCC
CCTTCTTTTCTATGGTTAAGTTCATGTCAT
AGGAAGGGGAGAAGTAACAGGGTACAGTTT
AGAATGGGAAACAGACGAATGATTGCATCA
GTGTGGAAGTCTCAGGATCGTTTTAGTTTC
TTTTATTTGCTGTTCATAACAATTGTTTTC
TTTTGTTTAATTCTTGCTTTCTTTTTTTTT
CTTCTCCGCAATTTTTACTATTATACTTAA
TGCCTTAACATTGTGTATAACAAAAGGAAA
TATCTCTGAGATACATTAAGTAACTTAAAA
AAAAACTTTACACAGTCTGCCTAGTACATT
ACTATTTGGAATATATGTGTGCTTATTTGC
ATATTCATAATCTCCCTACTTTATTTTCTT
TTATTTTTAATTGATACATAATCATTATAC
ATATTTATGGGTTAAAGTGTAATGTTTTAA
TATGTGTACACATATTGACCAAATCAGGGT
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TTATTTCTAATACTTTCCCTAATCTCTTTC
TTTCAGGGCAATAATGATACAATGTATCAT
GCCTCTTTGCACCATTCTAAAGAATAACAG
TGATAATTTCTGGGTTAAGGCAATAGCAAT
ATTTCTGCATATAAATATTTCTGCATATAA
ATTGTAACTGATGTAAGAGGTTTCATATTG
CTAATAGCAGCTACAATCCAGCTACCATTC
TGCTTTTATTTTATGGTTGGGATAAGGCTG
GATTATTCTGAGTCCAAGCTAGGCCCTTTT
GCTAATCATGTTCATACCTCTTATCTTCCT
CCCACAGCTCCTGGGCAACGTGCTGGTCTG
TGTGCTGGCCCATCACTTTGGCAAAGAATT
CACCCCACCAGTGCAGGCTGCCTATCAGAA
AGTGGTGGCTGGTGTGGCTAATGCCCTGGC
CCACAAGTATCACTAAGCTCGCTTTCTTGC
TGTCCAATTTCTATTAAAGGTTCCTTTGTT
CCCTAAGTCCAACTACTAAACTGGGGGATA
TTATGAAGGGCCTTGAGCATCTGGATTCTG
```

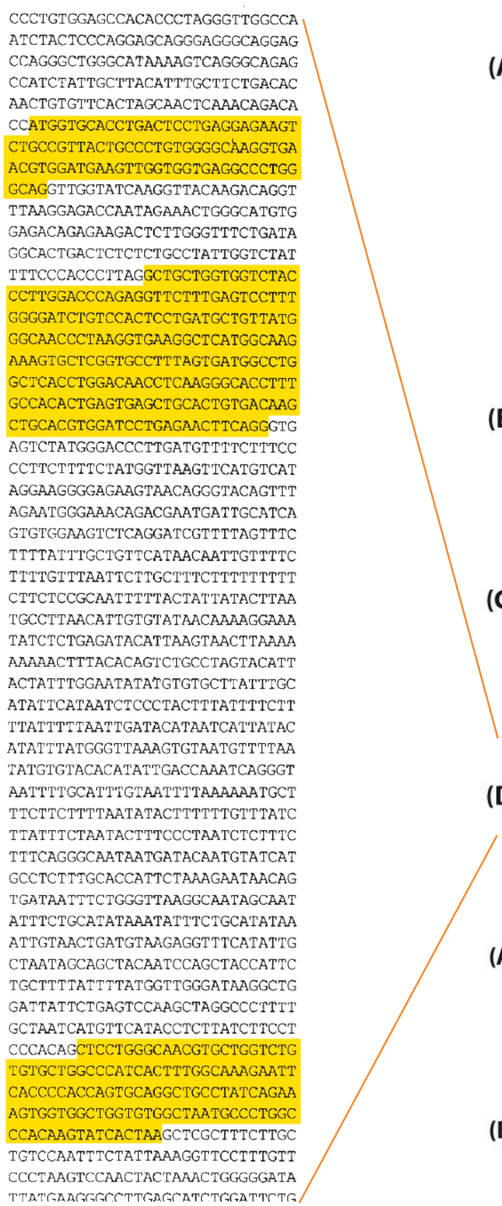
TABLE 4–1 Some Vital Statistics for the Human Genome

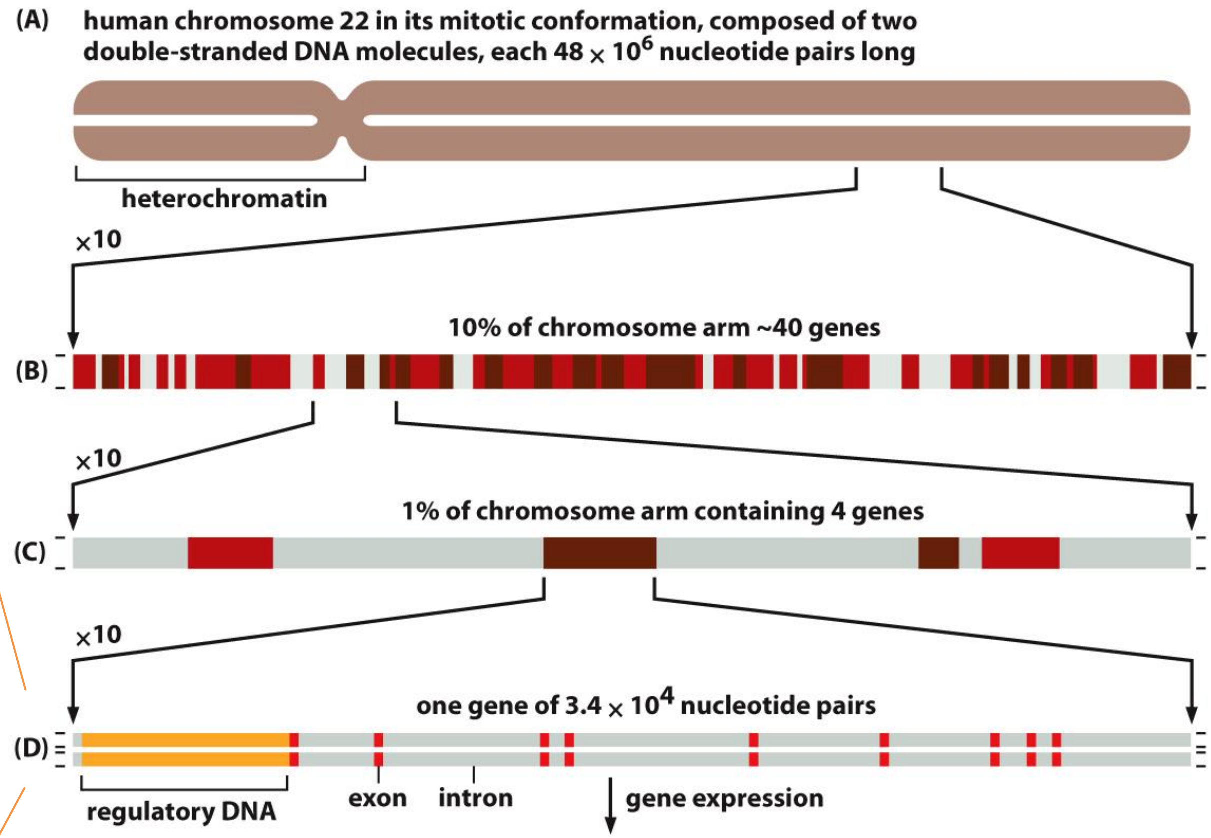
Human genome	
DNA length	3.2×10^9 nucleotide pairs*
Number of genes coding for proteins	Approximately 21,000
Largest gene coding for protein	2.4×10^6 nucleotide pairs
Mean size for protein-coding genes	27,000 nucleotide pairs
Smallest number of exons per gene	1
Largest number of exons per gene	178
Mean number of exons per gene	10.4
Largest exon size	17,106 nucleotide pairs
Mean exon size	145 nucleotide pairs
Number of noncoding RNA genes	Approximately 9000**
Number of pseudogenes***	More than 20,000
Percentage of DNA sequence in exons (protein-coding sequences)	1.5%
Percentage of DNA in other highly conserved sequences****	3.5%
Percentage of DNA in high-copy-number repetitive elements	Approximately 50%

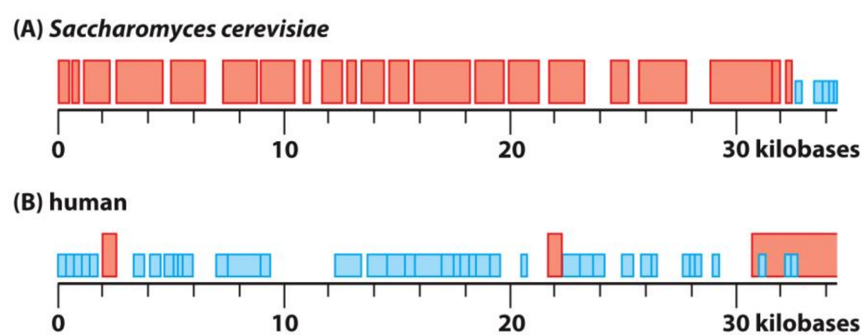
* The sequence of 2.85 billion nucleotides is known precisely (error rate of only about 1 in 100,000 nucleotides). The remaining DNA primarily consists of short sequences that are tandemly repeated many times over, with repeat numbers differing from one individual to the next. These highly repetitive blocks are hard to sequence accurately.
** This number is only a very rough estimate.
*** A pseudogene is a DNA sequence closely resembling that of a functional gene, but containing numerous mutations that prevent its proper expression or function. Most pseudogenes arise from the duplication of a functional gene followed by the accumulation of damaging mutations in one copy.
**** These conserved functional regions include DNA encoding 5′ and 3′ UTRs (untranslated regions of mRNA), DNA specifying structural and functional RNAs, and DNA with conserved protein-binding sites.

Chromosomes Contain Long Strings of Genes

Chromosomes carry genes

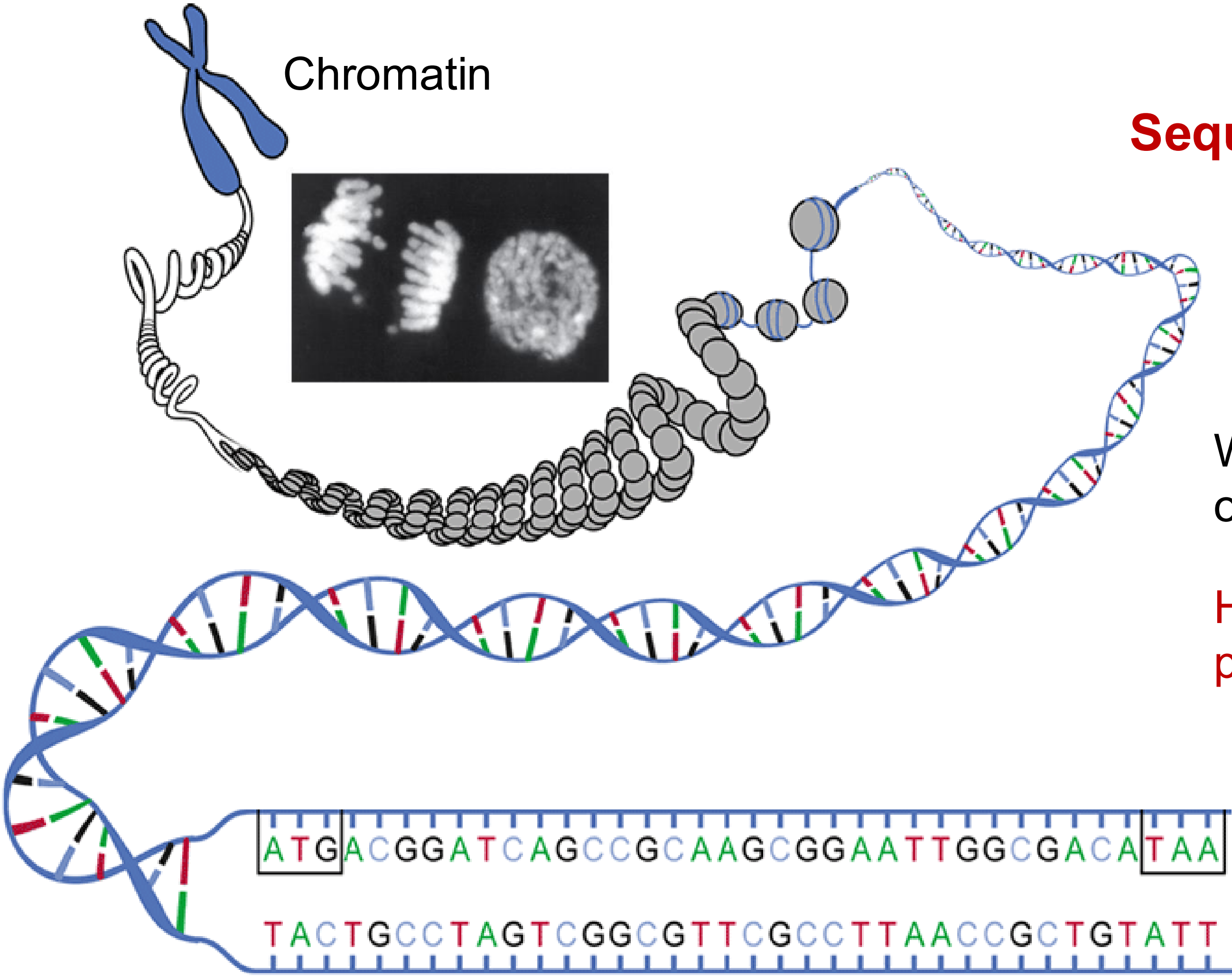






200 times larger

Eukaryotic DNA Is Packaged into a Set of Chromosomes



genome.gov

Sequence and Structure

What's the DNA sequence of a human genome?

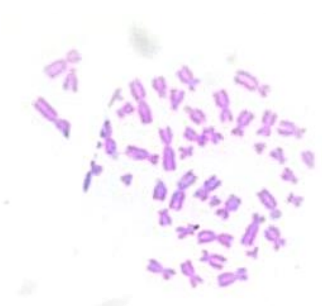
How the DNA is packed in the nucleus?

CHROMOSOME PACKAGING

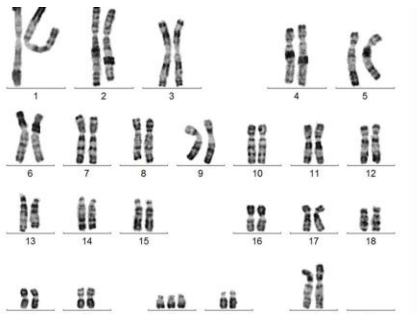
- Eukaryotic DNA Is Packaged into a Set of Chromosomes
- Chromosomes Contain Long Strings of Genes.
- Nucleosomes Are a Basic Unit of Eukaryotic Chromosome Structure.

Complete Set of Human Chromosomes - Karyotype

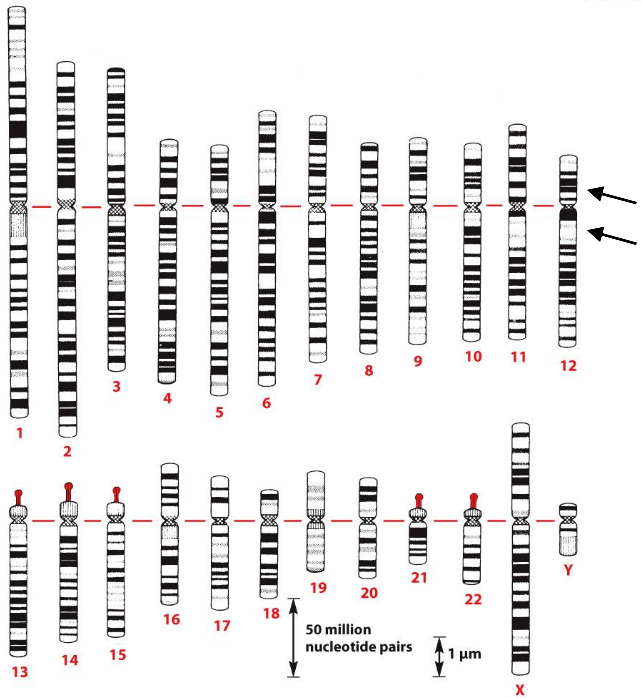
Giemsa banding
（G-banding）



Karyotype



- The pattern of bands on chromosomes is unique.
- It provided the initial means to identify and number each human chromosome reliably.
- Chromosomes form in pairs are called homologous chromosomes (homologs).



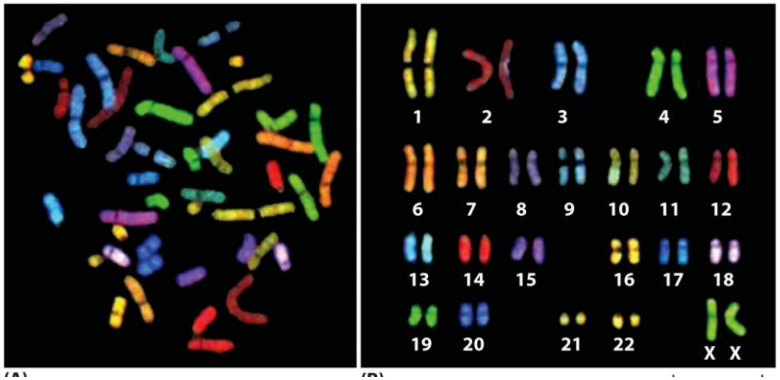
Human genome contains 46 chromosomes—22 pairs autosomes plus two sex chromosomes.

The sex chromosomes are nonhomologous in males.

Figure 4-10, 11 *Molecular Biology of the Cell* (© Garland Science 2015)

Aberrant Human Chromosomes

Spectral karyotyping



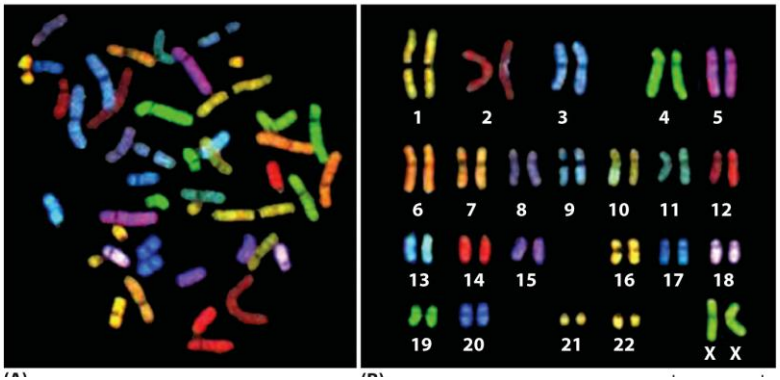
chromosome painting（Multiplex-FISH）

<u>Karyotyping is used to detect:</u>
Aneuploid
Trisomy
Structure variations

Figure 4-11, 12 *Molecular Biology of the Cell* (© Garland Science 2015)

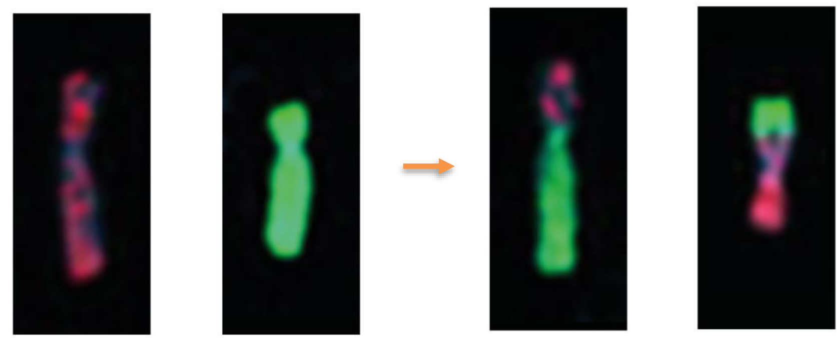
Aberrant Human Chromosomes

Spectral karyotyping

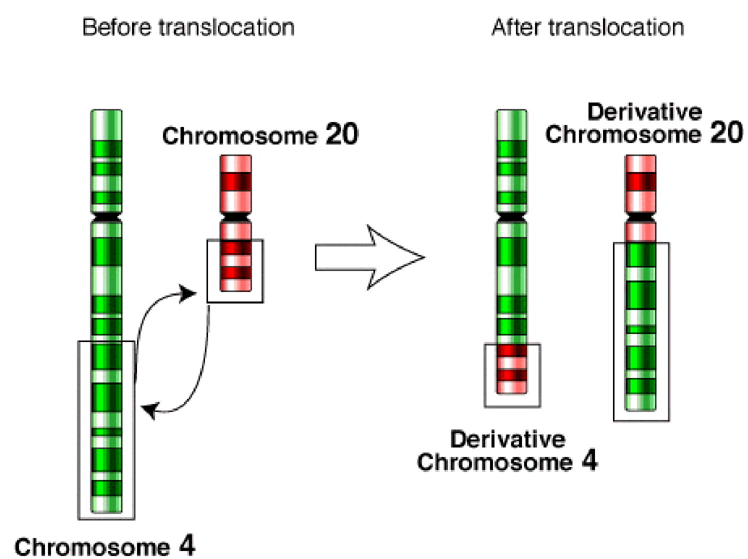


chromosome painting（Multiplex-FISH）

chromosomal translocation



chromosome 6 chromosome 4 reciprocal chromosomal translocation



<u>Karyotyping is used to detect:</u>

Aneuploid

Trisomy

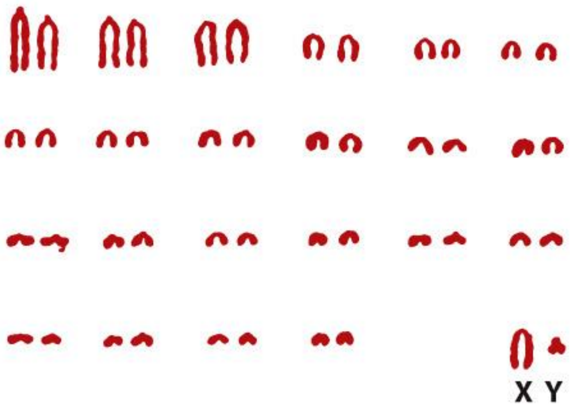
Structure variations

Figure 4-11, 12 *Molecular Biology of the Cell* (© Garland Science 2015)

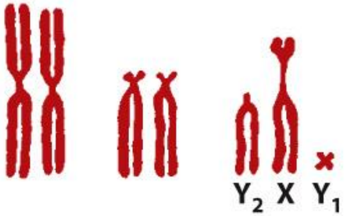
Chromosomes Differ Between Eukaryotic Species

Common Name	Species	Diploid number	Common Name	Species	Diploid number
Animals (2n)			Plants (2n)		
Human	*Homo sapiens*	46	Corn	*Zea mays*	20
Monkey	*Macaca mulatta*	42	Potato	*S. tuberosum*	48
Dog	*Canis familiaris*	78	Green algae	*A. mediterranea*	20
Cat	*Felis domesticus*	38			

Two closely related species of muntjac with very different chromosome numbers.



Chinese muntjac



Indian muntjac

Figure 4-14 Molecular Biology of the Cell 6e (© Garland Science 2015)

Chromosome number does not provide a valid indication of genetic complexity

Chromosome Must Contain a Centromere, Two Telomeres, and Replication Origins

telomere

To replicate and protect the end of the chromosome

replication origin

To start the duplication of the DNA

centromere

To separate two sister chromatids

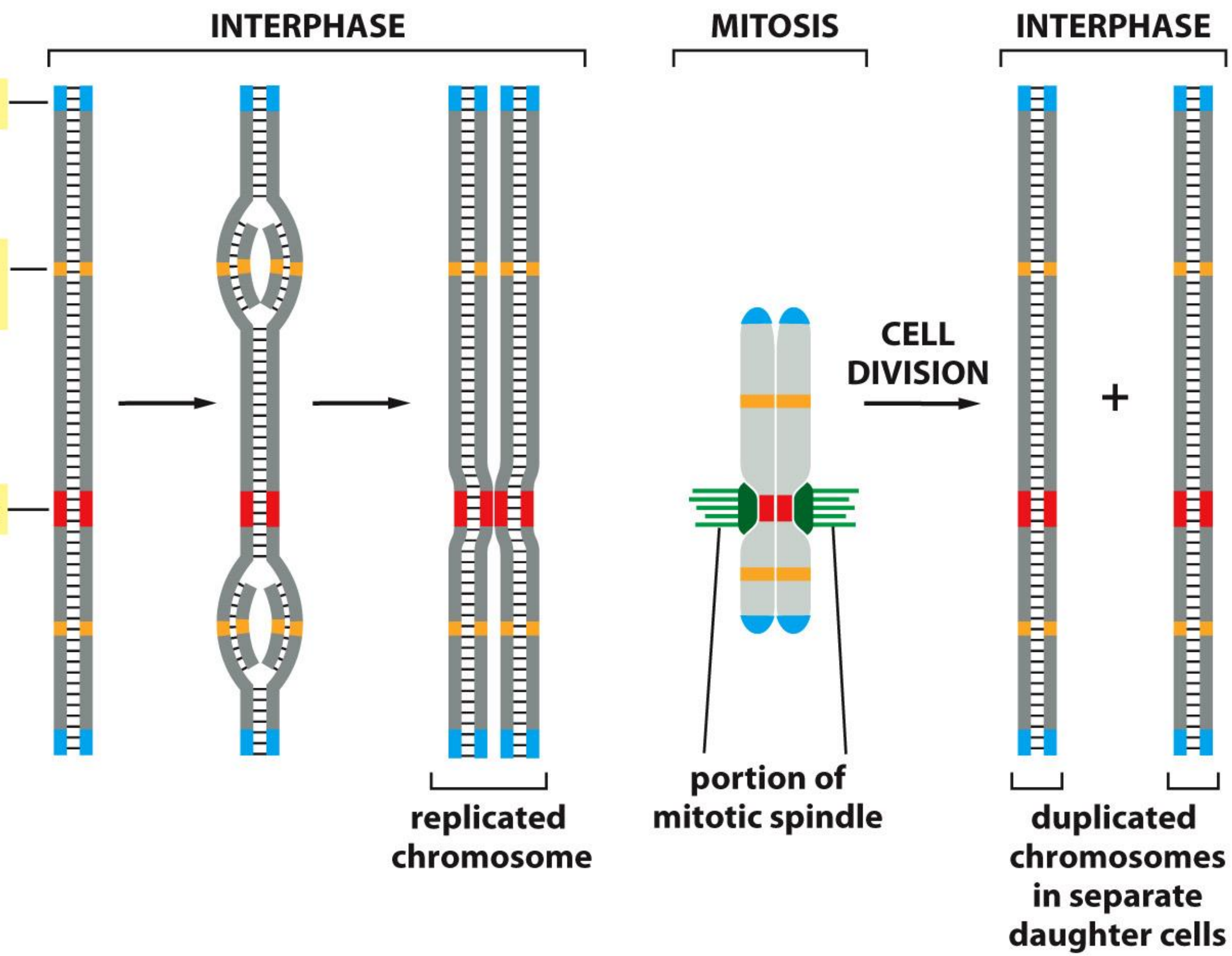
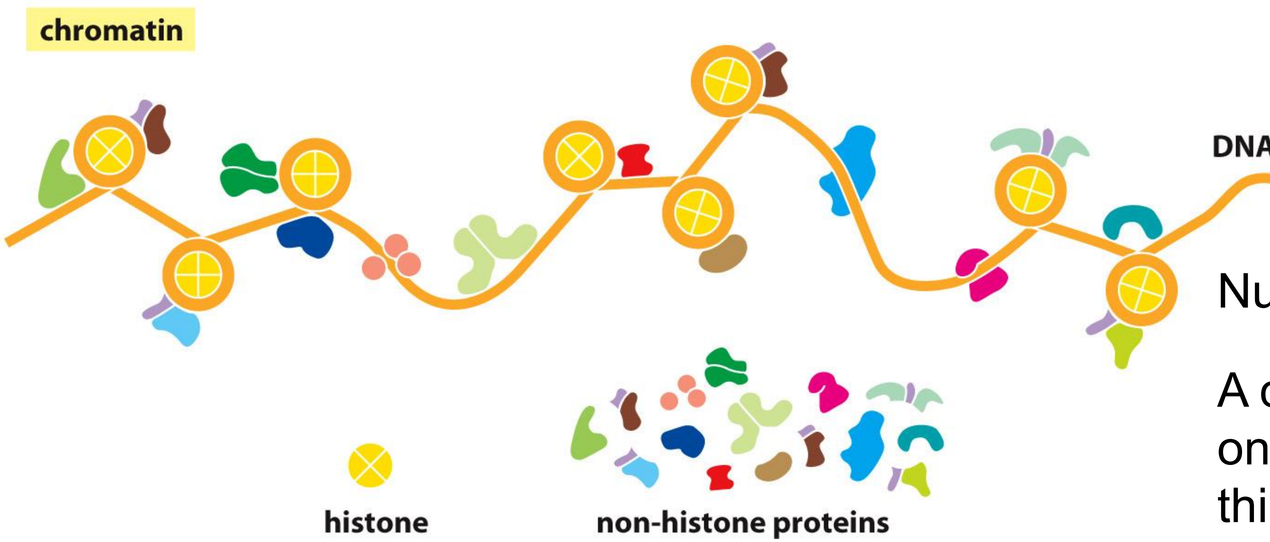


Figure 4-19 Molecular Biology of the Cell 6e (© Garland Science 2015)

Nucleosomes Are Basic Units of Eukaryotic Chromosome Structure

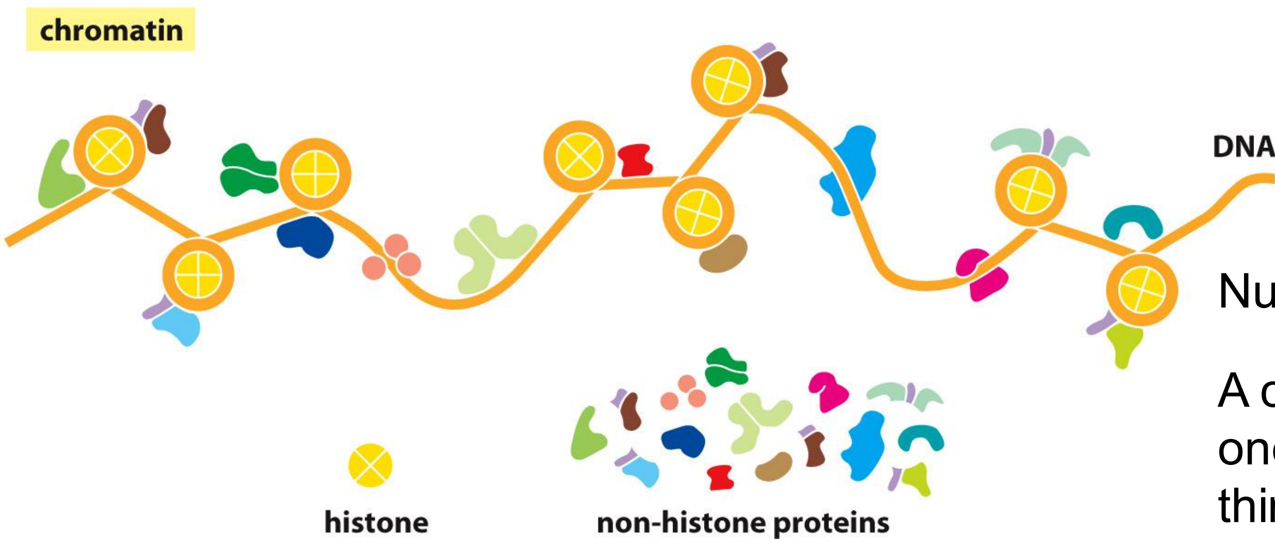


Nucleosome

A chromosome is about one-third DNA and two-thirds protein by mass.

Figure 4-20, 21 *Molecular Biology of the Cell* (© Garland Science 2015)

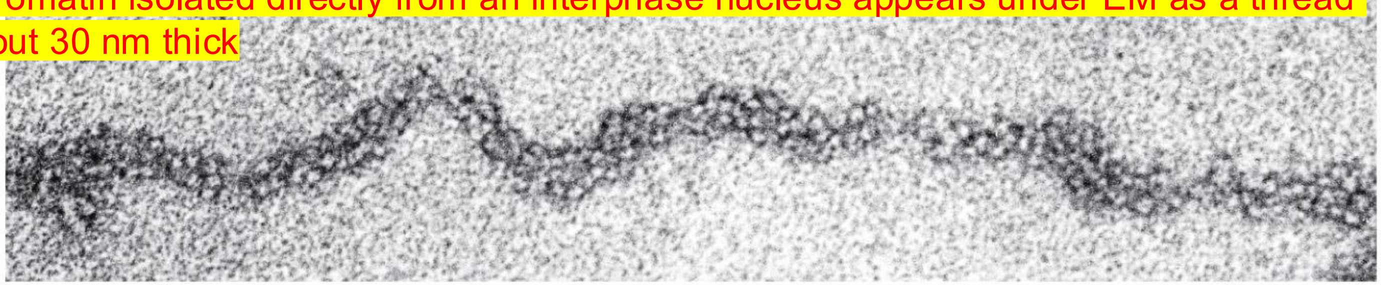
Nucleosomes Are Basic Units of Eukaryotic Chromosome Structure



Nucleosome

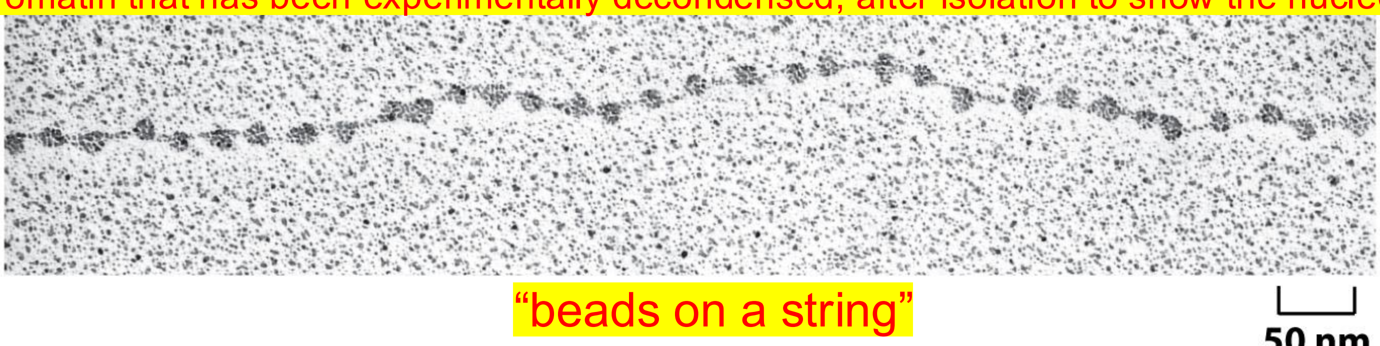
A chromosome is about one-third DNA and two-thirds protein by mass.

Chromatin isolated directly from an interphase nucleus appears under EM as a thread about 30 nm thick



30 nm

Chromatin that has been experimentally decondensed, after isolation to show the nucleosomes

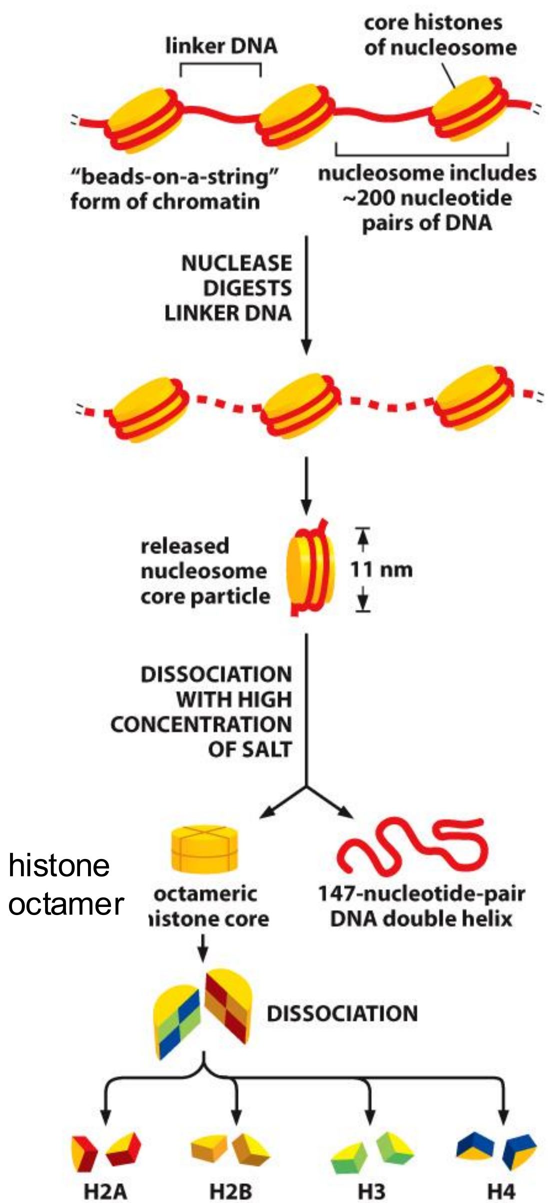


11 nm

"beads on a string"

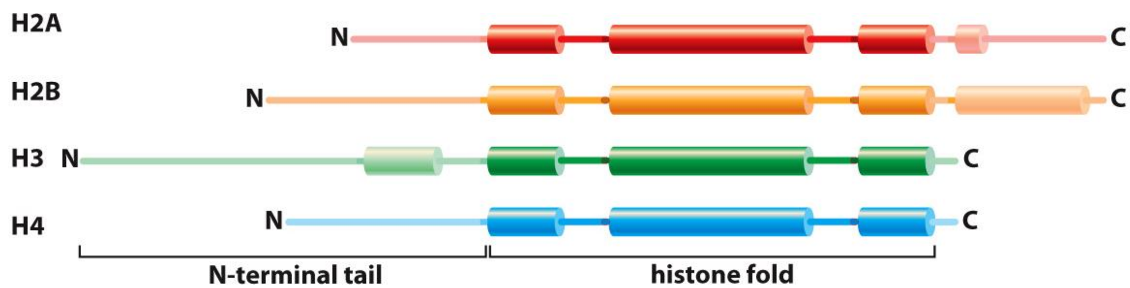
Figure 4-20, 21 *Molecular Biology of the Cell* (© Garland Science 2015)

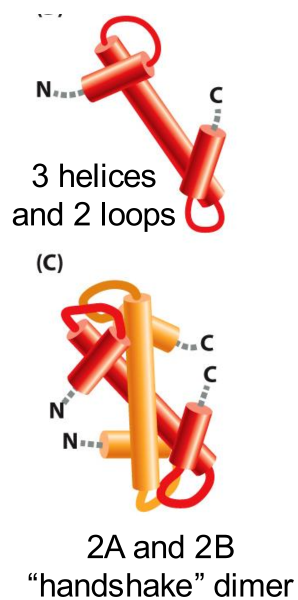
Nucleosomes Are Basic Units of Eukaryotic Chromosome Structure

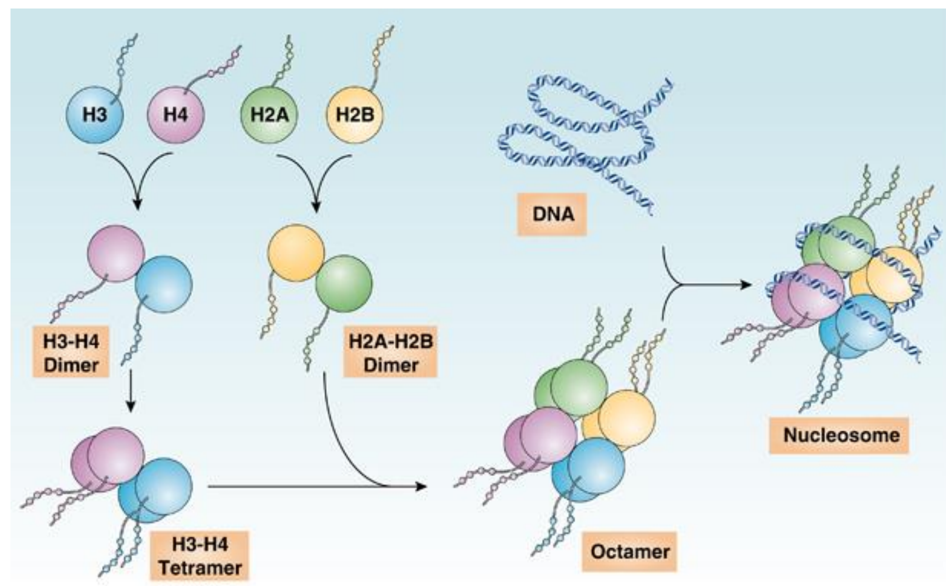


N-terminal tails are subjected to modifications.

Histones are relatively small proteins (102–135 amino acids) and among the most highly conserved eukaryotic proteins.





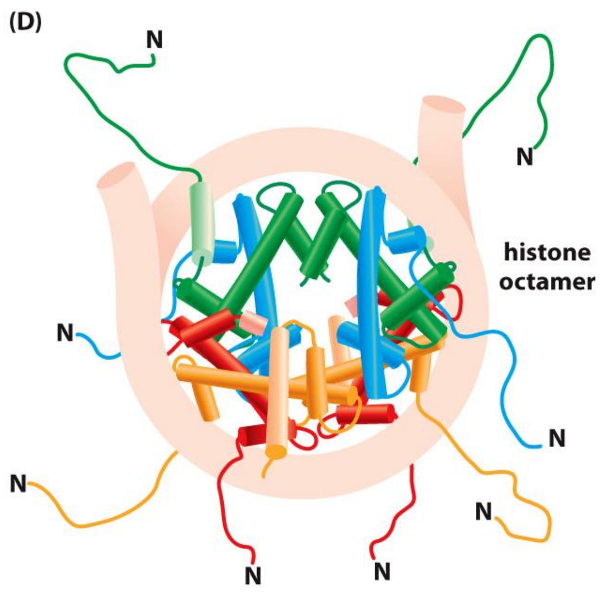


Chen et al, Cell Death and Disease, 2014

Figure 4-22, 24 *Molecular Biology of the Cell* (© Garland Science 2015)

Nucleosomes Are Basic Units of Eukaryotic Chromosome Structure

Histone Octamer



The Structure of the Nucleosome Core Particle

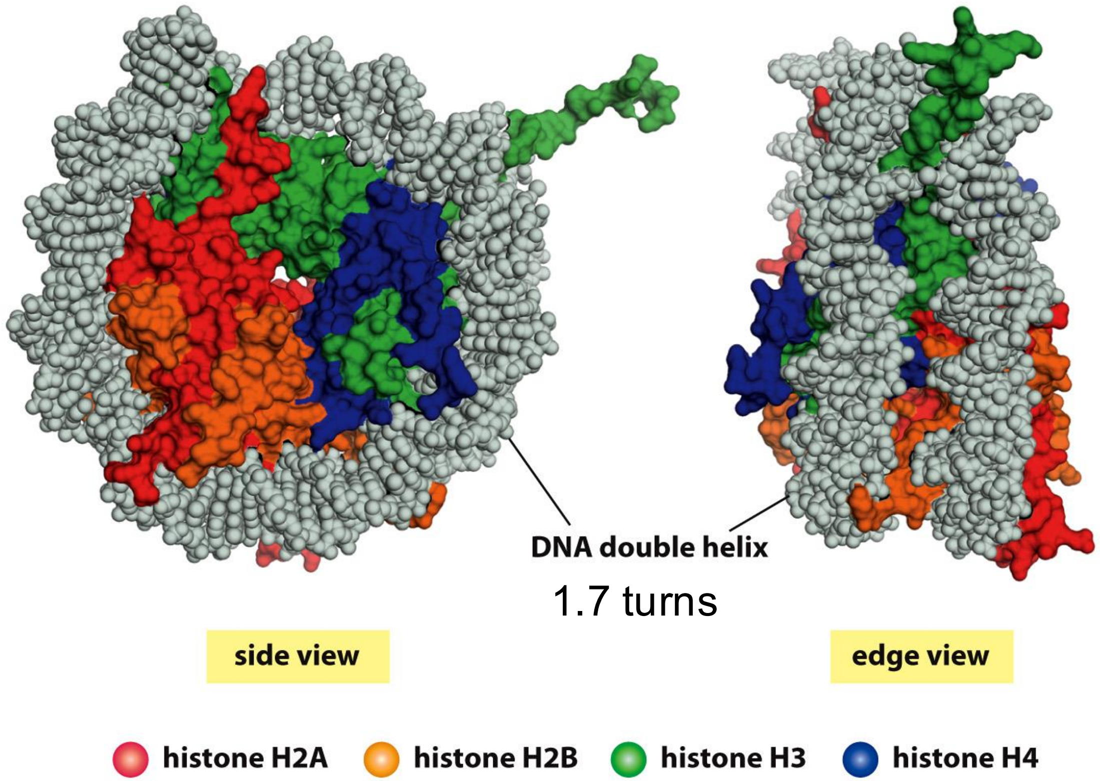
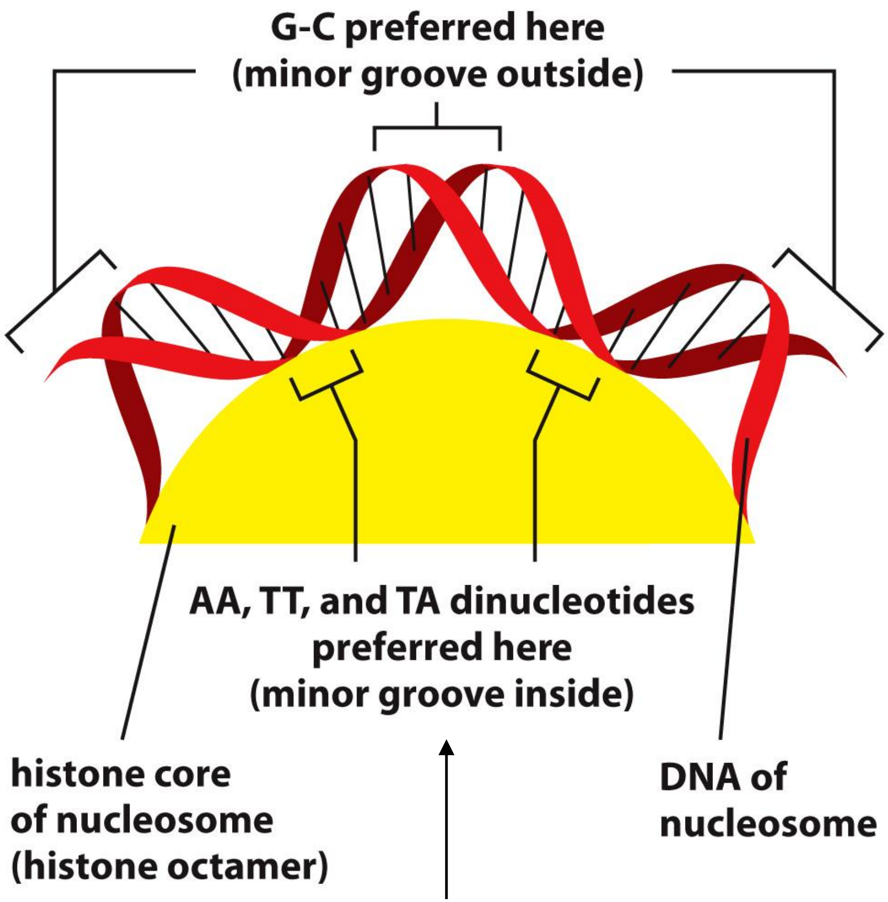


Figure 4-23 Molecular Biology of the Cell 6e (© Garland Science 2015)

A diploid human cell contains ~30 million nucleosomes. The formation of nucleosomes converts a DNA molecule into a chromatin thread about one-third of its initial length.

Figure 4-23, 24 *Molecular Biology of the Cell* (© Garland Science 2015)

The Bending of DNA in a Nucleosome



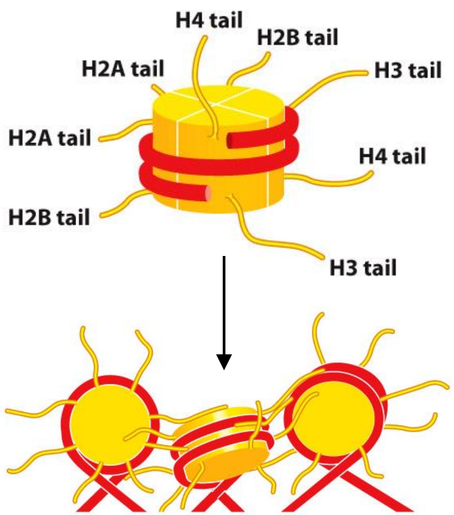
The minor groove is compressed on the inside of the turn.

- The DNA helix makes 1.7 tight turns around the histone octamer.
- 142 hydrogen bonds are formed between DNA and the histone core in each nucleosome.
- ~1/5 of the core histones are either lysine or arginine that are positively charged to neutralize negatively charged DNA.
- The bending requires a substantial compression of the inside minor groove of the DNA helix.
- The sequence preference of nucleosomes must be weak enough to allow other factors to dominate.

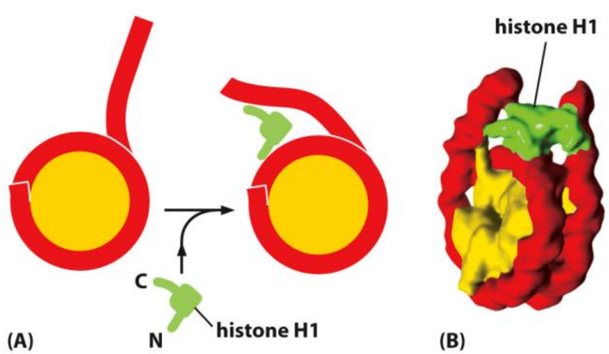
Figure 4-25 *Molecular Biology of the Cell* (© Garland Science 2015)

Nucleosomes Are Packed Together to Form Compact Chromatin Fiber

Histone tails in the compaction of chromatin.



Linker Histone H1 changes the exit path of DNA to help compaction.



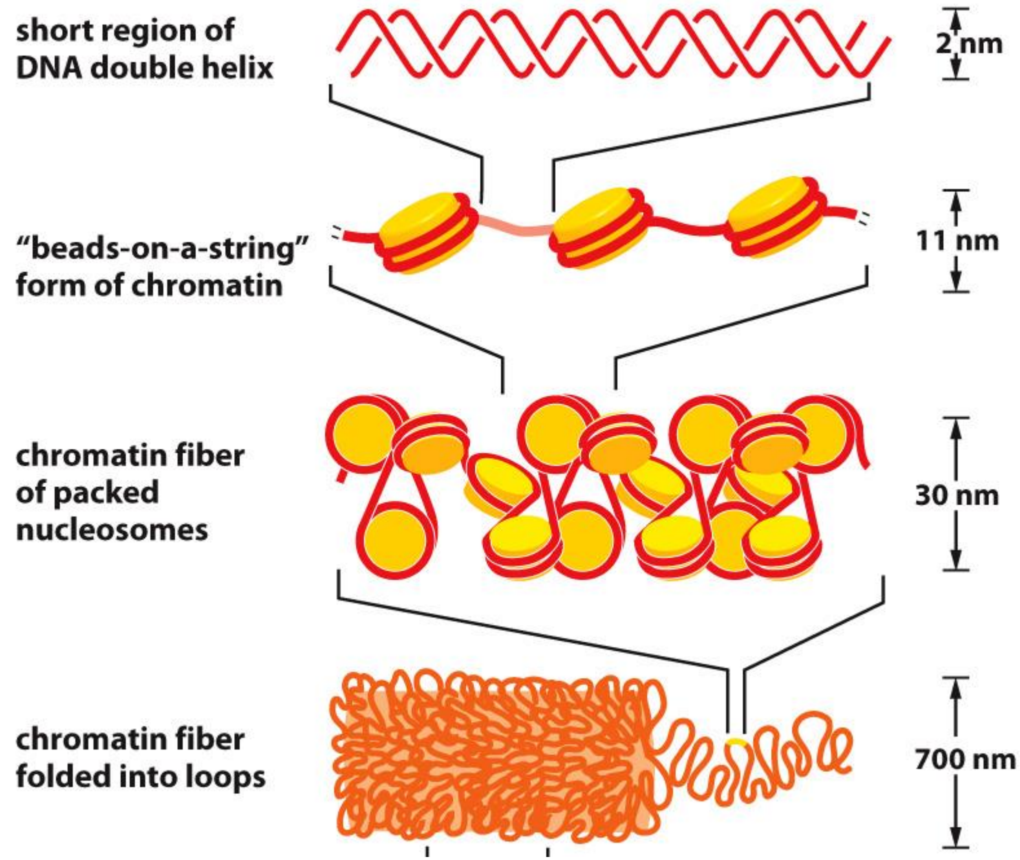
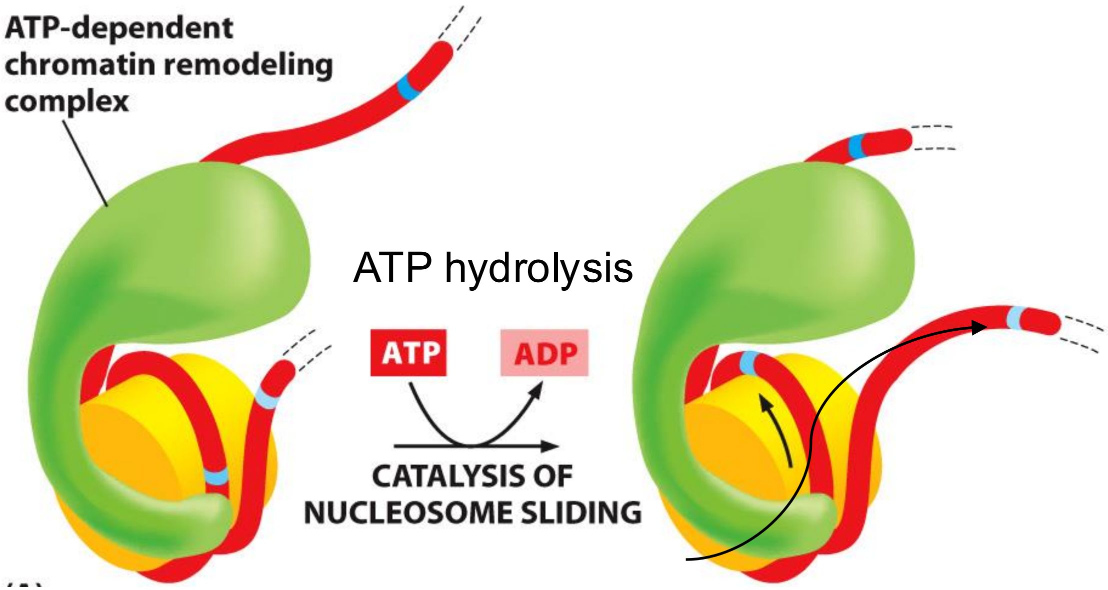


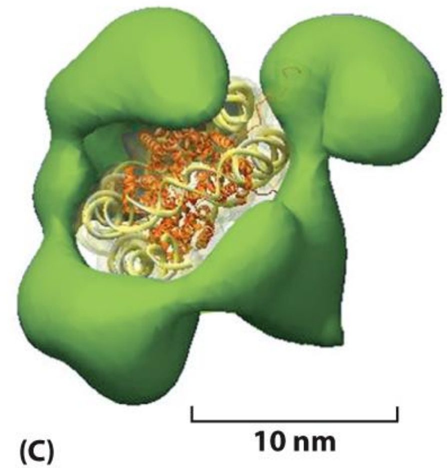
Figure 4-29, 30, 61 *Molecular Biology of the Cell* (© Garland Science 2015)

Nucleosomes Sliding Catalyzed by ATP-Dependent Chromatin Remodeling Complexes

DNA in nucleosomes should be available for binding other proteins.



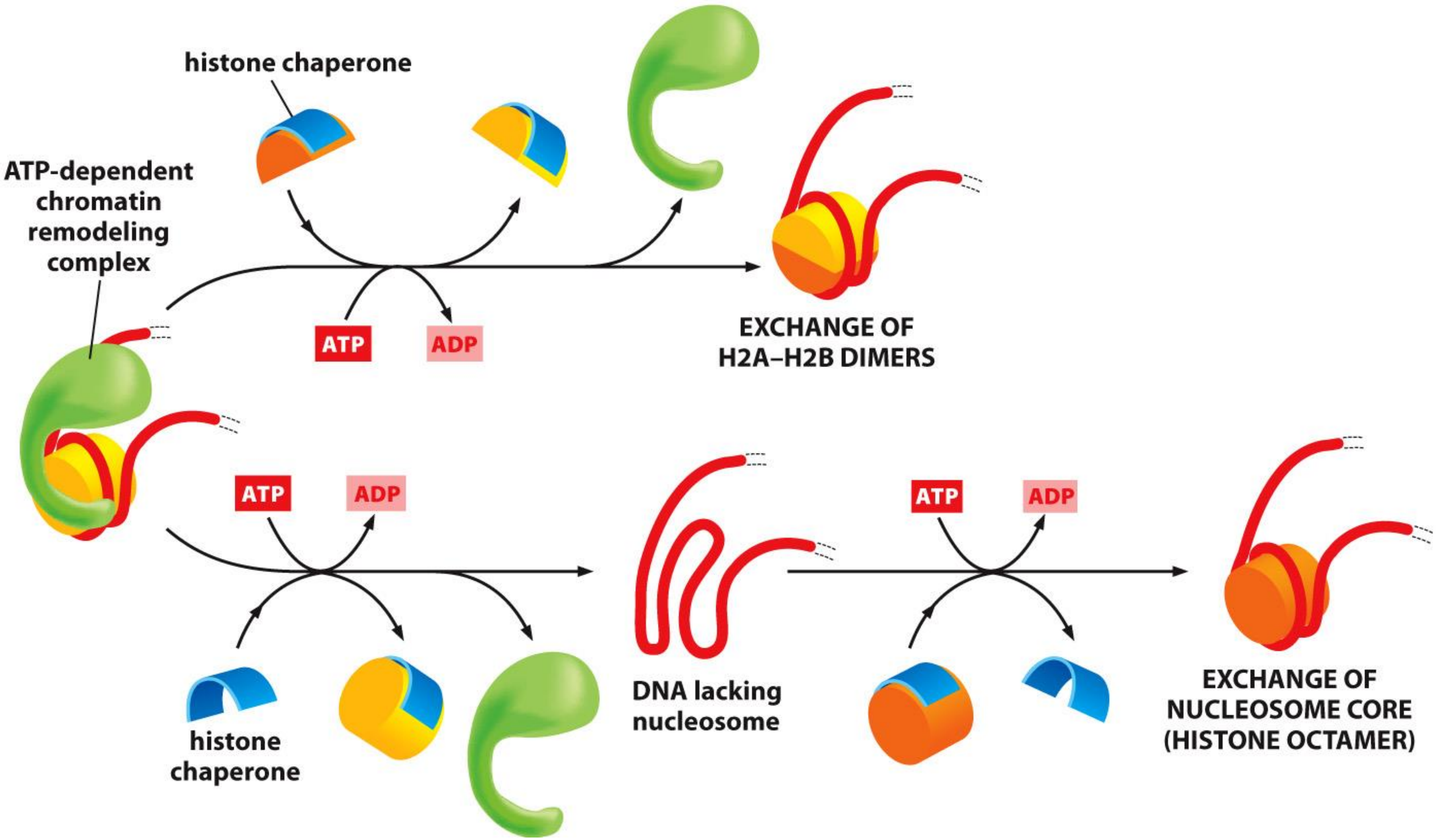
Yeast RSC complex



Chromatin remodeling complexes contain 10 or more subunits.

Figure 4-26 *Molecular Biology of the Cell* (© Garland Science 2015)

Nucleosome Removal and Histone exchange Catalyzed by ATP-Dependent Chromatin Remodeling Complexes



A typical nucleosome is replaced on the DNA every one or two hours inside the cell.

Figure 4-27 *Molecular Biology of the Cell* (© Garland Science 2015)

Chromatin Acquires Additional Variety Through the Site-Specific Insertion of a Small Set of Histone Variants

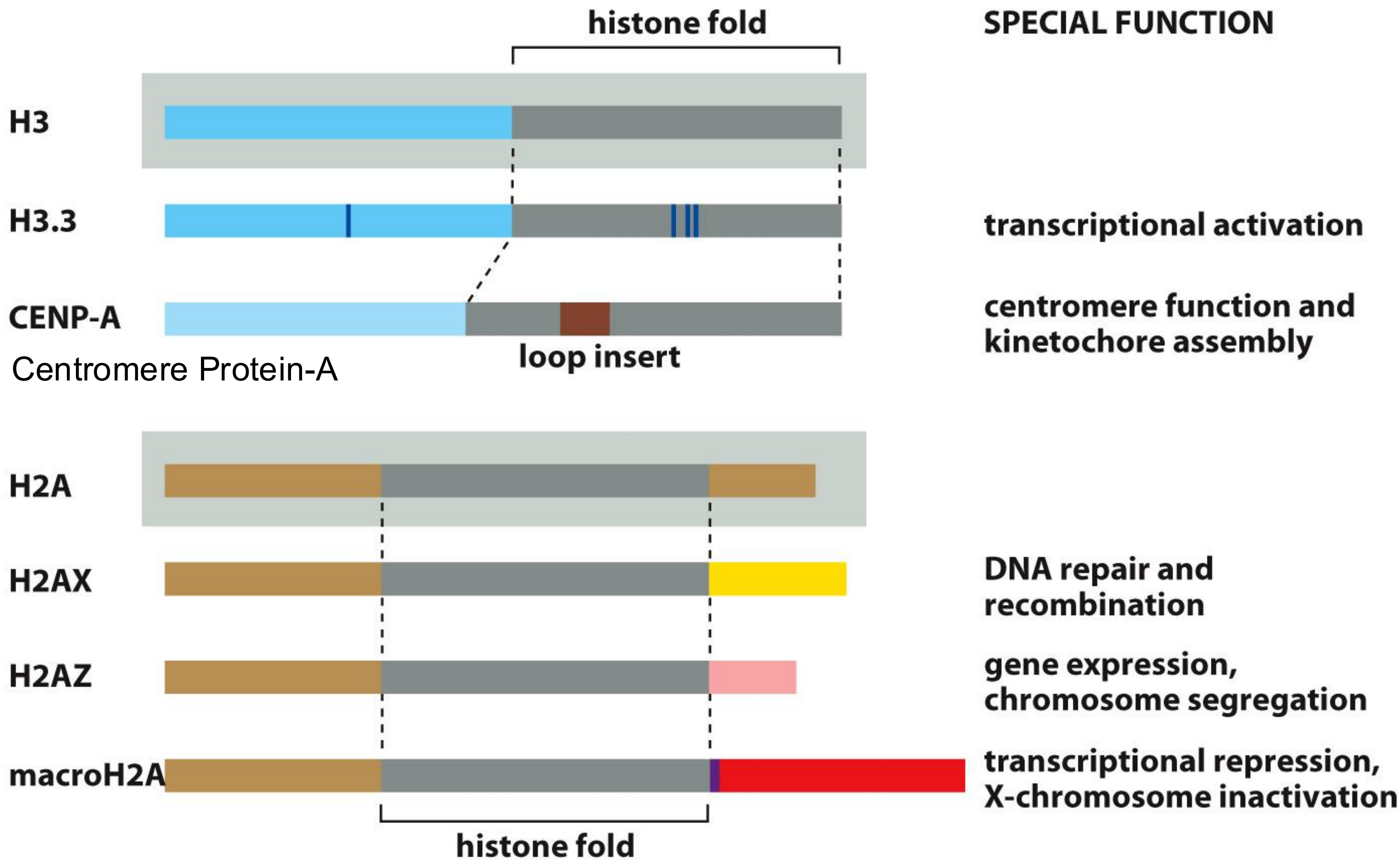
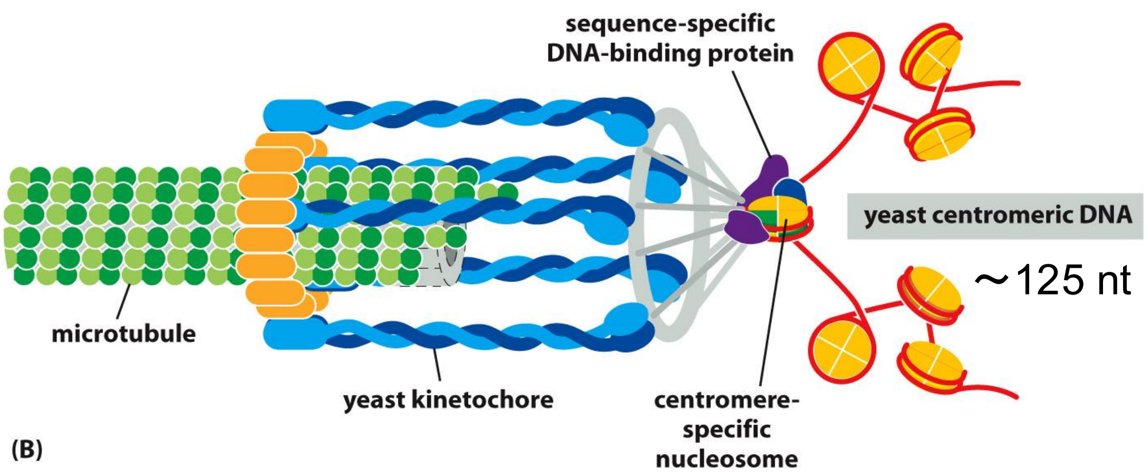


Figure 4-35 Molecular Biology of the Cell 6e (© Garland Science 2015)

The Chromatin in Centromeres Reveals How Histone Variants Can Create Special Structures

A simple centromere in the yeast

CENP-A: Centromere Protein-A



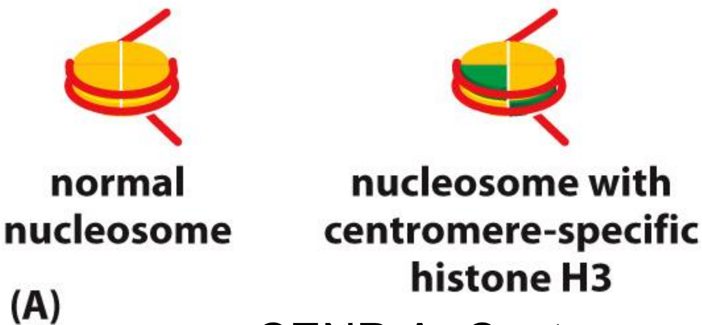
~125 nt

Figure 4-42, 43 *Molecular Biology of the Cell* (© Garland Science 2015)

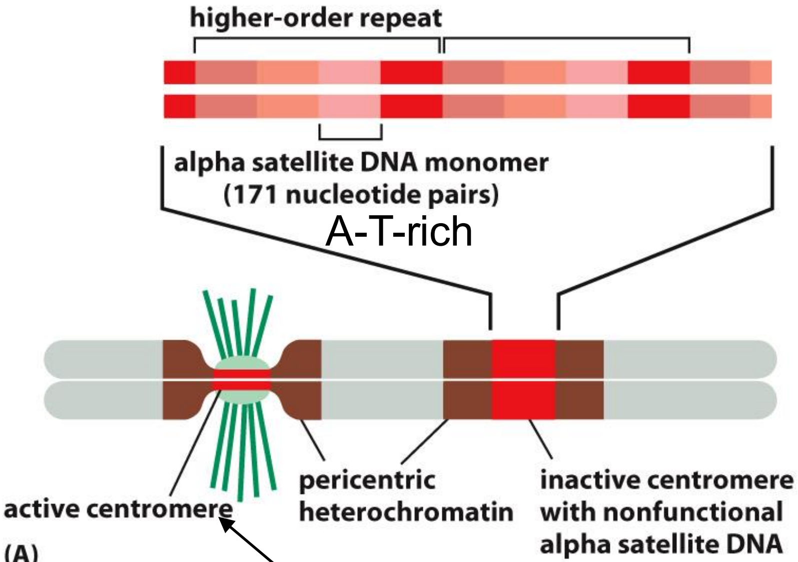
The Chromatin in Centromeres Reveals How Histone Variants Can Create Special Structures

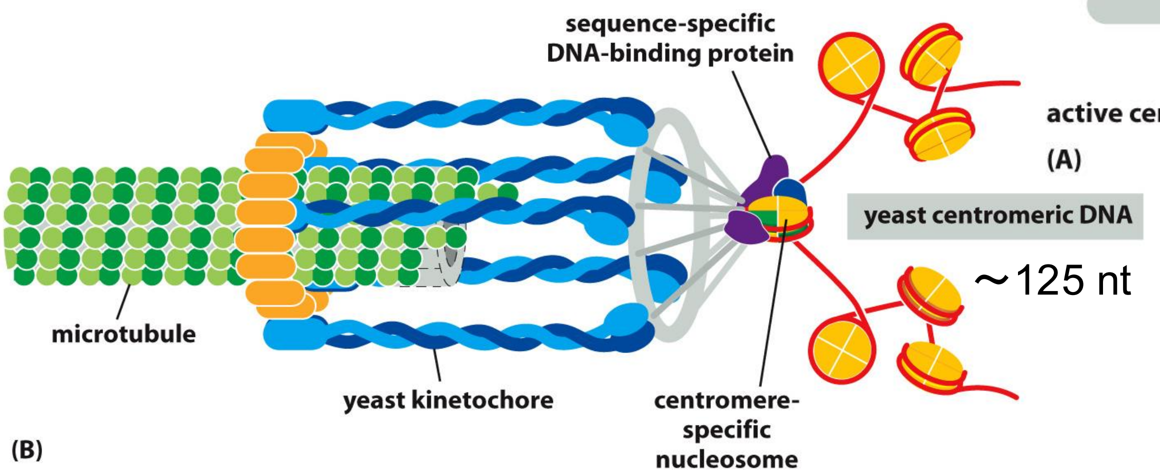
A simple centromere in the yeast

Plasticity of human centromere formation



CENP-A: Centromere Protein-A





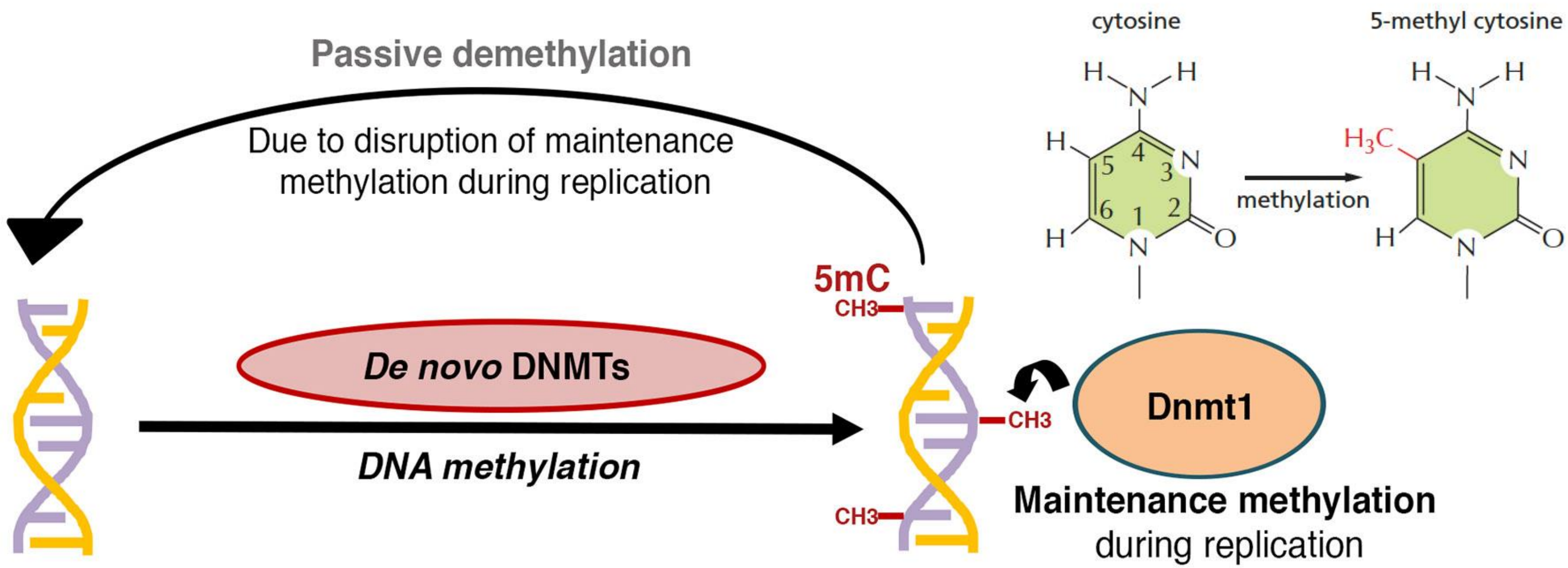
Stable inheritance requires that each chromosome should contain one centromere, and one only.

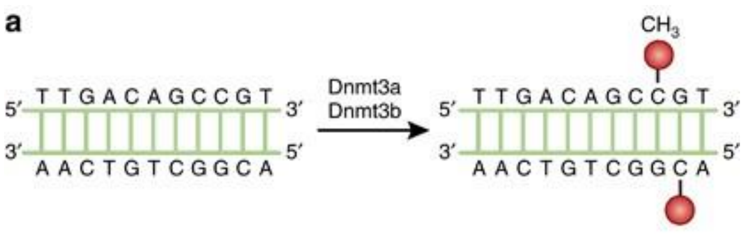
Figure 4-42, 43 *Molecular Biology of the Cell* (© Garland Science 2015)

CHROMOSOME MODIFICATIONS

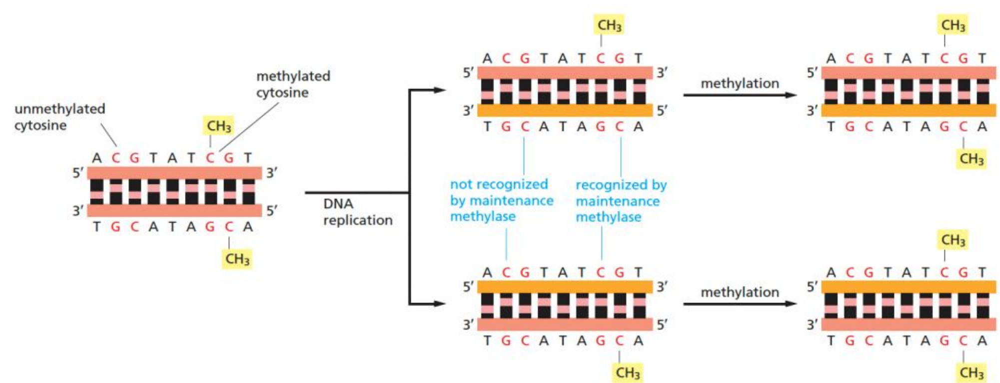
- Modifications can happen directly on DNA.
- The core histones are covalently modified at many different sites.
- Heterochromatin is highly organized and restricts gene expression.
- The heterochromatic state is self-propagating.

DNA Methylation





DNA Methylation Is Maintained In DNA Replication

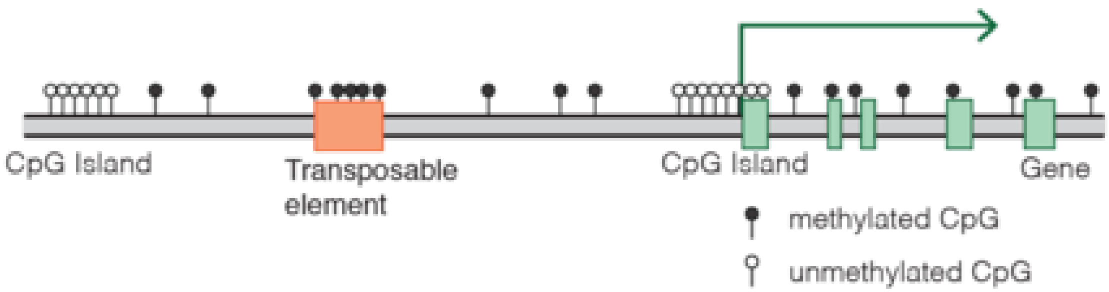


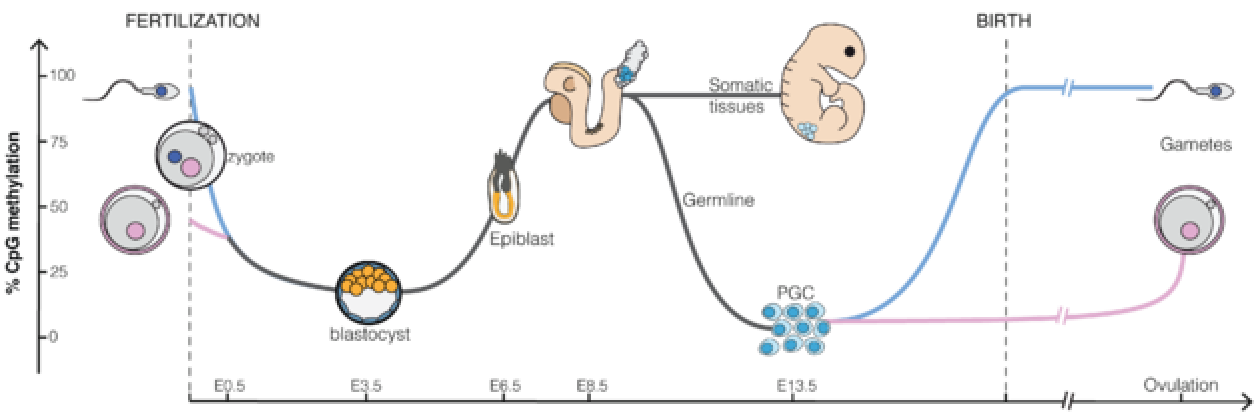
N. Parveen and S. Dhawan, Front. Endocrinol (2021)

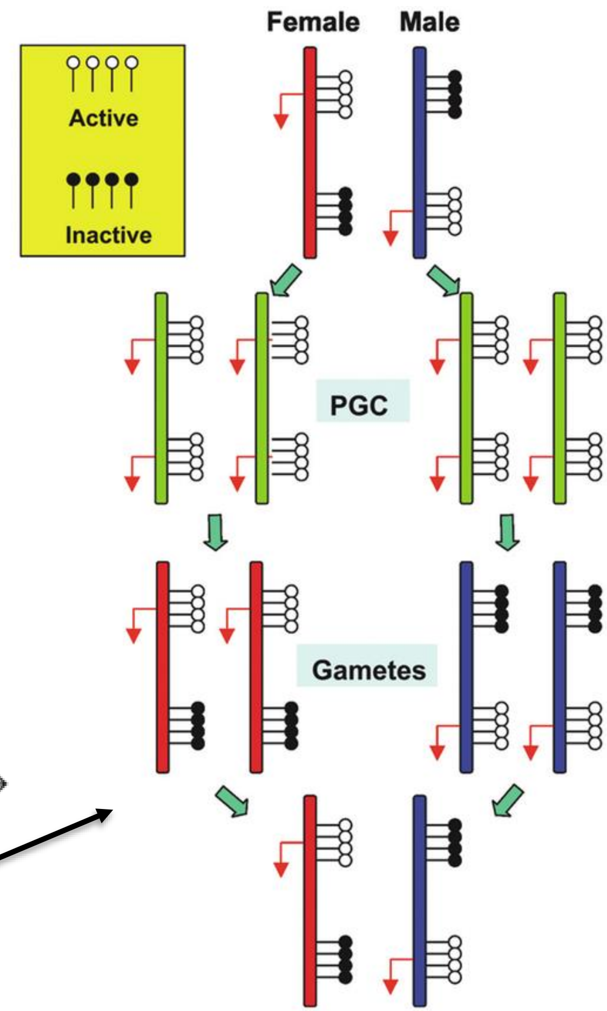
Figure 7-44 *Molecular Biology of the Cell* (© Garland Science 2015)

The Function of DNA Methylation

Typical Mammalian DNA Methylation landscape



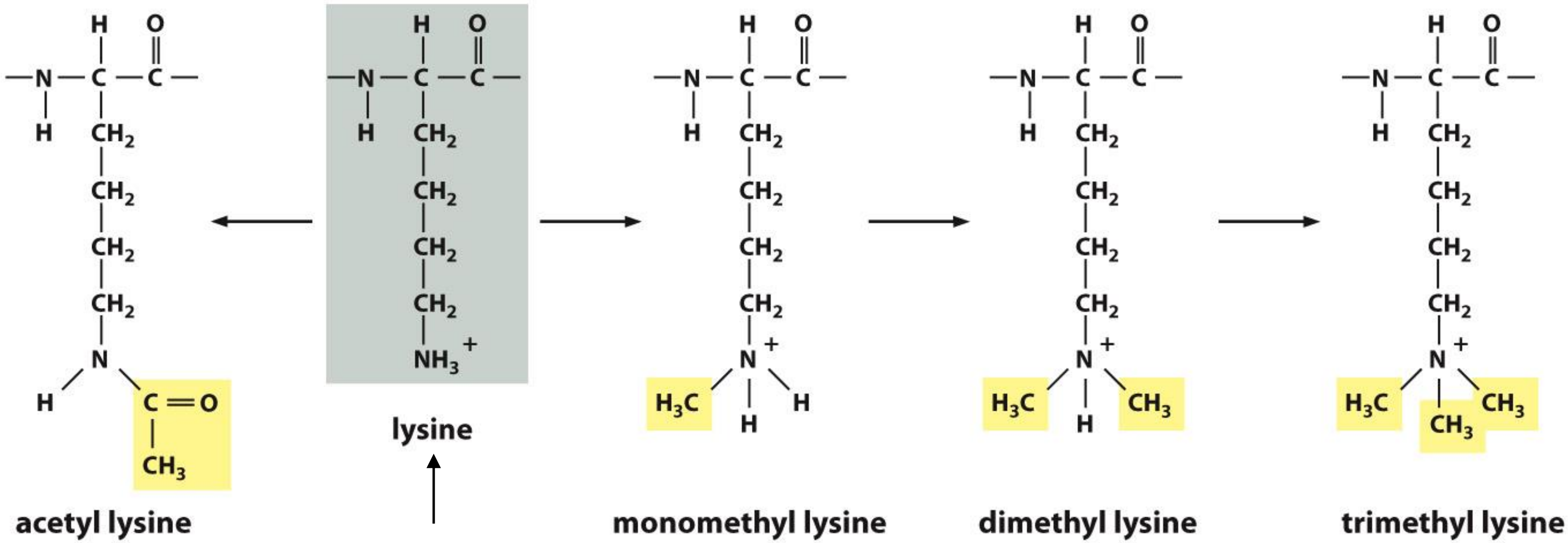




Establishing
Genomic Imprinting

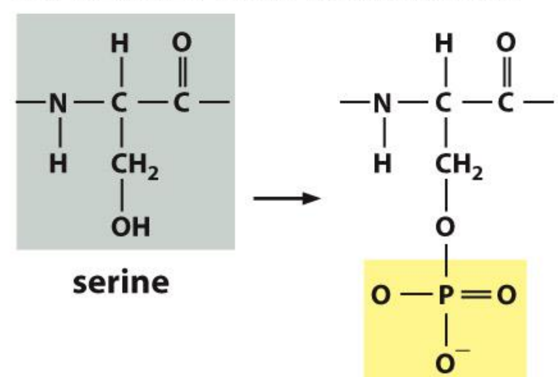
The Core Histones Are Covalently Modified at Many Different Sites

(A) LYSINE ACETYLATION AND METHYLATION ARE COMPETING REACTIONS



Acetylation removes the
plus charge on lysine.

(B) SERINE PHOSPHORYLATION



phosphoserine

phosphorylation adds a
negative charge to a histone

Reversible modifications with particular enzymes:
+ acetyl/methyl transferases
- deacetylase/demethylases

Figure 4-33 Molecular Biology of the Cell 6e (© Garland Science 2015)

The Core Histones Are Covalently Modified at Many Different Sites

"Histone Code"

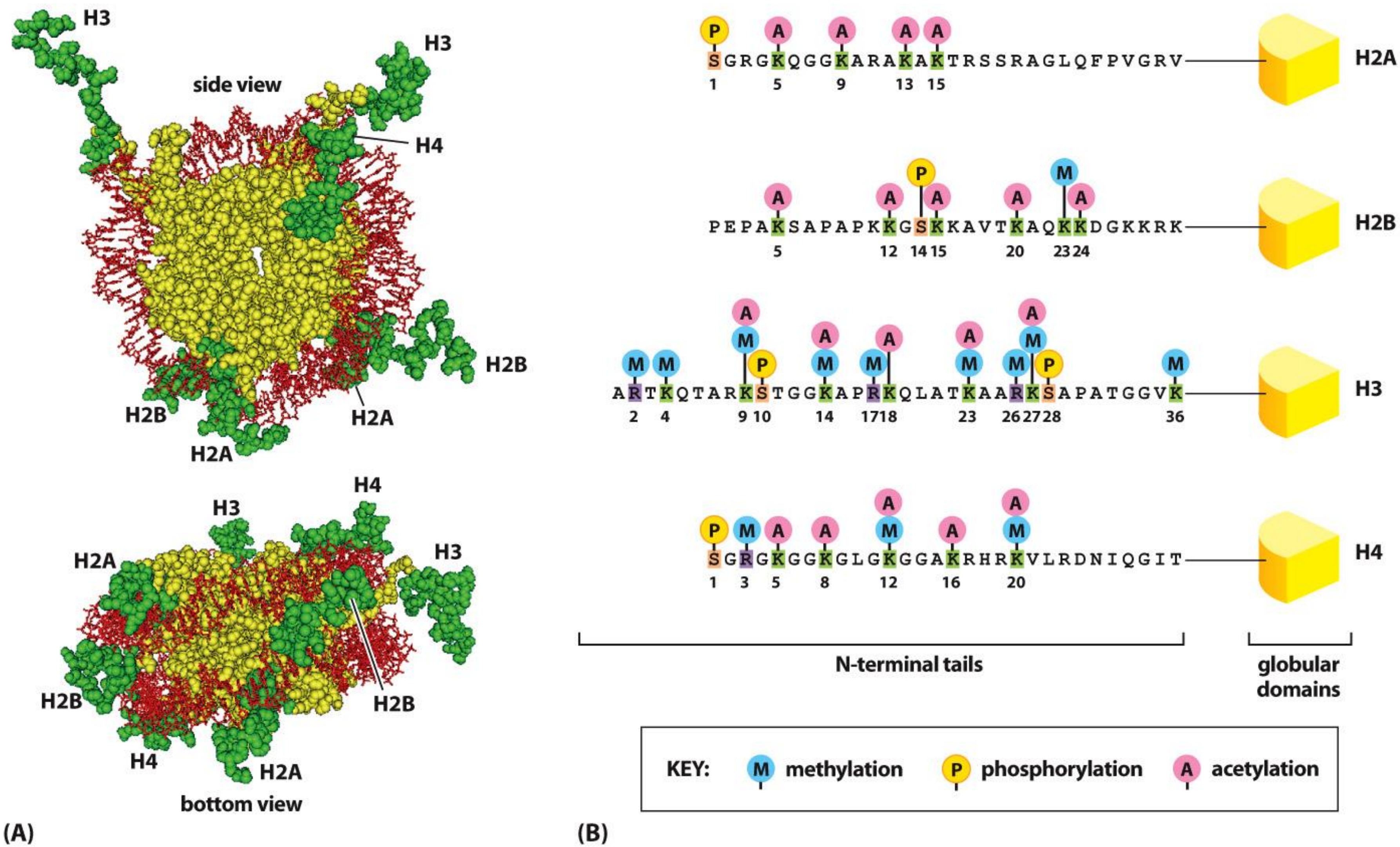


Figure 4-34 Molecular Biology of the Cell 6e (© Garland Science 2015)

Covalent Modifications – Some Specific Meanings

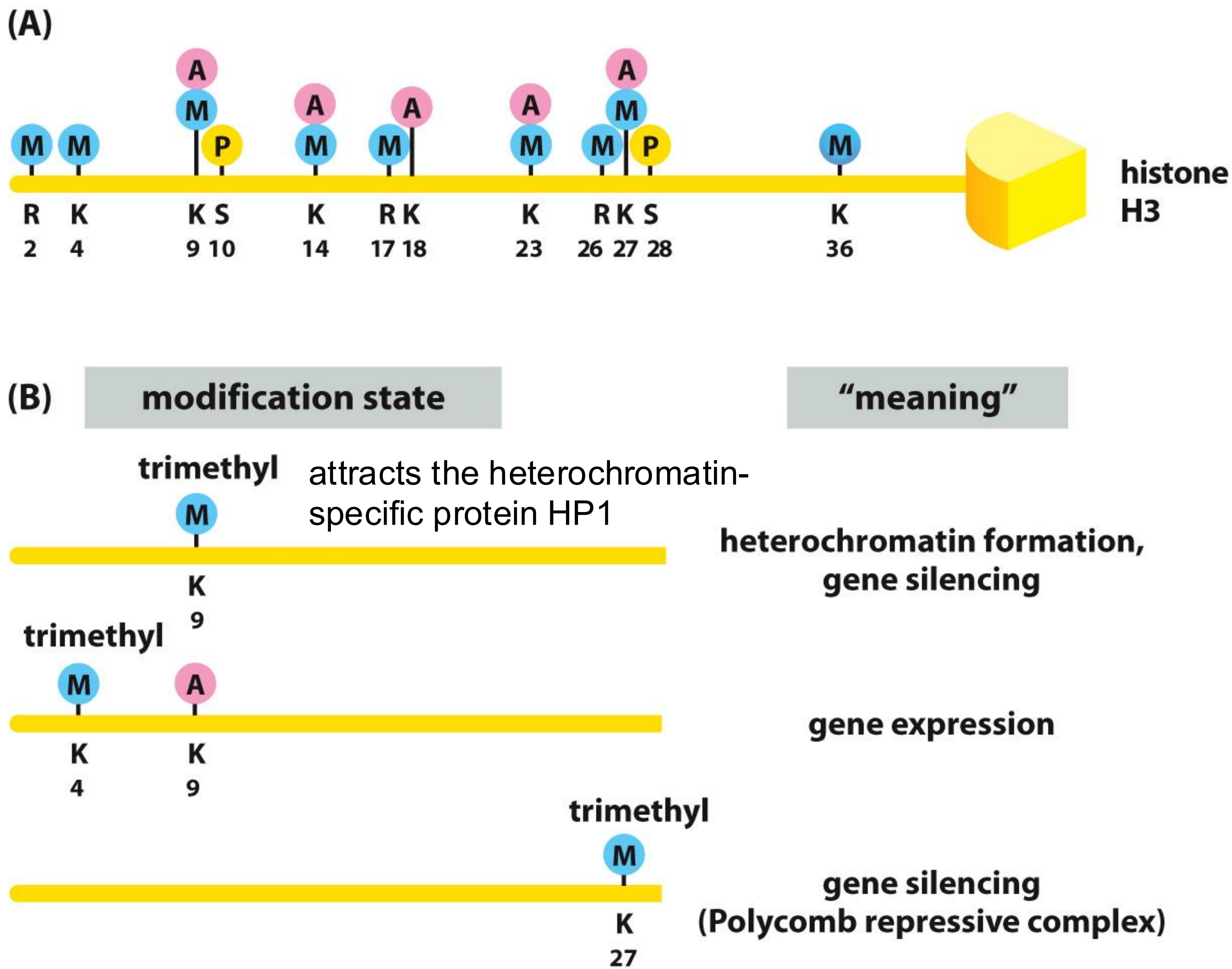
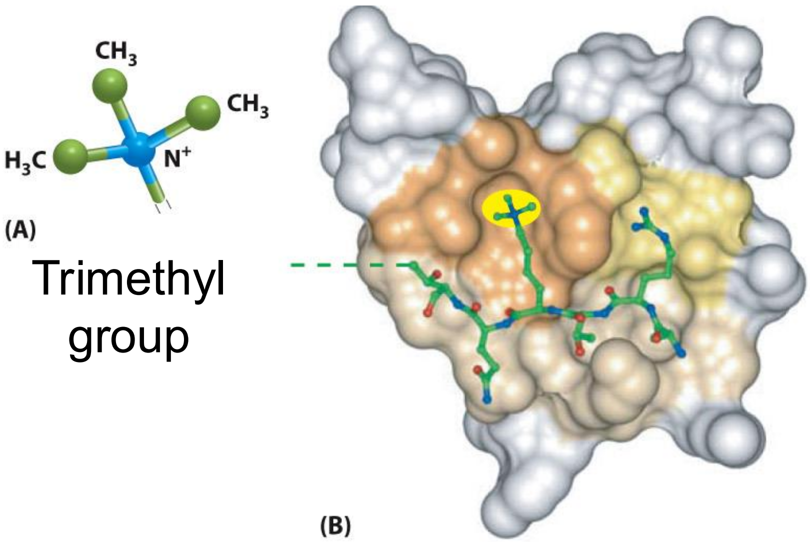


Figure 4-39 Molecular Biology of the Cell 6e (© Garland Science 2015)

Covalent Histone Modifications to Control Chromosome Functions

The most profound effects of the histone modifications lie in their ability to recruit specific other proteins to the modified stretch of chromatin.



Trimethyl group

Figure 4-36 Molecular Biology of the Cell 6e (© Garland Science 2015)

ING-PHD domain binding to methylated H3 histone

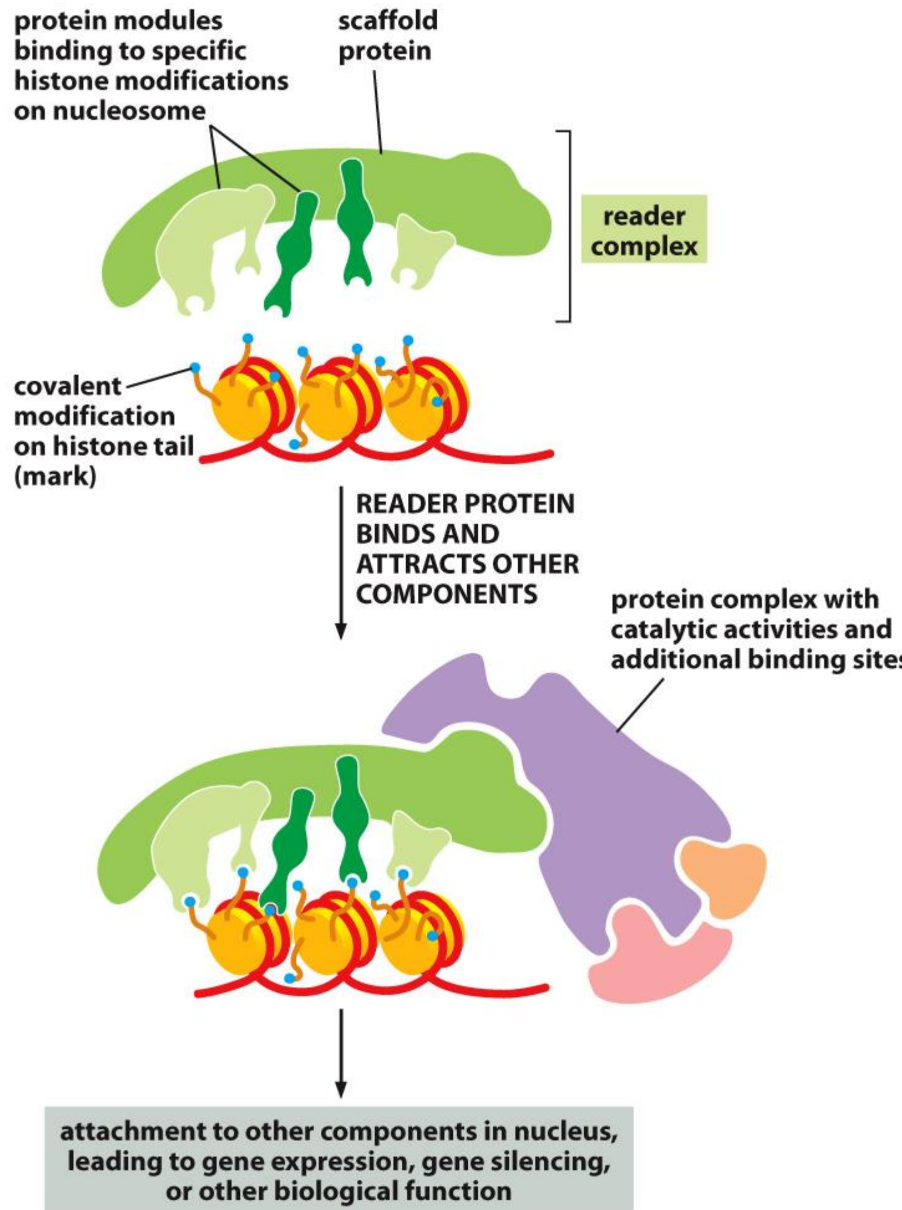


Figure 4-38 Molecular Biology of the Cell 6e (© Garland Science 2015)

The Spreading of Histone Modifications Along a Chromosome

A Complex of Reader and Writer Proteins Can Spread Specific Chromatin Modifications

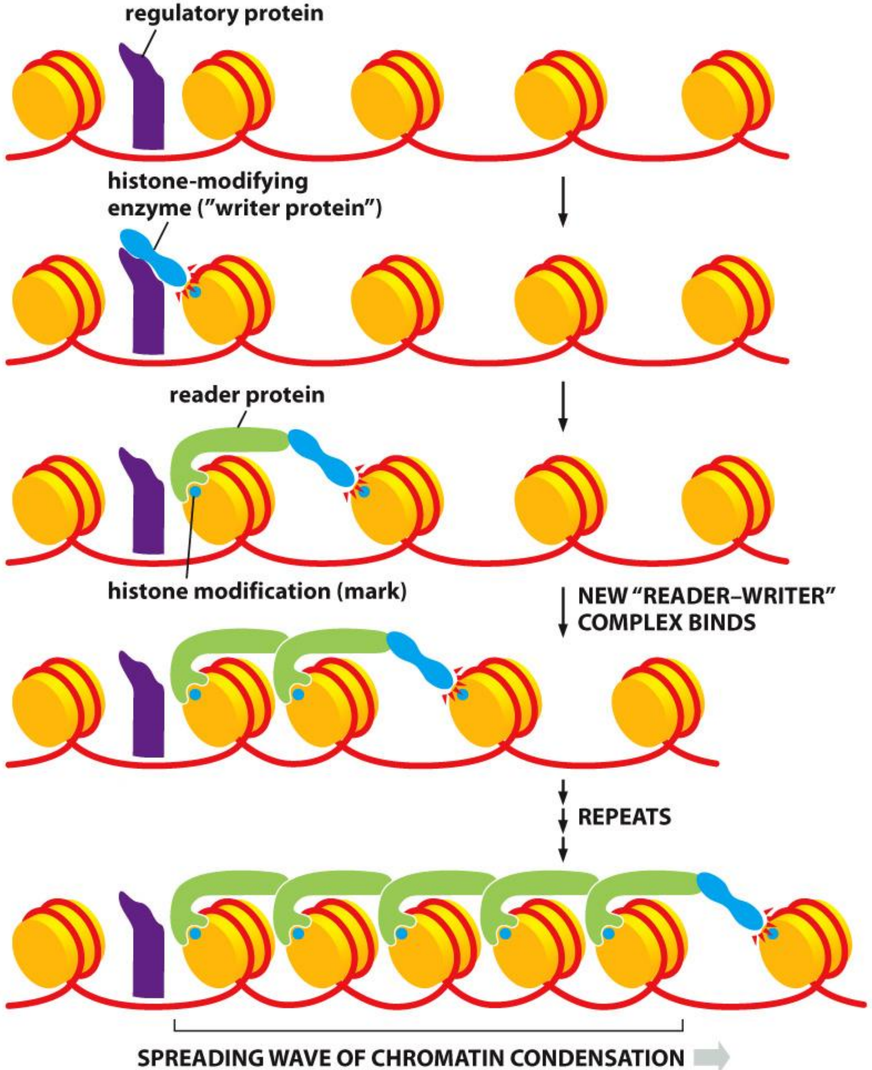


Figure 4-40 Molecular Biology of the Cell 6e (© Garland Science 2015)

The Spreading of Histone Modifications Along a Chromosome

A Complex of Reader and Writer Proteins Can Spread Specific Chromatin Modifications

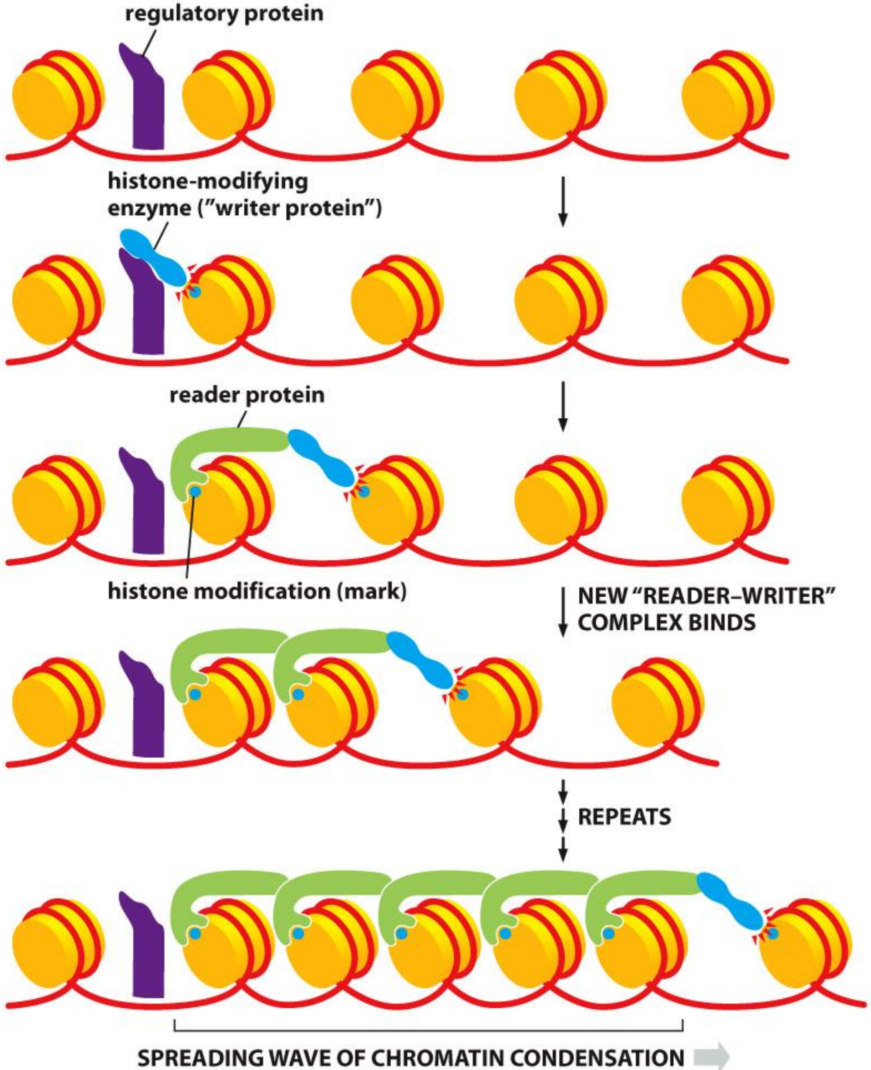


Figure 4-40 Molecular Biology of the Cell 6e (© Garland Science 2015)

Barrier DNA Sequences Block the Spread

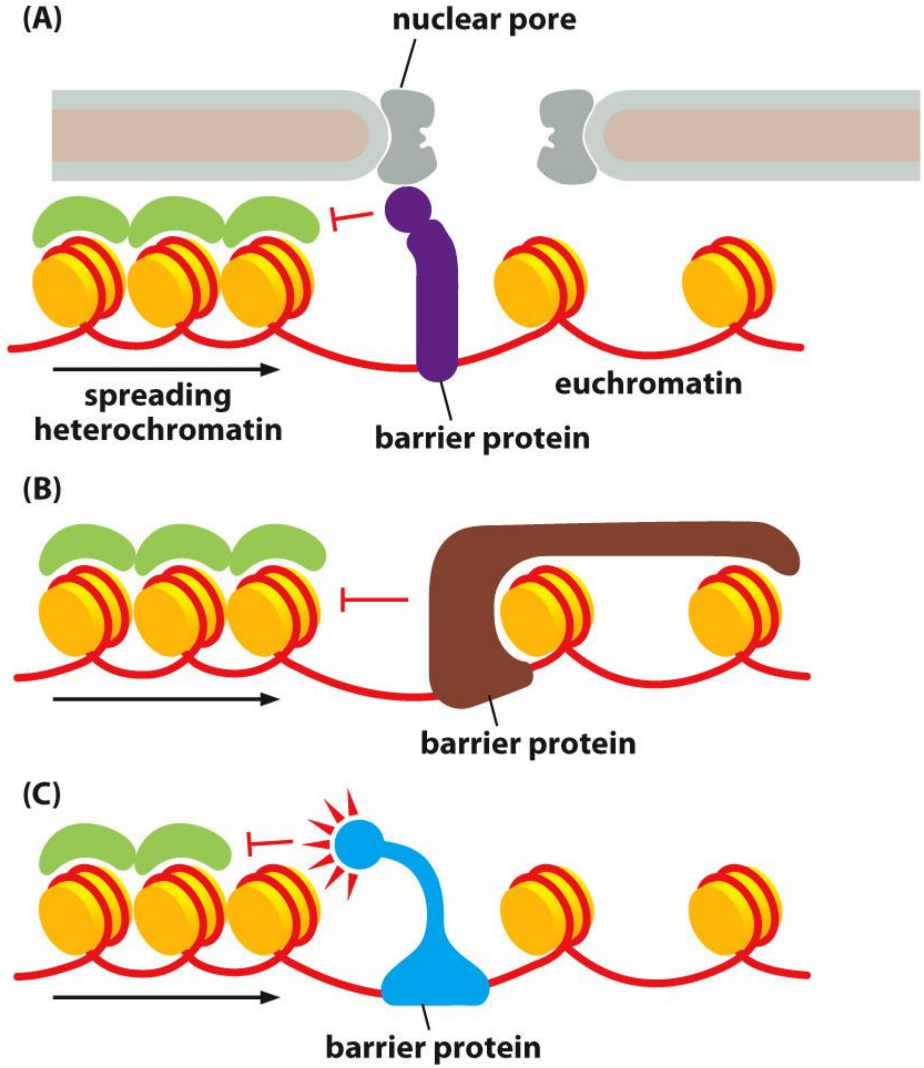
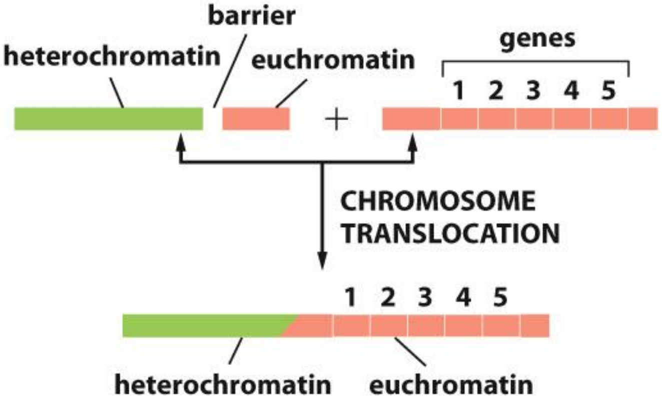


Figure 4-41 Molecular Biology of the Cell 6e (© Garland Science 2015)

The Heterochromatic State Is Self-Propagating

Position Effect



The cause of position effect variegation in *Drosophila*

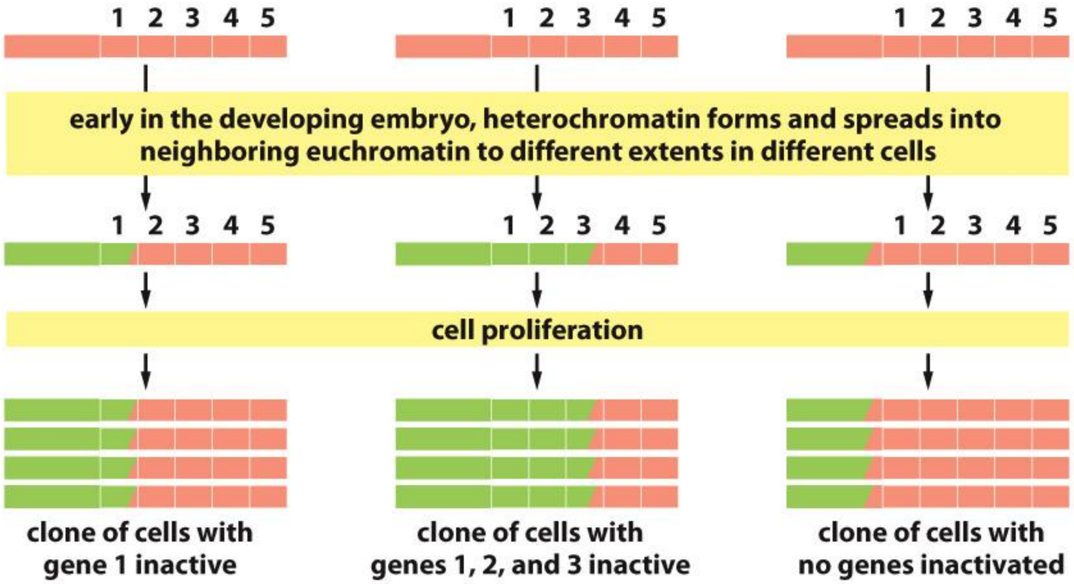
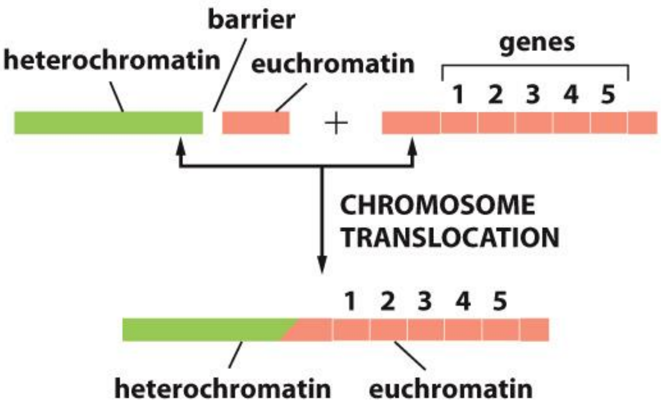


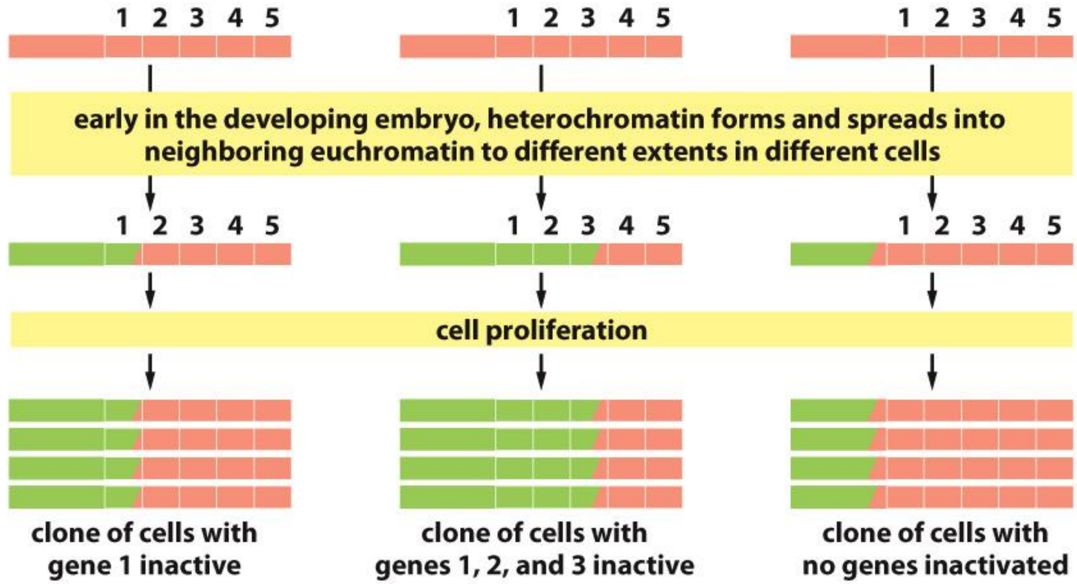
Figure 4-31, 32 *Molecular Biology of the Cell* (© Garland Science 2015)

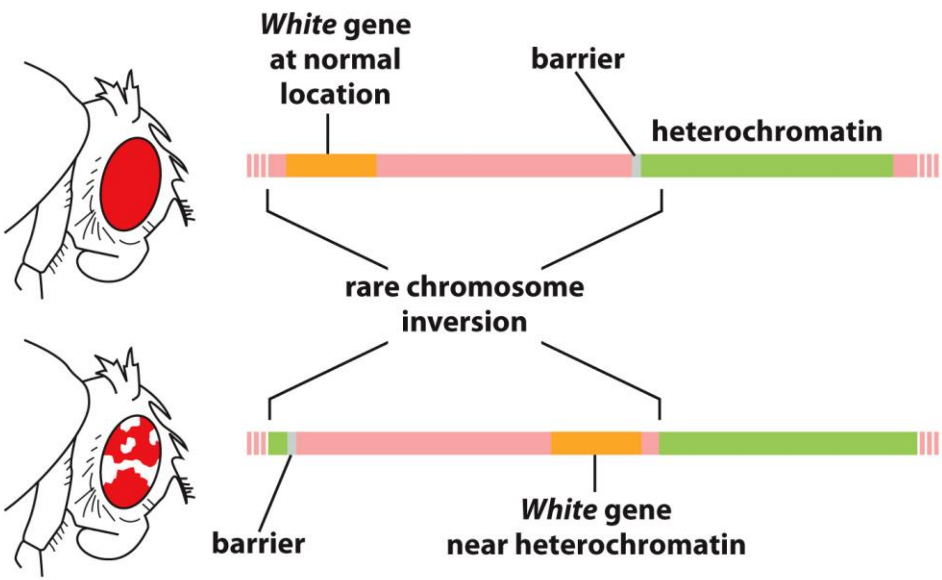
The Heterochromatic State Is Self-Propagating

The cause of position effect variegation in *Drosophila*

Position Effect







The White gene in the fruit fly Drosophila controls eye pigment production and is named after the mutation that first identified it.

Figure 4-31, 32 *Molecular Biology of the Cell* (© Garland Science 2015)

Some Chromatin Structures Can Be Directly Inherited

Chromatin structures can be inherited——epigenetic inheritance

Propagation of chromatin modifications across cell generations

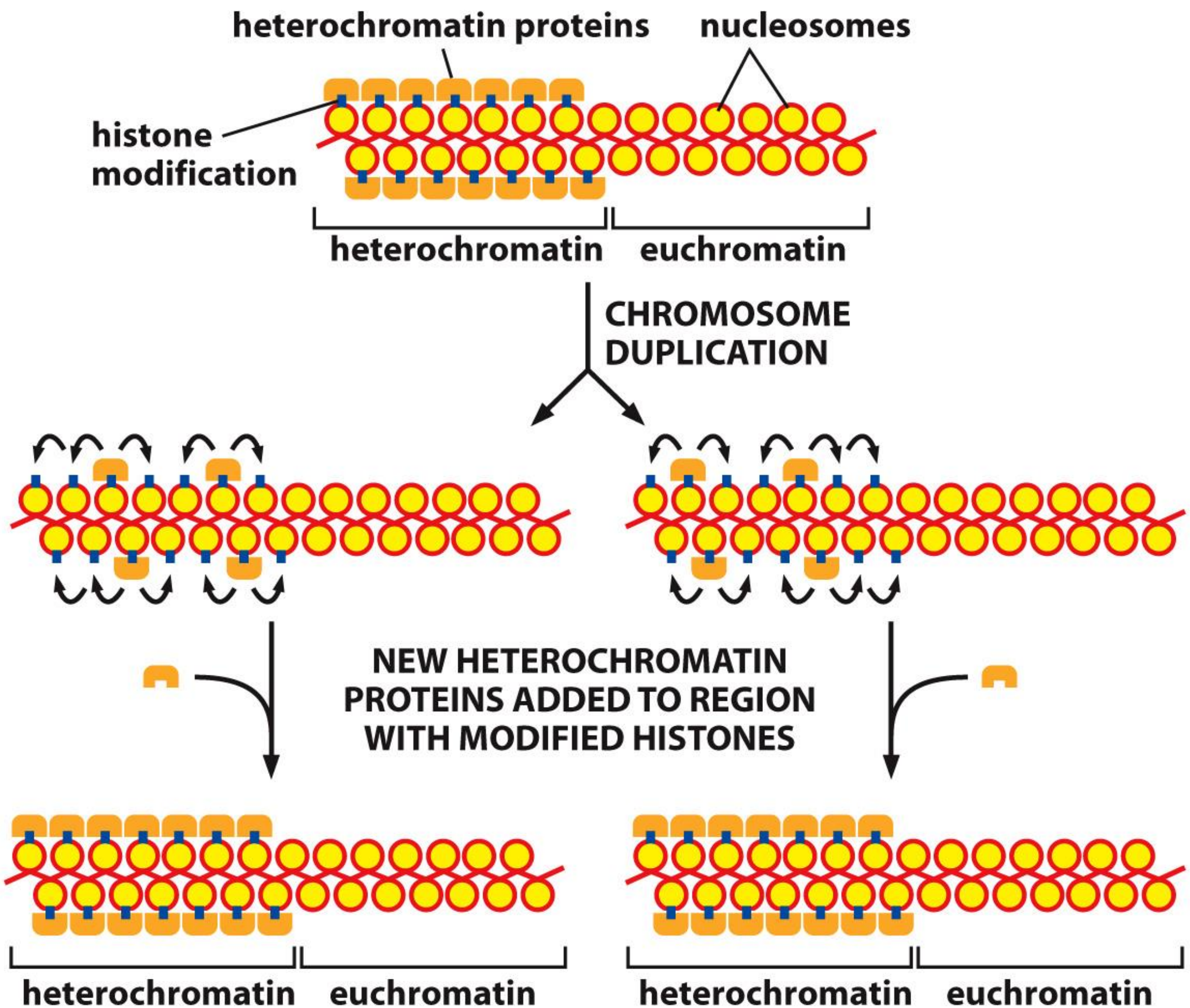


Figure 4-44 Molecular Biology of the Cell 6e (© Garland Science 2015)

Both Activating and Repressive Chromatin Structures Can Be Inherited Epigenetically

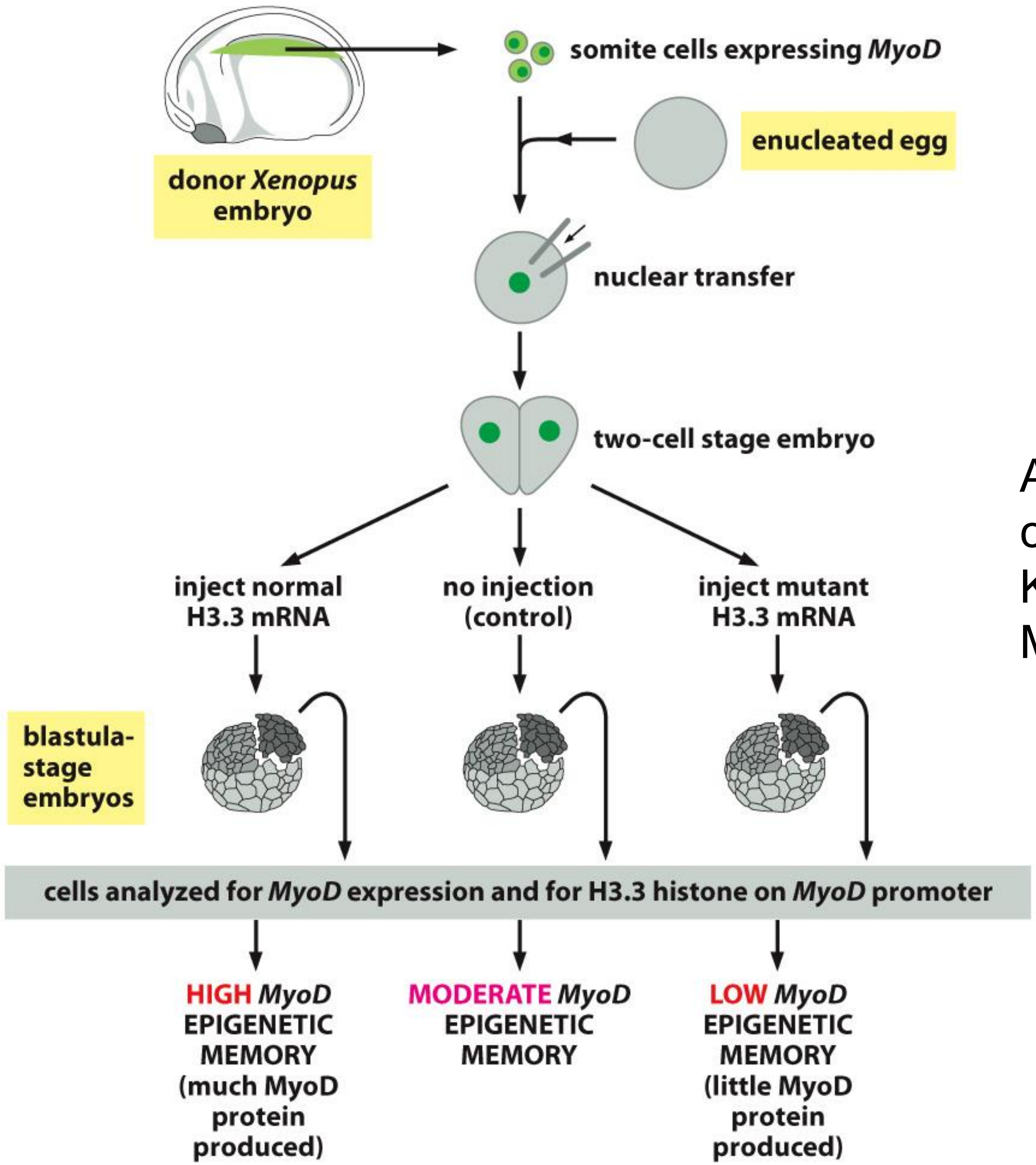


Figure 4-45 Molecular Biology of the Cell 6e (© Garland Science 2015)

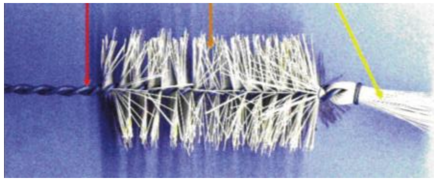
A mutant form of H3.3 that cannot be methylated at K4 reduces the epigenetic MyoD production.

CHROMOSOME HIGH-ORDER STRUCTURE

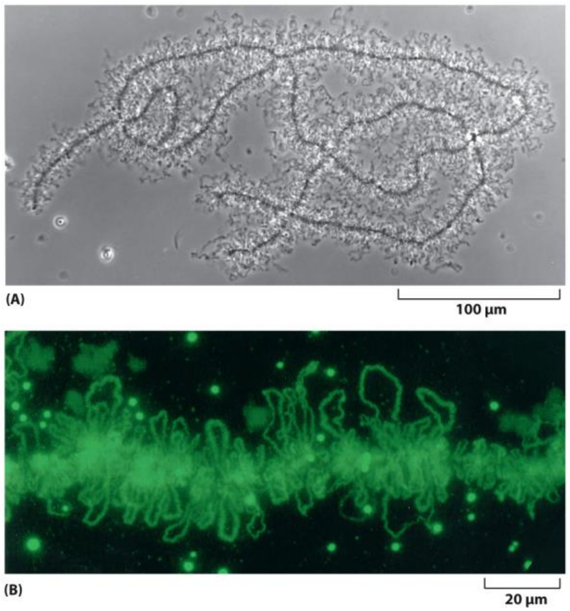
- High-order structure involves the folding of the chromatin into a series of loops and coils.

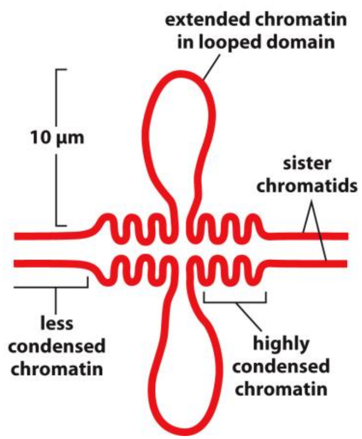
- High-order structure is fluid and hierarchical.

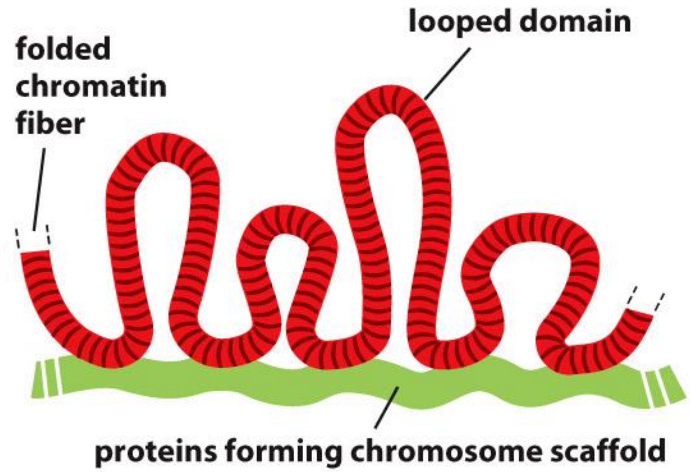
Chromosomes Are Folded into Large Loops of Chromatin



Lampbrush chromosomes
in an amphibian oocyte







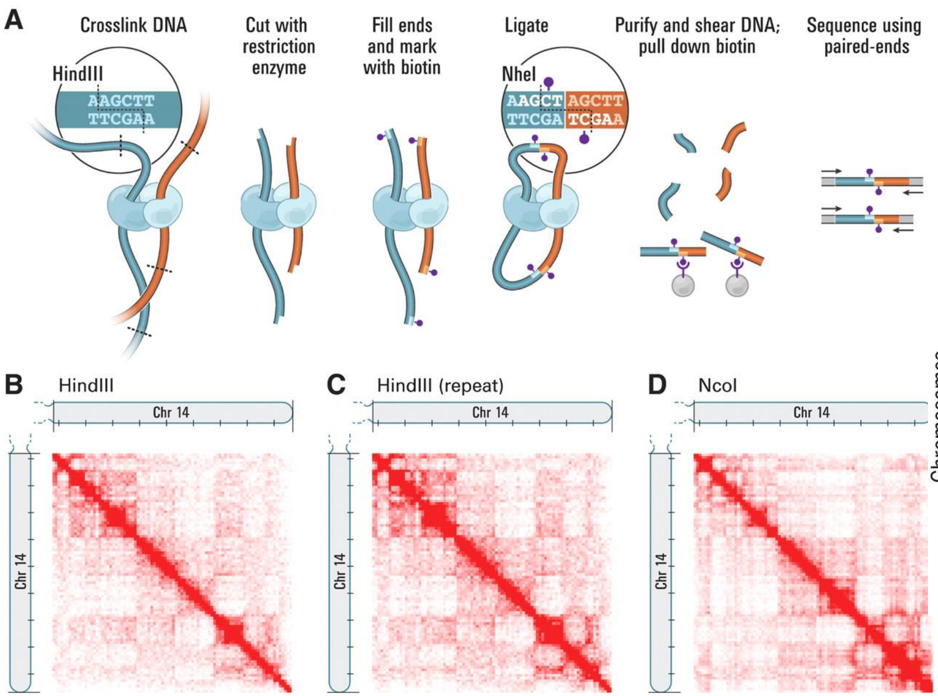
The interphase chromosomes of all eukaryotes are similarly arranged in loops but are too small and fragile to be easily observed in a light microscope.

A typical loop might contain between 50,000 and 200,000 nucleotide pairs of DNA.

Figure 4-46, 47, 49 *Molecular Biology of the Cell* (© Garland Science 2015)

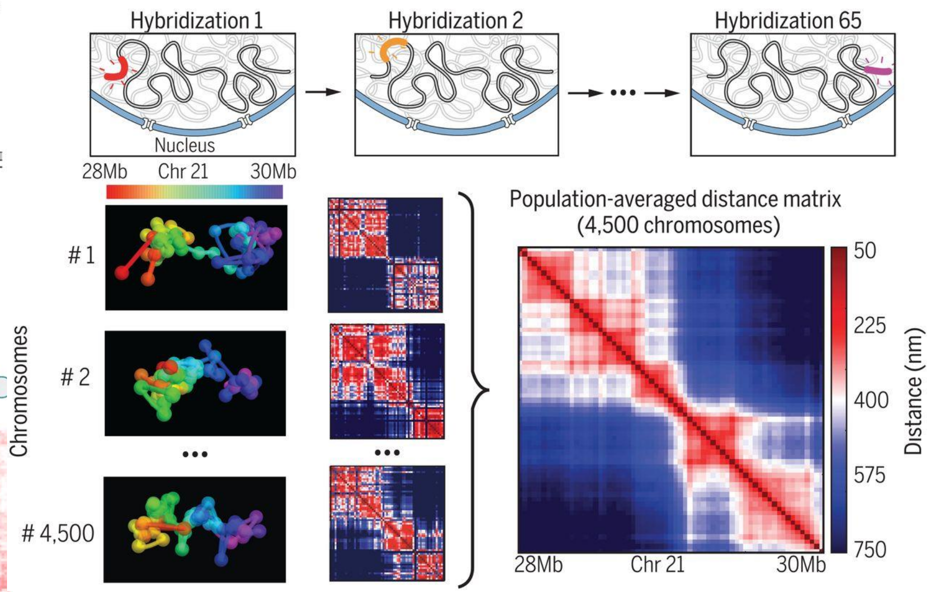
Methods to Dissect Chromosome High-order Structure

Chromosome conformation capture (3C)



Lieberman-Aiden et al., Science, 2009

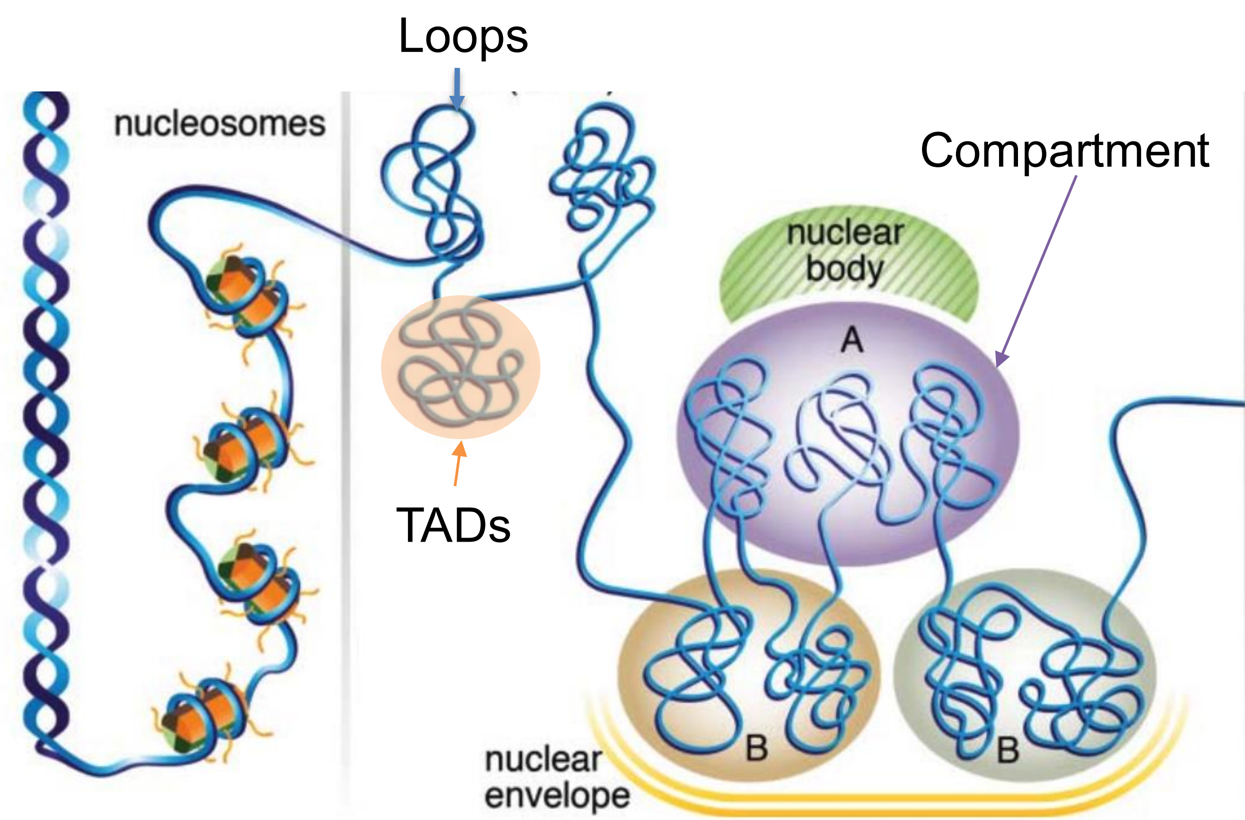
Chromosome Tracing



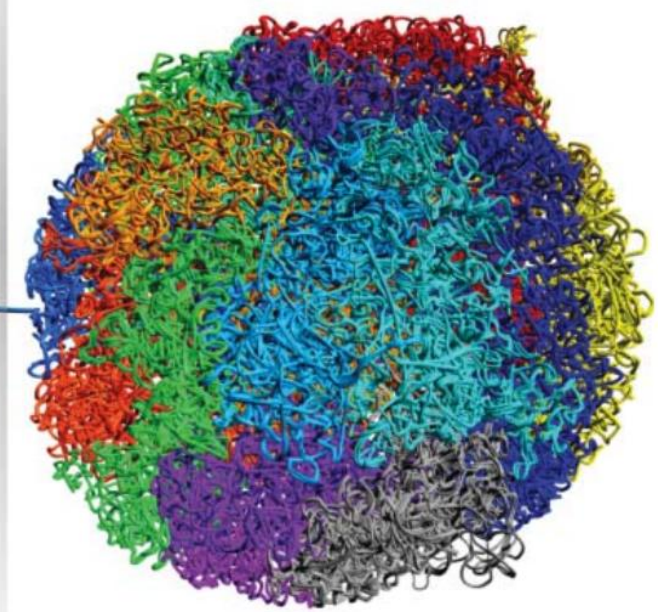
Bintu et al., Science, 2018

Other Methods:
SPRITE, *Quinodoz et al., Cell, 2018*
GAM, *Beagrie et al., Nature, 2017*

High-order Chromosome Conformation



Fractal Globule
Chromosome Territories



Hansen et al., Nucleus, 2018

Homologous chromosomes are not in general co-located.

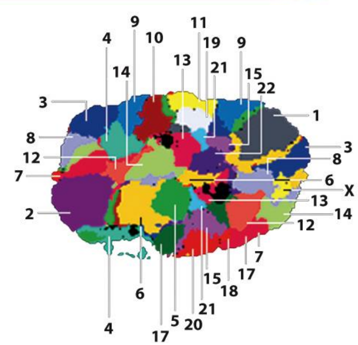
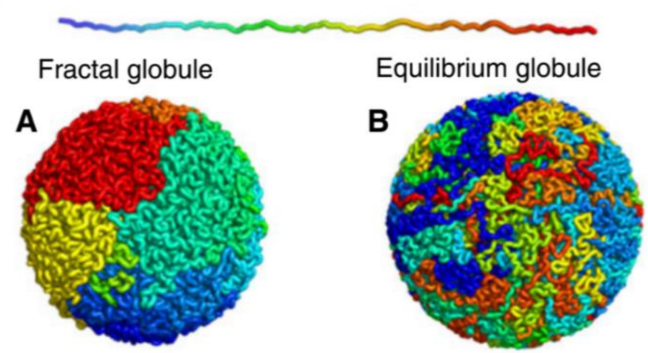


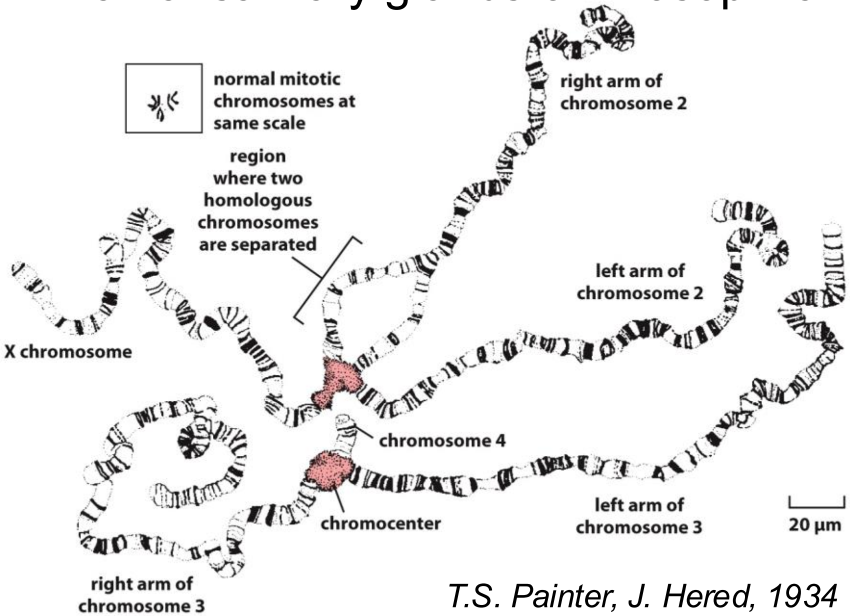
Figure 4-53 Molecular Biology of the Cell 6e (© Garland Science 2015)



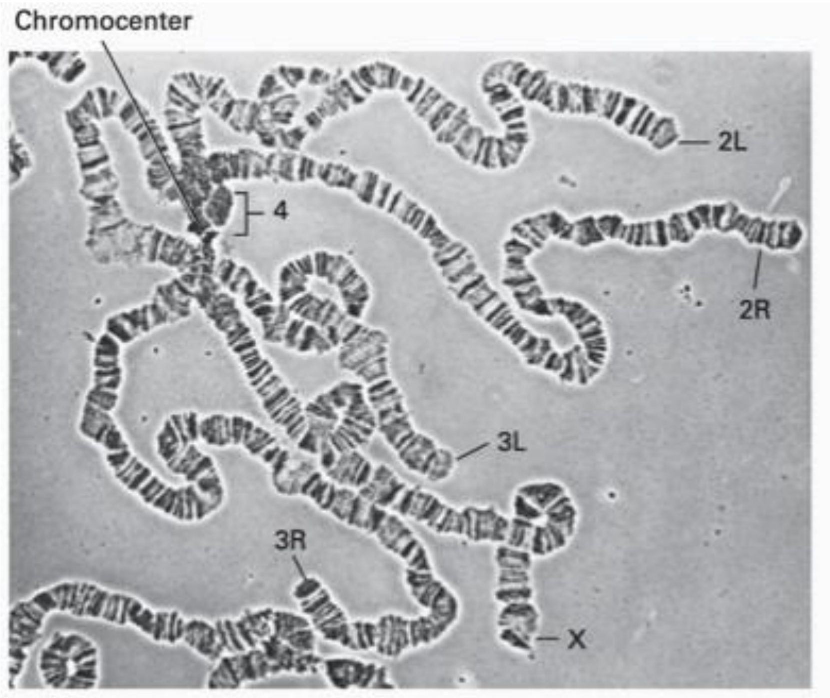
Polytene Chromosomes Are Uniquely Useful for Visualizing Chromatin Structures

Polyploid: multiple cycles of DNA synthesis without cell division

“larval salivary glands of Drosophila”



T.S. Painter, J. Hered, 1934



The four polytene chromosomes are normally linked together by heterochromatic regions near their centromeres that aggregate to create a single large **chromocenter.**

Polytene Chromosomes Are Uniquely Useful for Visualizing Chromatin Structures

Light Micrograph

Electron Micrograph

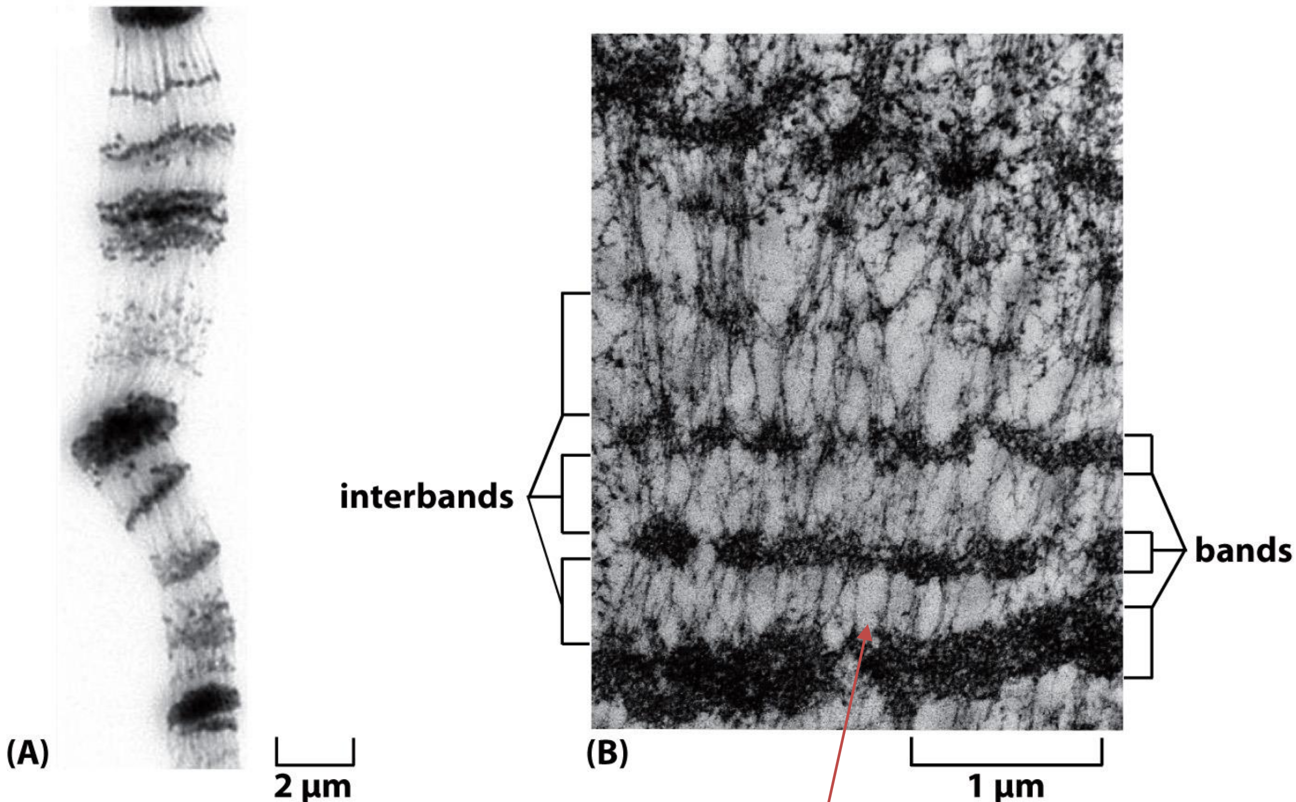
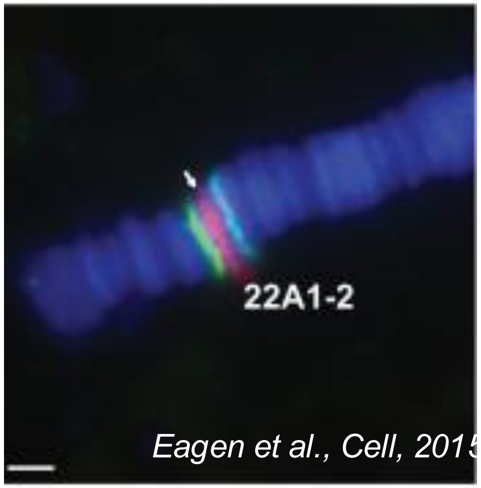


Figure 4-51 Molecular Biology of the Cell 6e (© Garland Science 2015)

FISH



Scale bar, 2 µm

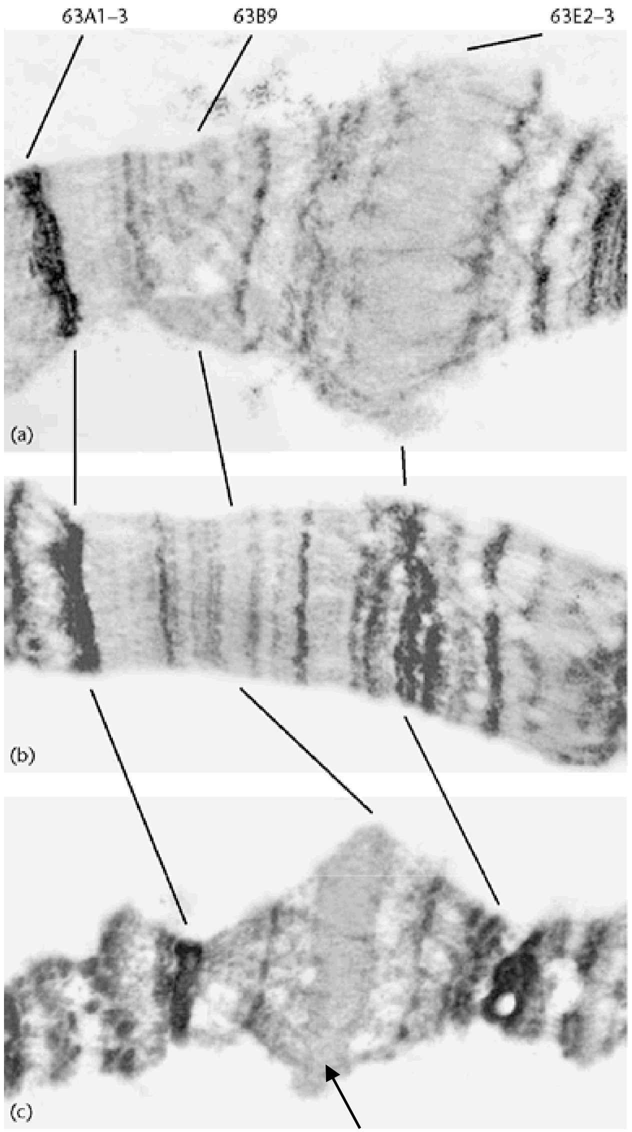
hundreds or thousands of copies of the genome

individual chromatin fibers are aligned side by side

Polytene Chromosome Puffs to Promote Gene Expression

Electron microscopic view

induction of ecdysone



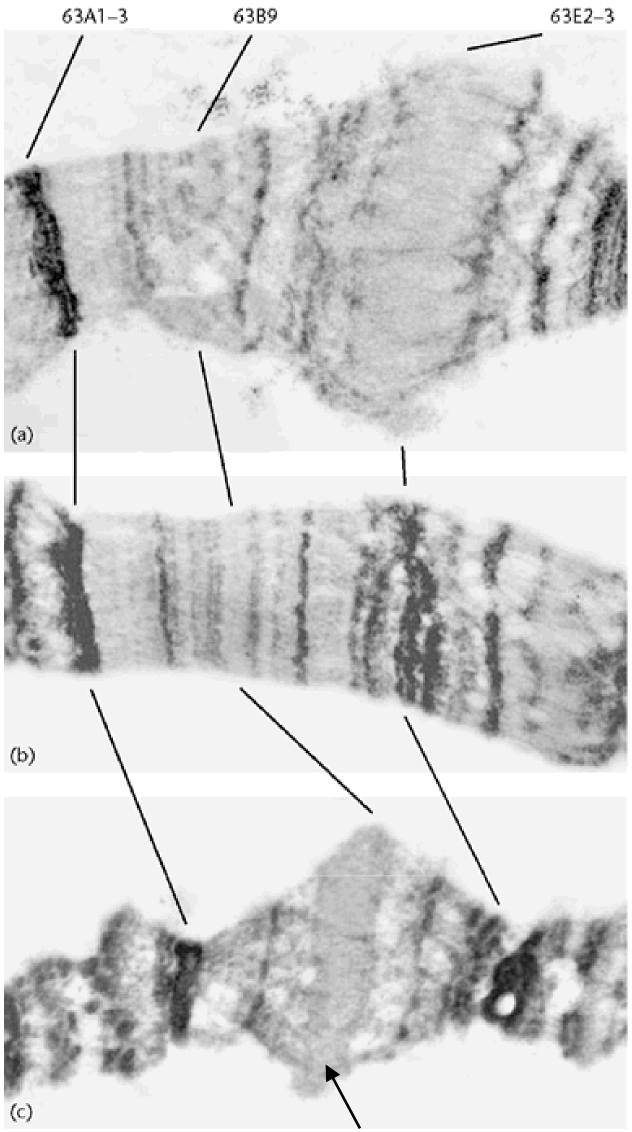
heat shock

chromosome puff

Polytene Chromosome Puffs to Promote Gene Expression

Electron microscopic view

induction of ecdysone



heat shock

chromosome puff

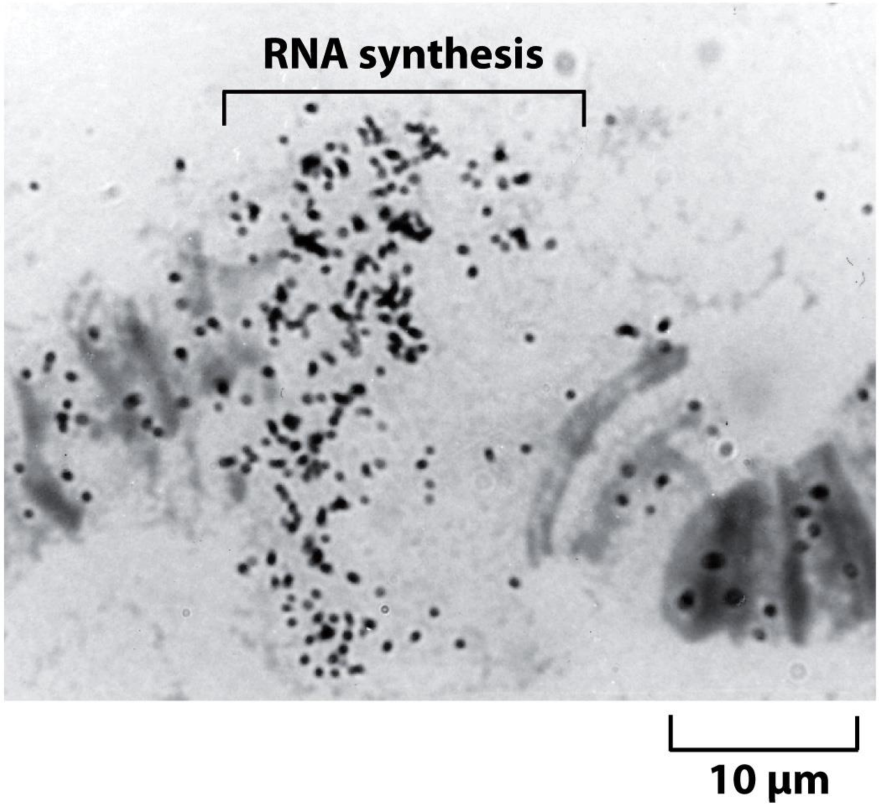
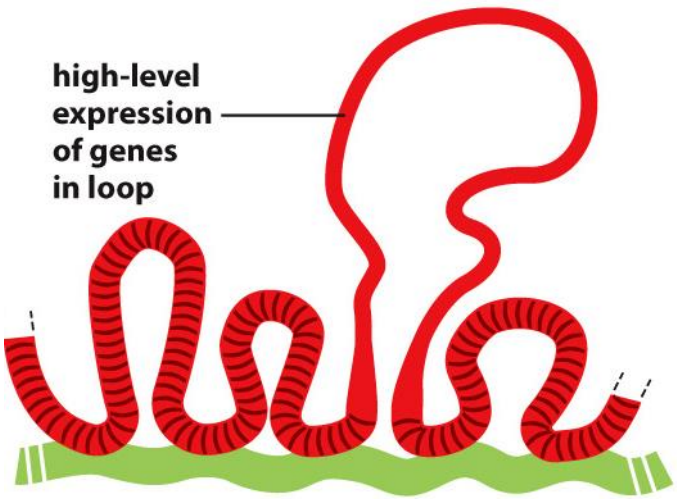


Figure 4-52 Molecular Biology of the Cell 6e (© Garland Science 2015)

The decondensed portion of the chromosome is undergoing RNA synthesis and has become labeled with 3H-uridine, an RNA precursor molecule that is incorporated into growing RNA chains.

Chromatin Loops Decondense When the Genes Within Them Are Expressed

The region of the chromosome adjacent to the gene (red) is seen to leave its chromosomal territory (green) only when it is highly active.



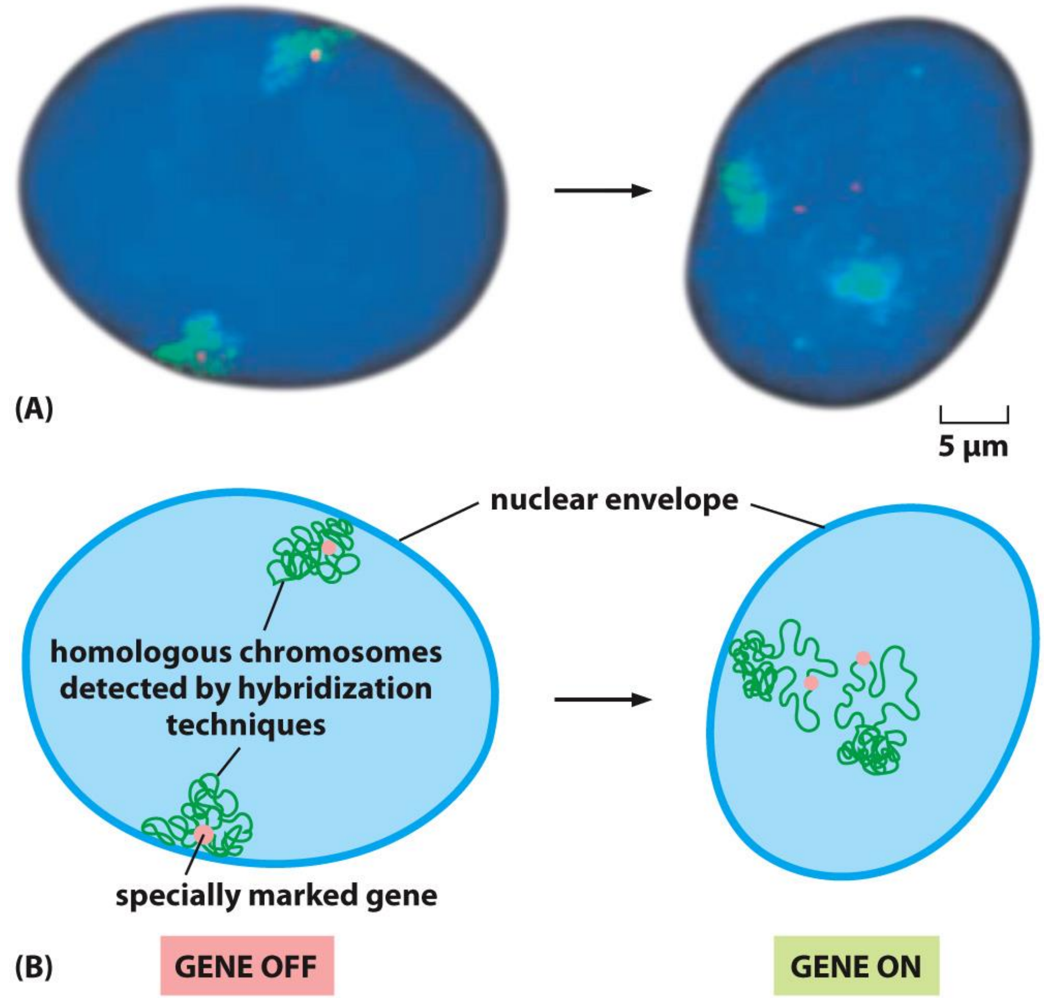


Figure 4-49, 56 *Molecular Biology of the Cell* (© Garland Science 2015)

Mitotic Chromosomes Are Especially Highly Condensed

Additional 10-fold compaction in the passage from interphase to mitosis.

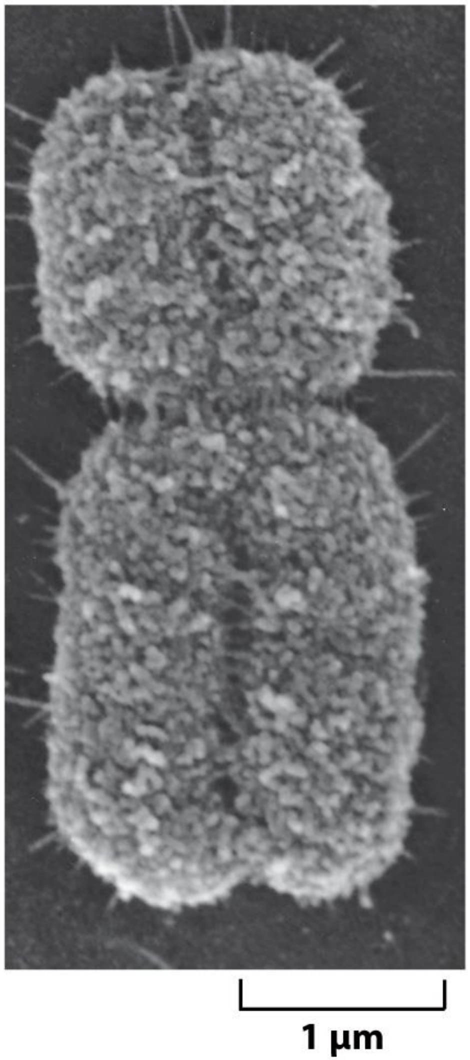


Figure 4-18 Molecular Biology of the Cell 6e (© Garland Science 2015)

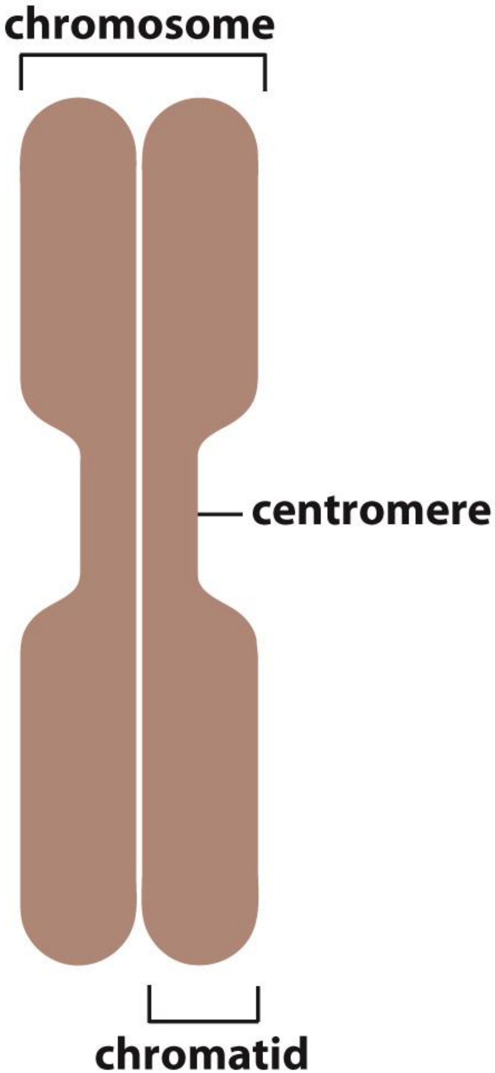


Figure 4-59 Molecular Biology of the Cell 6e (© Garland Science 2015)

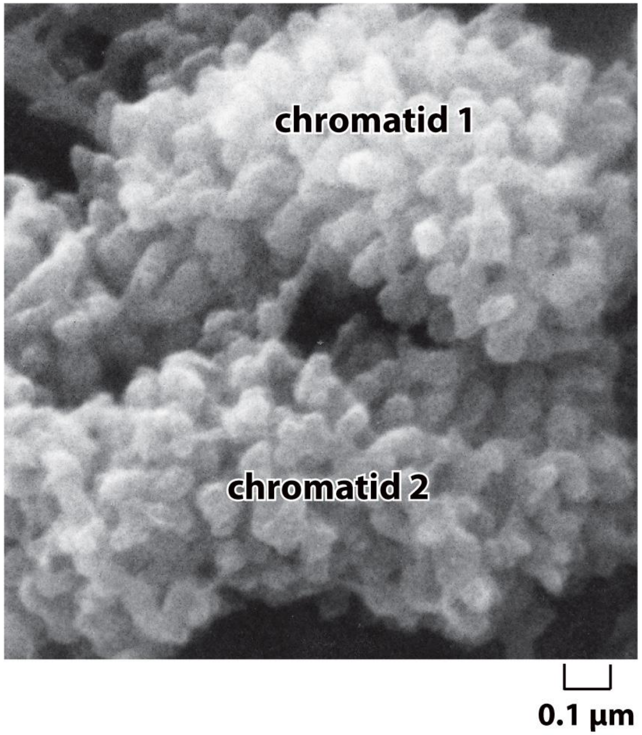


Figure 4-60 Molecular Biology of the Cell 6e (© Garland Science 2015)

HOW GENOMES EVOLVE

The genes and genomes have evolved over time to produce the vast diversity of modern-day life-forms on our planet.

- Sequence similarity has become a major tool for inferring gene and protein function.
- DNA sequences that have a function are much more likely to be conserved than those without a function.
- Some regions are conserved while the others are nonconserved.
- Homologous genes—genes that are similar in both their nucleotide sequence and function because of a common ancestry—can often be recognized across vast phylogenetic distances.

The Detection of Multispecies Conserved Sequences

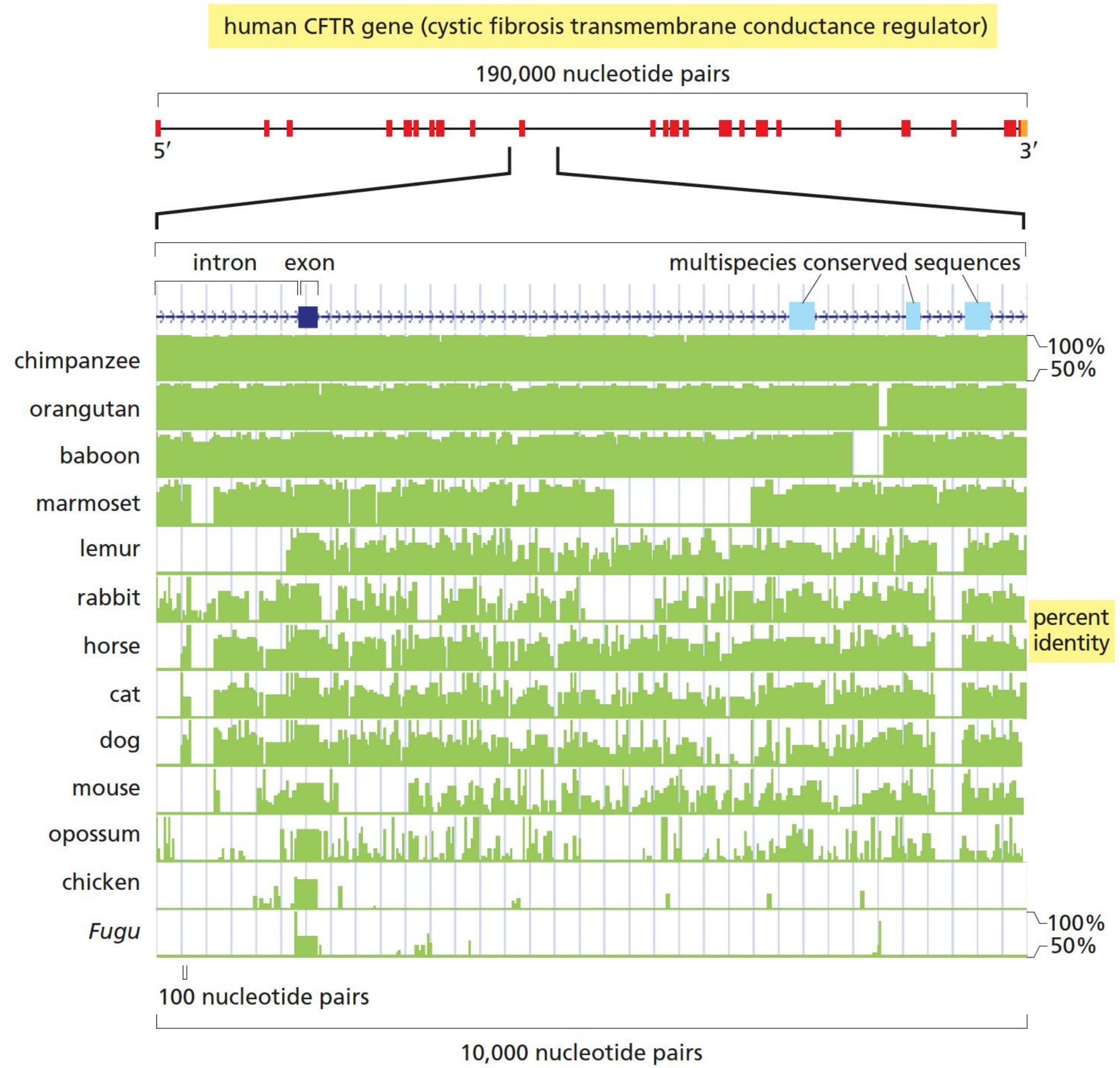


Figure 4-73 *Molecular Biology of the Cell* (© Garland Science 2015)

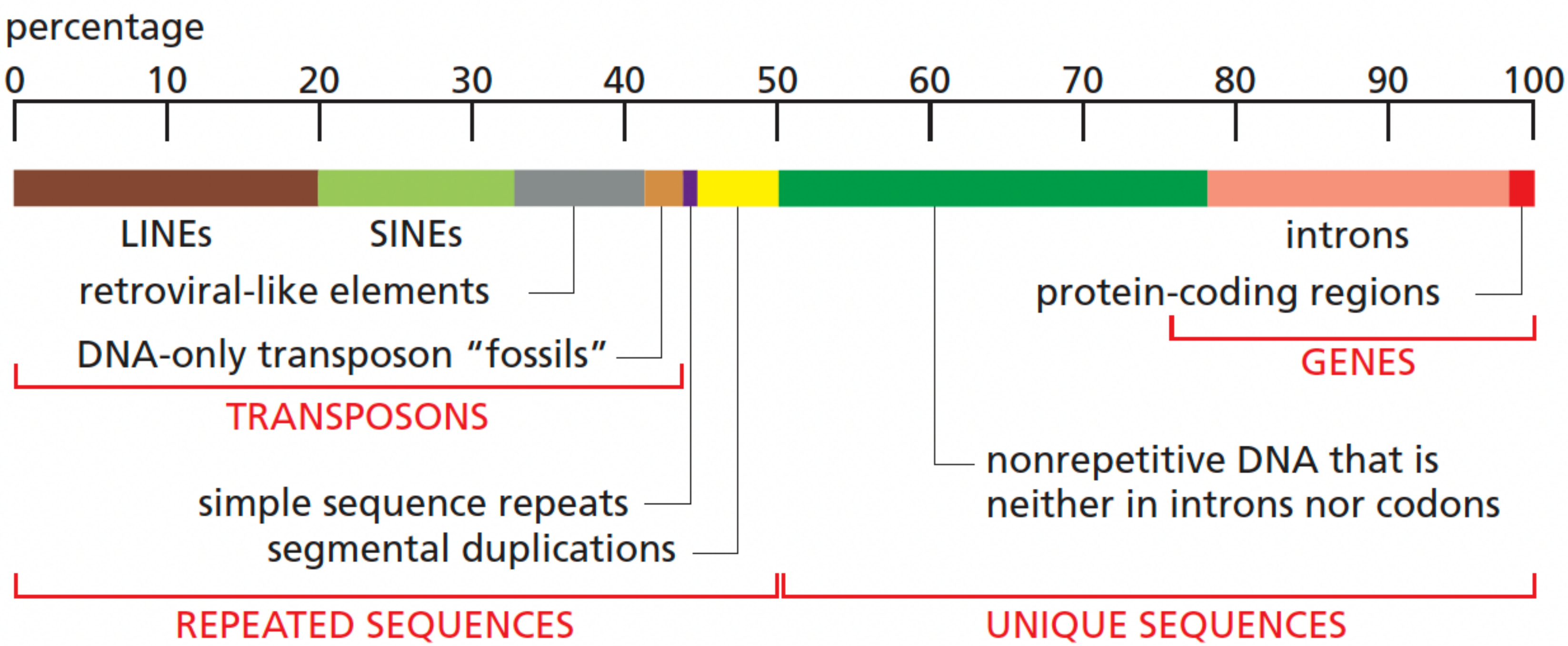
Conservation indicates functional importance.

Some of these code for protein, but most of them do not, and the function remains a mystery.

Roughly 5% of the human genome consists of “multispecies conserved sequences.”

Genome Alterations Are Caused by…

Errors in DNA replication, DNA repair, as well as by Transposable DNA Elements.



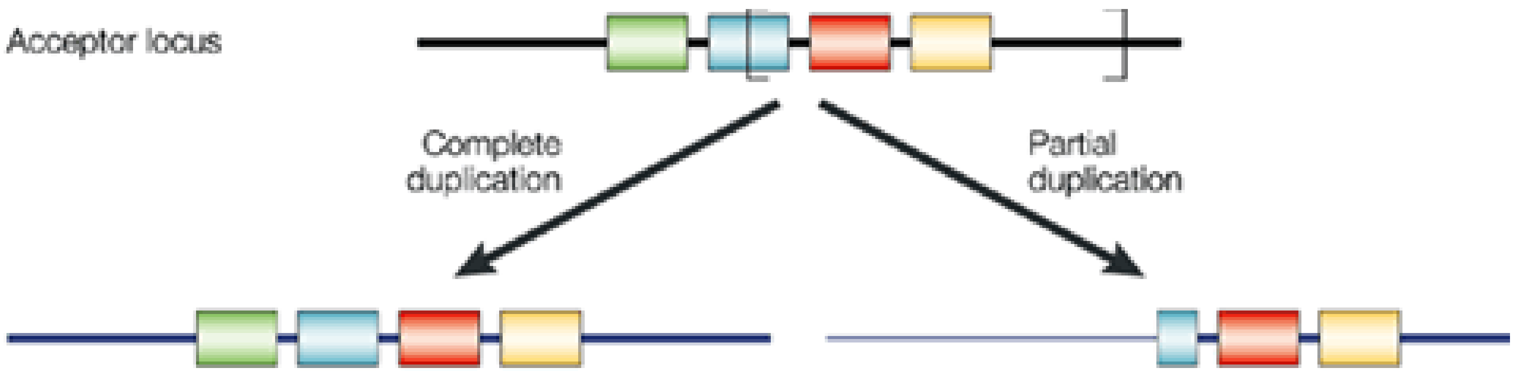
The LINEs (long interspersed nuclear elements), SINEs (short interspersed nuclear elements), retroviral-like elements, and DNA-only transposons are mobile DNA elements that have multiplied in our genome by replicating themselves and inserting the new copies in different positions.

Figure 4-62 *Molecular Biology of the Cell* (© Garland Science 2015)

Gene Duplication Provides Source of Genetic Novelty in Evolution

Evolution depends on the creation of new genes, as well as on the modification of those that already exist. Genes without homologous counterparts are relatively scarce.

Segmental Duplications (1 to 400 kb in length, “low-copy repeats”) have added about 5 million nucleotide pairs to each genome every million years.

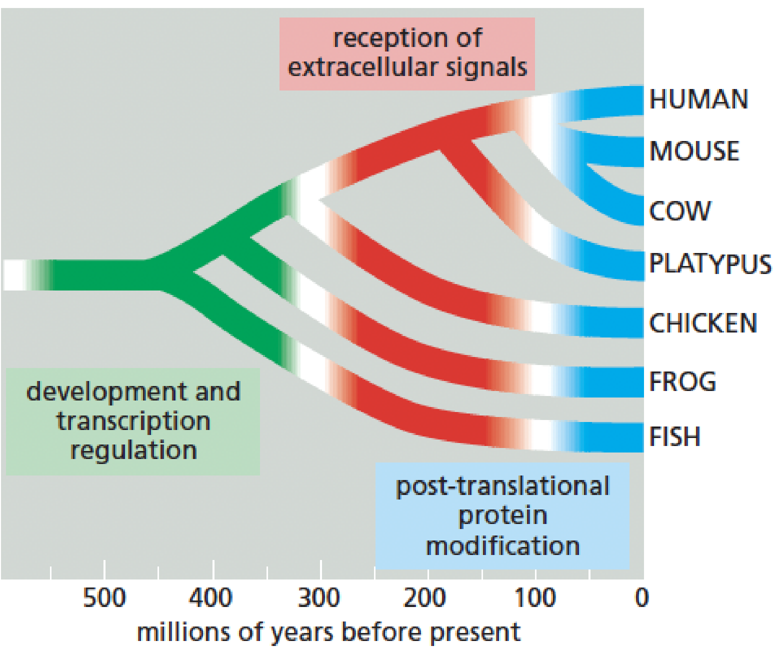


Samonte, R., Eichler, E. Nat Rev Genet (2002)

Many duplication events are likely to be followed by loss-of-function mutations in one or the other gene, which eventually becomes pseudogene.

Genetic Variations Contribute to Organism Evolution

Mutations in the DNA sequences that control gene expression have driven many of the evolutionary changes in vertebrates.



The evolution of the globin gene family shows how DNA duplications contribute to the evolution of organisms

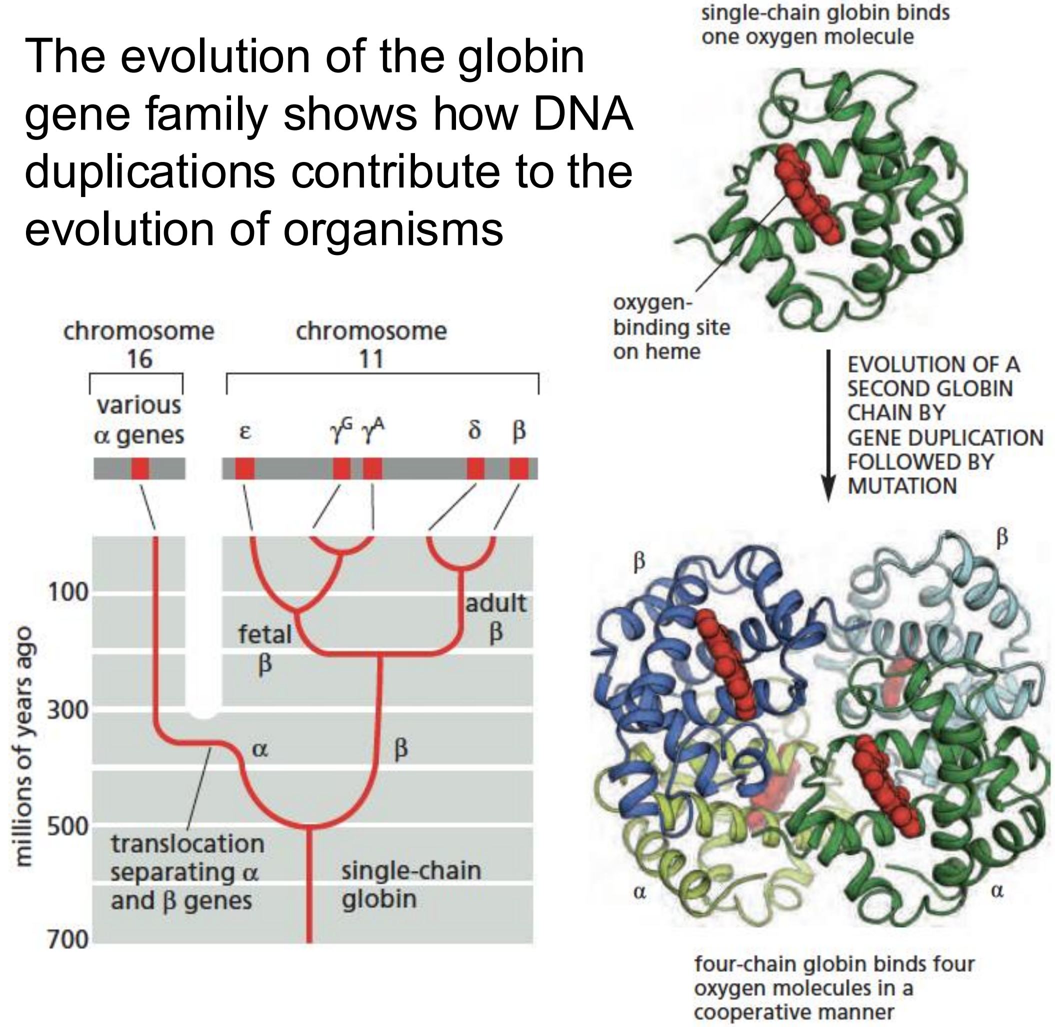
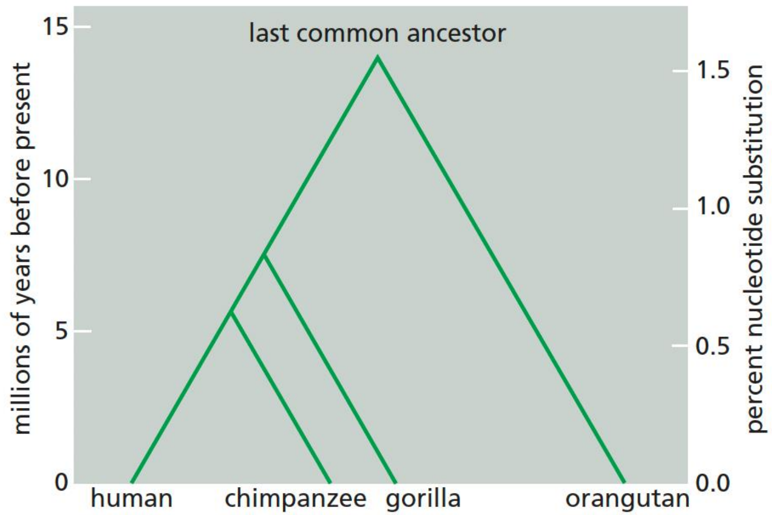


Figure 4-74, 75, 76 *Molecular Biology of the Cell* (© Garland Science 2015)

Genome Similarity is Corresponding to The Evolution Time

The basic organizing framework for comparative genomics is the phylogenetic tree, which provides a molecular clock for evolution.



divergences
~1.2%

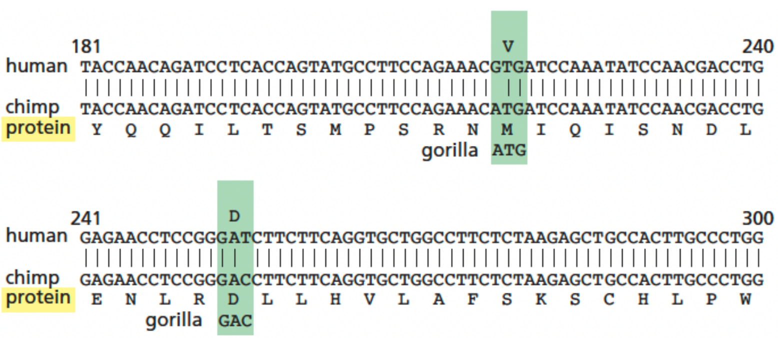
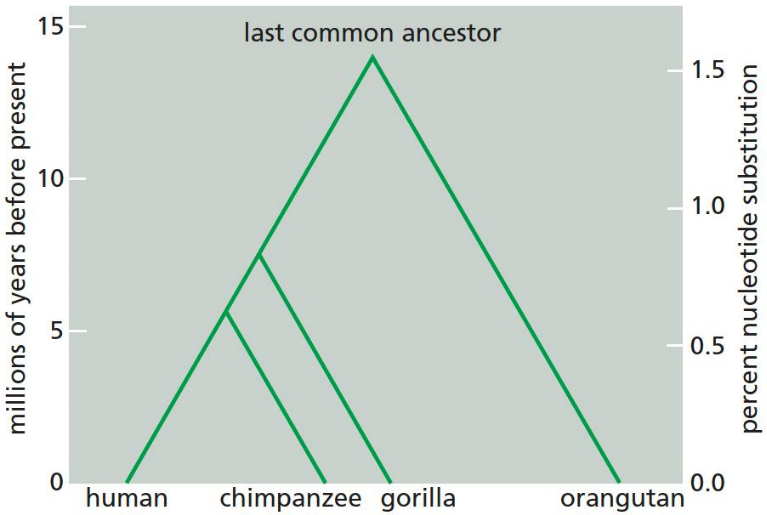


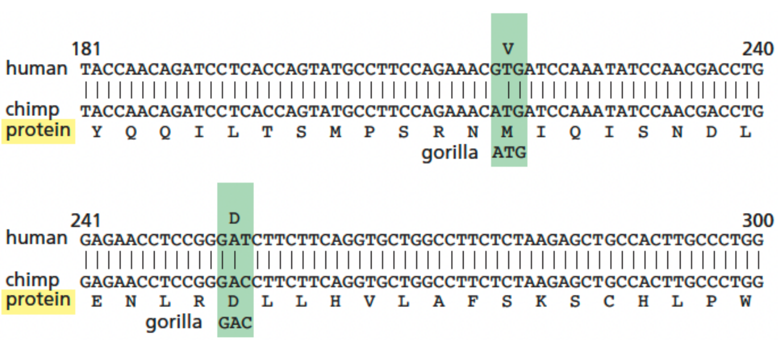
Figure 4-63, 64, 66 *Molecular Biology of the Cell* (© Garland Science 2015)

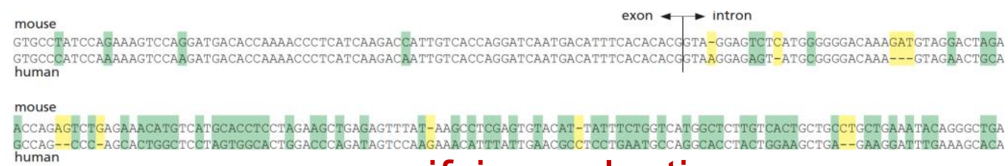
Genome Similarity is Corresponding to The Evolution Time

The basic organizing framework for comparative genomics is the phylogenetic tree, which provides a molecular clock for evolution.

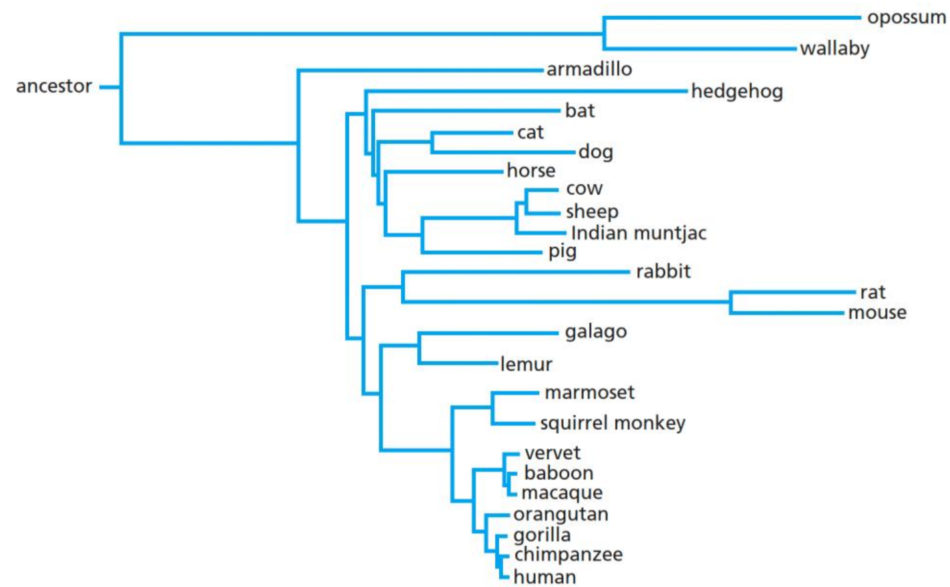


divergences
~1.2%





purifying selection



The molecular clock for evolution is determined not only by the degree of purifying selection but also by the mutation rate.

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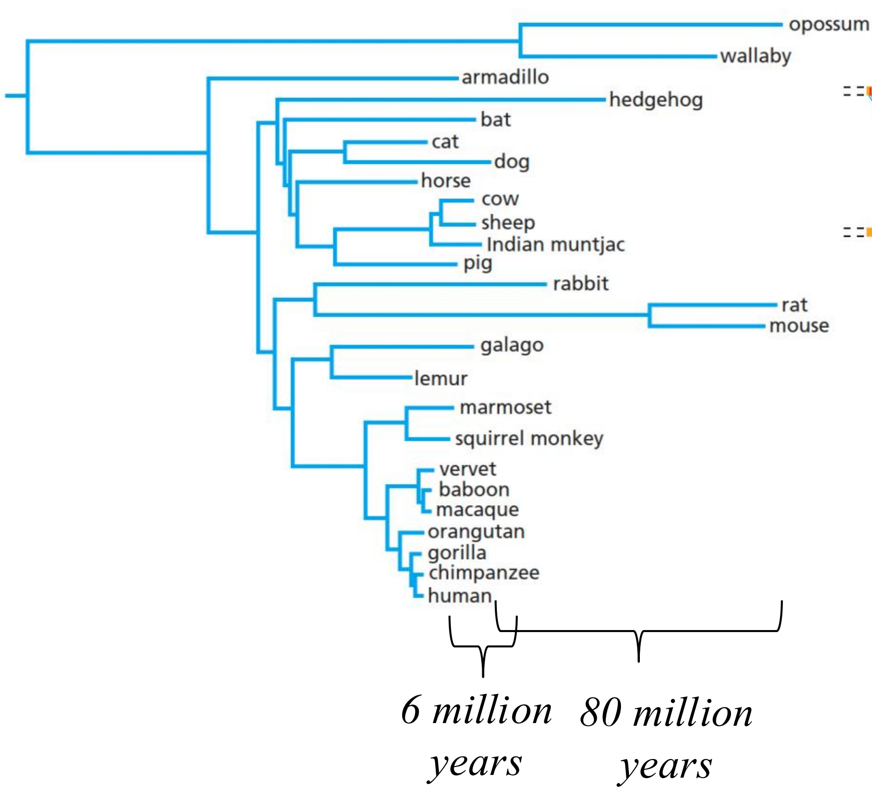
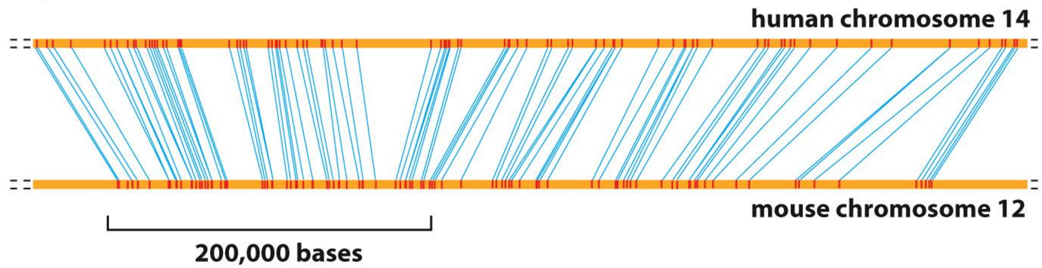
The Divergence of Genome Structures

Comparison of Human and Mouse Genomes——*roughly the same size and contain nearly identical sets of genes*

According to rough estimates, a total of about 180 breakage-and-rejoining events have occurred in the human and mouse lineages.





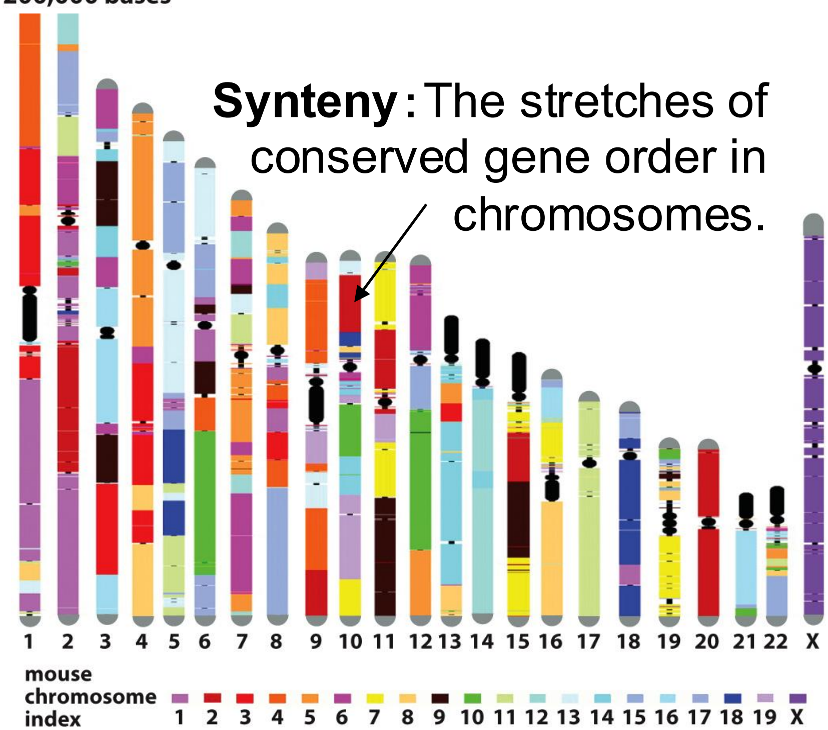
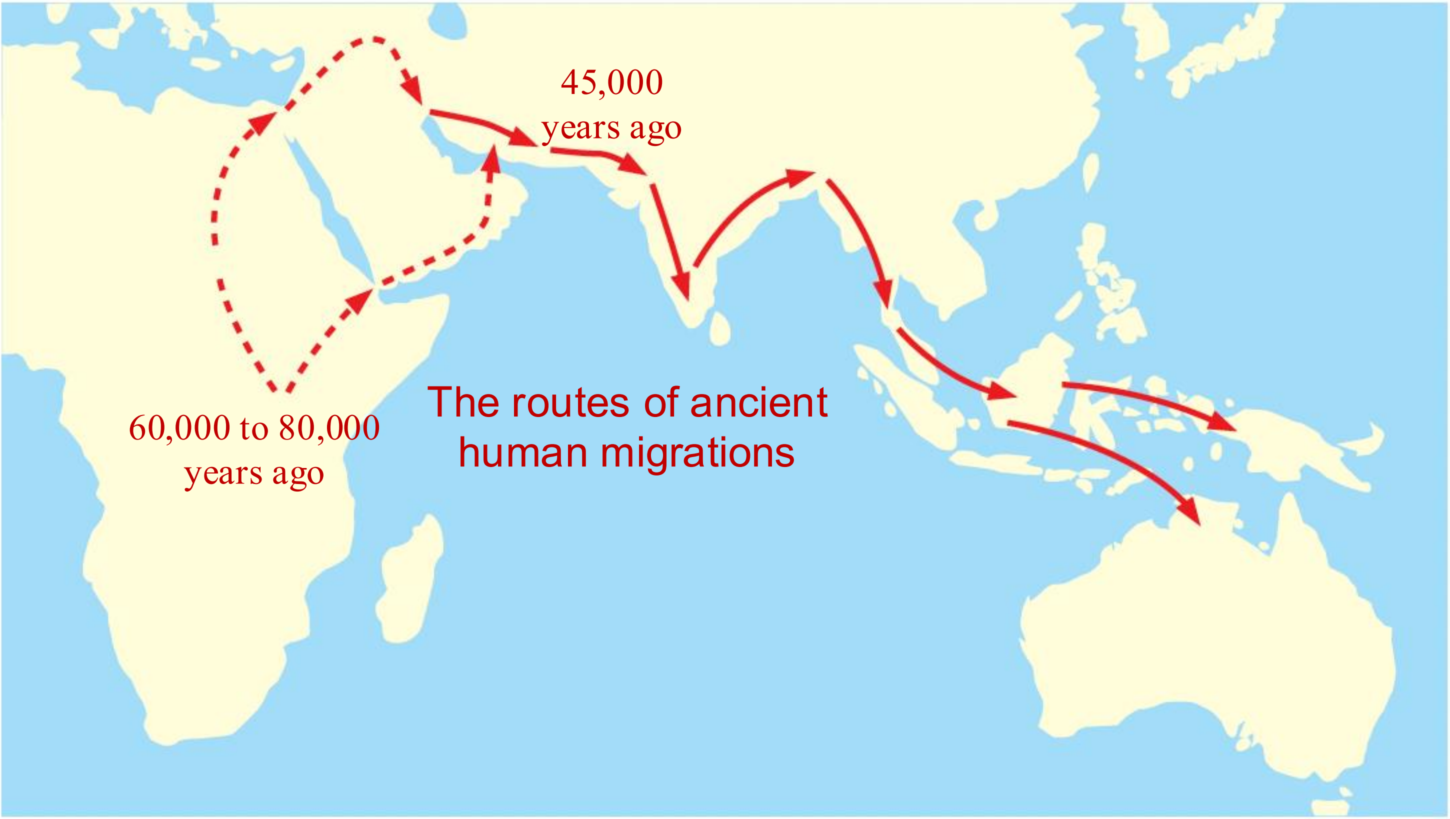


Figure 4-66, 67, 68 *Molecular Biology of the Cell* (© Garland Science 2015)

Variation Among Humans——Population Genetics

Tracing the course of human history by analyses of genome sequences.



Copy Number Variations (CNVs)

Single-nucleotide Polymorphisms (SNPs)

Have substantial effects on human health, physiology, behavior, and physique.

Figure 4-79 *Molecular Biology of the Cell* (© Garland Science 2015)